# Optimization — Exam Study Agenda

Statements (without solutions), pseudocode and validation tables

IMCA — Master's in Mathematical and Computational Modeling

Scope: `ch1`, `ch2-v2`, `ch3`, `ch3_13-15`, `ch4`, `ch5`, `ch6`

**About this document.** Exam preparation for ch1–ch6: every proof, counterexample and exercise of the notes, plus added practice, organized by unit. Only statements are given; solutions are developed separately. Codes: numbered items come from the notes (* = proposed exercise); **P** = added practice; **A** = pseudocode; **V** = validation table to check an implementation row by row. Each item is self-contained, and a **Note** flags a gap or error in the notes. Click a unit in the index to jump to it; "↑Index" brings you back.

## Index


| Unit | Topic | Sequence | Page |
|---|---|---|---|
| U0 | Taylor's theorem | 0.1 | 2 |
| U1 | Convexity | 1.5, 1.1, 1.2, 1.9* | 2 |
| U2 | Optimality conditions | 1.6, 1.3, 1.7, 1.4, 1.8*, 1.10* | 3 |
| U3 | Armijo and Wolfe | 2.10, 2.14*, P1, A1, V1, P2, 2.1, A2, P3, V2, 2.13* | 3 |
| U4 | Zoutendijk | 2.2, 2.11 | 6 |
| U5 | Strict and strong convexity | 2.3, 2.4, 2.5 | 7 |
| U6 | Rate of steepest descent | P4, A3, V3, V4, 2.6, 2.7, 2.8, 2.12, 2.9, 2.15* | 7 |
| U7 | Newton and modified Hessian | 3.16, A4, V5, 3.1, 3.12, 3.13, 3.20*, 3.17, 3.2, 3.21*, P6, V6 | 10 |
| U8 | DFP and BFGS formulas | 3.14, 3.15, 3.3, 3.5, 3.7, 3.18, 3.4, 3.6, 3.8, 3.22*, A5 | 12 |
| U9 | Quasi-Newton convergence | P5, V7, 3.10, 3.9, 3.11, 3.19, V8 | 14 |
| U10 | Optimality over convex sets | P7, 4.1, 4.2, 4.30*, P9, A6, V9 | 16 |
| U11 | Tangent cone and qualification | 4.15, 4.3, 4.4, 4.5, 4.21, 4.6, 4.26*, 4.27*, 4.7, 4.24, 4.8, 4.29* | 18 |
| U12 | Farkas and KKT | 4.16, 4.9, 4.10, 4.20, P10, A7, V10, 4.25*, 4.31*, 4.17, 4.18, 4.19, 4.32*, 4.33* | 20 |
| U13 | Lagrangian and duality | P8, 4.11, 4.22, 4.12, 4.13, 4.28*, 4.23, 4.14, 4.34* | 22 |
| U14 | Trust region: radius | 5.16, P12, 5.25* | 24 |
| U15 | Exact subproblem | 5.9, 5.1, 5.18*, 5.13, 5.2, A9, V13, 5.10, V12, 5.23*, 5.24*, 5.3, 5.17, P13, V14, 5.20* | 24 |
| U16 | Cauchy point and Dogleg | 5.11, 5.14, 5.4, 5.5, 5.21*, 5.12, 5.15, 5.6, 5.19*, P11, A8, V11, 5.7, 5.22* | 29 |
| U17 | Global convergence (TR) | A10, P14, V15, 5.8, 5.26*, 5.27* | 33 |
| U18 | Conjugate directions and linear CG | 6.9, P15, V16, 6.1, 6.15*, 6.2, 6.4, 6.3, 6.14*, 6.16*, 6.8, P16, A11, V17 | 35 |
| U19 | Krylov subspaces and rate of CG | P17, V18, 6.5, 6.7, 6.6, 6.20*, 6.22*, P18, V19 | 39 |
| U20 | Nonlinear CG and L-BFGS | 6.10, 6.17*, P19, 6.18*, 6.23*, 6.21*, 6.25*, A12, P20, V20, V21, 6.19*, 6.11, 6.12, 6.24* | 42 |

## U0. Taylor's theorem

↑Index

**0.1** Proof *ch2-v2, Theorem 1.1, p. 1*

Let $f:\mathbb{R}^n\to\mathbb{R}$ be of class $C^1$ and $p\in\mathbb{R}^n$. Prove that

$$f(x+p)=f(x)+\int_0^1 \nabla f(x+tp)^{\mathsf{T}}p\,dt, \tag{T1}$$

$$f(x+p)=f(x)+\nabla f(x+tp)^{\mathsf{T}}p \quad \text{for some } t\in(0,1). \tag{T2}$$

If moreover $f$ is of class $C^2$, prove that

$$\nabla f(x+p)=\nabla f(x)+\int_0^1 \nabla^2 f(x+tp)\,p\,dt, \tag{T3}$$

$$f(x+p)=f(x)+\nabla f(x)^{\mathsf{T}}p+\int_0^1 (1-t)\,p^{\mathsf{T}}\nabla^2 f(x+tp)\,p\,dt, \tag{T4}$$

$$f(x+p)=f(x)+\nabla f(x)^{\mathsf{T}}p+\tfrac{1}{2}\,p^{\mathsf{T}}\nabla^2 f(x+tp)\,p \quad \text{for some } t\in(0,1). \tag{T5}$$

Explain also why (T3) does not admit, in general, a mean-value form analogous to (T2).

## U1. Convexity

↑Index

**1.5** Counterexample *ch1, p. 3 (version corrected in class)*

Show that the conclusion "every local minimum is global" fails if convexity of $f$ or of the domain is removed.

a) $f(x)=x^3-3x$ on $\mathbb{R}$: prove that $x^*=1$ is a local minimum and that $f$ has no global minimum.

b) Square donut: $\Omega=\{x\in\mathbb{R}^2:\ 1\le\|x\|_\infty\le 2\}$ and $f(x_1,x_2)=x_1$. Prove that $f$ is convex, that $\Omega$ is not convex, that $(1,0)$ is a local minimum of $f$ on $\Omega$, and that it is not a global minimum.

**1.1** Proof *ch1, Theorem 2.1, p. 2*

Let $C\subseteq\mathbb{R}^n$ be convex and nonempty and $f:C\to\mathbb{R}$ convex. Prove that if $x^*\in C$ is a local minimum of $f$ on $C$, then it is a global minimum of $f$ on $C$.

**1.2** Proof *ch1, Proposition 2.1, pp. 3–4*

Let $C\subseteq\mathbb{R}^n$ be open and convex and $f:C\to\mathbb{R}$ of class $C^1$.

a) Prove that $f$ is convex on $C$ if and only if $f(y)\ge f(x)+\nabla f(x)^{\mathsf{T}}(y-x)$ for all $x,y\in C$.

b) If $f$ is $C^2$, prove that $f$ is convex on $C$ if and only if $\nabla^2 f(x)\succeq 0$ for all $x\in C$.

Indicate where in the proof the openness of $C$ is used.

**1.9*** Proposed exercise *ch1, p. 8*

For $\delta>0$, the Huber loss is

$$\rho_\delta(t)=\begin{cases}\tfrac{1}{2}t^2, & |t|\le\delta,\\ \delta|t|-\tfrac{1}{2}\delta^2, & |t|>\delta.\end{cases}$$

Prove that $\rho_\delta$ is convex and of class $C^1$ on $\mathbb{R}$, and find $\rho_\delta'(t)$.

## U2. Unconstrained optimality conditions

↑Index

**1.6** Counterexample *ch1, p. 6*

Show that $\nabla f(x^*) = 0$ is not sufficient for a local minimum:

a) $f(x) = x^3$ at $x^* = 0$.

b) $f(x_1, x_2) = x_1^2 - x_2^2$ at $x^* = (0, 0)$ (saddle point).

**1.3** Proof *ch1, Theorem 3.1 (FONC), p. 5*

Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuously differentiable on an open neighbourhood of $x^*$. Prove that if $x^*$ is a local minimum of $f$, then $\nabla f(x^*) = 0$.

**1.7** Counterexample *ch1, p. 7*

Show that $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \succeq 0$ do not imply a local minimum:

a) $f(x) = x^3$ at $x^* = 0$.

b) $f(x) = -x^4$ at $x^* = 0$ (strict local maximum).

**1.4** Proof *ch1, Theorem 3.2, pp. 6–7*

Let $f : \mathbb{R}^n \to \mathbb{R}$ be of class $C^2$ on a neighbourhood of $x^*$.

a) (SONC) If $x^*$ is a local minimum, then $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \succeq 0$.

b) (SOSC) If $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*) \succ 0$, then $x^*$ is a strict local minimum.

**1.8*** Proposed exercise *ch1, p. 8*

Let $f(x_1, x_2) = 2x_1^2 + x_2^2 - 2x_1x_2 + 2x_1 - 4x_2$.

a) Write $f(x) = \frac{1}{2}x^\mathsf{T}Ax + b^\mathsf{T}x$.

b) Compute $\nabla f(x)$ and $\nabla^2 f(x)$.

c) Find all stationary points and classify them with SONC/SOSC.

**1.10*** Proposed exercise *ch1, p. 8*

Let $f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$ (Rosenbrock).

a) Prove that $x^* = (1, 1)^\mathsf{T}$ is the only stationary point.

b) Verify $\nabla^2 f(x^*) \succ 0$ and conclude that $x^*$ is a strict global minimum.

c) Show that the Hessian is indefinite or singular in some regions of the plane.

**Note.** In b), SOSC only gives a strict local minimum; the global character needs a separate argument.

## U3. Line search: Armijo and Wolfe

↑Index

**2.10** Counterexample *ch2-v2, p. 4*

Let $f(x) = x^2$, $x_0 = 2$, $p_k = -1$ and $\alpha_k = (1/2)^{k+1}$, with $x_{k+1} = x_k + \alpha_k p_k$.

a) Prove that $p_k$ is a descent direction and that $f(x_{k+1}) < f(x_k)$ for all $k$.

b) Prove that $x_k \to 1$, which is not a stationary point.

c) Determine which of the conditions (Armijo, curvature) these steps violate. With $\phi(\alpha) = f(x_k + \alpha p_k)$ and $0 < c_1 < c_2 < 1$: Armijo is $\phi(\alpha) \le \phi(0) + c_1\alpha\,\phi'(0)$; weak curvature, $\phi'(\alpha) \ge c_2\,\phi'(0)$; strong curvature, $|\phi'(\alpha)| \le c_2|\phi'(0)|$. Weak Wolfe = Armijo + weak curvature; strong Wolfe = Armijo + strong curvature.

**2.14*** PROPOSED EXERCISE *ch2-v2, p. 16*

Let $f(x) = x^2$, $p_k = -\operatorname{sign}(x_k)$ and $\alpha_k = 2 + 3 \cdot 2^{-(k+1)}$.

a) Show that $p_k$ is a descent direction for all $k$.

b) Prove that the sequence oscillates and does not converge to $x^* = 0$, and explain which condition fails. With $\phi(\alpha) = f(x_k + \alpha p_k)$ and $0 < c_1 < c_2 < 1$: Armijo is $\phi(\alpha) \le \phi(0) + c_1\alpha\,\phi'(0)$; weak curvature, $\phi'(\alpha) \ge c_2\,\phi'(0)$; strong curvature, $|\phi'(\alpha)| \le c_2|\phi'(0)|$. Weak Wolfe = Armijo + weak curvature; strong Wolfe = Armijo + strong curvature.

**Note.** The original statement does not fix $x_0$; take $x_0 = 2$.

**P1** PRACTICE *added*

*Backtracking with Armijo.* Let $f(x_1, x_2) = 10x_1^2 + x_2^2$, $x_0 = (1, 1)^\mathsf{T}$, $p_0 = -\nabla f(x_0)$, $c_1 = 10^{-4}$, initial step $\alpha = 1$ and contraction factor $\rho = \frac{1}{2}$.

a) Write $\phi(\alpha)$ and $\phi'(0)$.

b) Run the backtracking by hand: list each $\alpha$ tried, the value $\phi(\alpha)$, the Armijo bound and the decision (reject or accept).

c) For the accepted step, decide whether it satisfies weak Wolfe and strong Wolfe with $c_2 = 0.9$ and with $c_2 = 0.1$. With $\phi(\alpha) = f(x_k + \alpha p_k)$ and $0 < c_1 < c_2 < 1$: Armijo is $\phi(\alpha) \le \phi(0) + c_1\alpha\,\phi'(0)$; weak curvature, $\phi'(\alpha) \ge c_2\,\phi'(0)$; strong curvature, $|\phi'(\alpha)| \le c_2|\phi'(0)|$. Weak Wolfe = Armijo + weak curvature; strong Wolfe = Armijo + strong curvature.

**A1** PSEUDOCODE *line search: backtracking with Armijo*

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**Backtracking**$(f,\, x,\, p,\, g;\ \alpha_0, \rho, c_1)$ with $g = \nabla f(x)$, $g^\mathsf{T} p < 0$

---

1: $\alpha \leftarrow \alpha_0$
2: **while** $f(x + \alpha p) > f(x) + c_1\,\alpha\,g^\mathsf{T} p$ **do** $\alpha \leftarrow \rho\,\alpha$
3: **return** $\alpha$

---

Values used in the tables: $\alpha_0 = 1$, $\rho = \frac{1}{2}$, $c_1 = 10^{-4}$.

**V1** VALIDATION TABLE *A1; solves P1*

$f(x) = 10x_1^2 + x_2^2$, $\nabla f(x) = (20x_1, 2x_2)^\mathsf{T}$, $c_1 = 10^{-4}$, $\rho = \frac{1}{2}$, $\alpha_0 = 1$, $x = (1, 1)^\mathsf{T}$, $p = -\nabla f(x) = (-20, -2)^\mathsf{T}$, $\phi(0) = 11$, $\phi'(0) = g^\mathsf{T} p = -404$, $\phi(\alpha) = 10(1 - 20\alpha)^2 + (1 - 2\alpha)^2$.

| trial | $\alpha$ | $\phi(\alpha)$ | bound $\phi(0) + c_1\alpha\phi'(0)$ | decision |
|---|---|---|---|---|
| 1 | 1 | 3611 | 10.9596 | reject |
| 2 | 0.5 | 810 | 10.9798 | reject |
| 3 | 0.25 | 160.25 | 10.9899 | reject |
| 4 | 0.125 | 23.0625 | 10.99495 | reject |
| 5 | 0.0625 | 1.390625 | 10.997475 | **accept** |

Curvature check at $\alpha = 0.0625$: $\phi'(\alpha) = 96.5$. With weak curvature $\phi'(\alpha) \ge c_2\phi'(0)$ and strong curvature $|\phi'(\alpha)| \le c_2|\phi'(0)|$: weak Wolfe holds for every $c_2$ ($96.5 \ge c_2 \cdot (-404)$); strong Wolfe holds with $c_2 = 0.9$ ($96.5 \le 363.6$) and fails with $c_2 = 0.1$ ($96.5 > 40.4$).

**What it shows.** Armijo only controls steps that are too long; the accepted step can violate strong Wolfe when $c_2$ is small.

**P2** Practice *added*

Let $\phi(\alpha) = \phi(0) + \phi'(0)\alpha + \frac{1}{2}a\alpha^2$ with $a > 0$ and $\phi'(0) < 0$, and let $\alpha^*$ be its minimizer. With $\phi(\alpha) = f(x_k + \alpha p_k)$ and $0 < c_1 < c_2 < 1$: Armijo is $\phi(\alpha) \le \phi(0) + c_1\alpha\,\phi'(0)$; weak curvature, $\phi'(\alpha) \ge c_2\,\phi'(0)$; strong curvature, $|\phi'(\alpha)| \le c_2|\phi'(0)|$. Weak Wolfe = Armijo + weak curvature; strong Wolfe = Armijo + strong curvature.

a) Find, in terms of $c_1$, $c_2$ and $\alpha^*$, the set of $\alpha > 0$ satisfying Armijo; the set satisfying weak curvature; and the set satisfying strong curvature.

b) Deduce for which values of $c_1$ the exact step $\alpha^*$ satisfies Armijo.

c) Deduce that the set of Wolfe steps is nonempty for every $0 < c_1 < c_2 < 1$.

**2.1** Proof *ch2-v2, Lemma 2.1, pp. 3–4*

Let $f : \mathbb{R}^n \to \mathbb{R}$ be of class $C^1$, bounded below on $\{x_k + \alpha p_k : \alpha > 0\}$, $p_k$ a descent direction and $0 < c_1 < c_2 < 1$. Prove that there is an interval of steps $\alpha > 0$ satisfying the weak and the strong Wolfe conditions. With $\phi(\alpha) = f(x_k + \alpha p_k)$ and $0 < c_1 < c_2 < 1$: Armijo is $\phi(\alpha) \le \phi(0) + c_1\alpha\,\phi'(0)$; weak curvature, $\phi'(\alpha) \ge c_2\,\phi'(0)$; strong curvature, $|\phi'(\alpha)| \le c_2|\phi'(0)|$. Weak Wolfe = Armijo + weak curvature; strong Wolfe = Armijo + strong curvature.

**A2** Pseudocode *line search: weak Wolfe by bisection*

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**WeakWolfe**$(f,\ \nabla f,\ x,\ p,\ g;\ \alpha_{\text{ini}}, c_1, c_2)$

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```
1: a ← 0;   b ← +∞;   α ← α_ini
2: repeat
3:     if f(x + αp) > f(x) + c1 α g^T p then          (Armijo fails: step too long)
4:         b ← α;   α ← (a + b)/2
5:     else if ∇f(x + αp)^T p < c2 g^T p then        (curvature fails: step too short)
6:         a ← α;   if b = +∞ then α ← 2a else α ← (a + b)/2
7:     else return α
```

---

Values used in the tables: $c_1 = 10^{-4}$, $c_2 = 0.9$.

**P3** Practice *added*

*Weak Wolfe search by bisection.* Let $f(x_1, x_2) = 10x_1^2 + x_2^2$, $x_0 = (1, 1)^\mathsf{T}$, $p_0 = -\nabla f(x_0)$, $\phi(\alpha) = f(x_0 + \alpha p_0)$, $c_1 = 10^{-4}$ and $c_2 = 0.9$. The weak Wolfe conditions are $\phi(\alpha) \le \phi(0) + c_1\alpha\,\phi'(0)$ (Armijo) and $\phi'(\alpha) \ge c_2\,\phi'(0)$ (curvature).

a) Run A2 by hand from $\alpha_{\text{ini}} = 1$, recording $[a, b]$, $\alpha$ and the failing condition at each iteration.

b) Repeat from $\alpha_{\text{ini}} = 10^{-3}$. Which condition drives the search in each case?

c) Prove that A2 stops after finitely many steps whenever $f$ is of class $C^1$, bounded below along $\{x_0 + \alpha p_0 : \alpha > 0\}$, $p_0$ is a descent direction and $0 < c_1 < c_2 < 1$.

**V2** Validation table *A2; solves P3 a) and b)*

$f(x) = 10x_1^2 + x_2^2$, $x = (1,1)^\mathsf{T}$, $p = -\nabla f(x) = (-20,-2)^\mathsf{T}$, $c_1 = 10^{-4}$, $c_2 = 0.9$; $\phi(\alpha) = 10(1-20\alpha)^2 + (1-2\alpha)^2$, $\phi(0) = 11$, $\phi'(0) = -404$, $c_2\phi'(0) = -363.6$ and $\phi'(\alpha) = -400(1-20\alpha) - 4(1-2\alpha)$. Columns $a$, $b$ show the interval *before* evaluating $\alpha$.

| it. | $a$ | $b$ | $\alpha$ | $\phi(\alpha)$ | $\phi'(\alpha)$ | result |
|---|---|---|---|---|---|---|
| | | | From $\alpha_{\text{ini}} = 1$ | | | |
| 1 | 0 | $\infty$ | 1 | 3611 | 7604 | Armijo fails |
| 2 | 0 | 1 | 0.5 | 810 | 3600 | Armijo fails |
| 3 | 0 | 0.5 | 0.25 | 160.25 | 1598 | Armijo fails |
| 4 | 0 | 0.25 | 0.125 | 23.0625 | 597 | Armijo fails |
| 5 | 0 | 0.125 | 0.0625 | 1.390625 | 96.5 | **accept** |
| | | | From $\alpha_{\text{ini}} = 10^{-3}$ | | | |
| 1 | 0 | $\infty$ | 0.001 | 10.600004 | $-395.992$ | curvature fails |
| 2 | 0.001 | $\infty$ | 0.002 | 10.208016 | $-387.984$ | curvature fails |
| 3 | 0.002 | $\infty$ | 0.004 | 9.448064 | $-371.968$ | curvature fails |
| 4 | 0.004 | $\infty$ | 0.008 | 8.024256 | $-339.936$ | **accept** |

**What it shows.** From an initial step that is too small, Armijo always holds, and it is the curvature condition that forces the step to grow.

**2.13*** Proposed exercise *ch2-v2, p. 15*

Let $f(x_1,x_2) = 10x_1^2 + x_2^2$, $x_0 = (1,1)^\mathsf{T}$ and $p_0 = -\nabla f(x_0)$.

a) Compute the exact step $\alpha^*$ minimizing $\phi(\alpha) = f(x_0 + \alpha p_0)$.

b) Prove that, for every $c_1 \in (0, 0.5)$ and $c_2 \in (c_1, 1)$, $\alpha^*$ satisfies the weak Wolfe conditions: $f(x_k + \alpha_k p_k) \le f(x_k) + c_1\alpha_k \nabla f(x_k)^\mathsf{T} p_k$ and $\nabla f(x_k + \alpha_k p_k)^\mathsf{T} p_k \ge c_2 \nabla f(x_k)^\mathsf{T} p_k$, with $0 < c_1 < c_2 < 1$ with $k = 0$.

## U4. Zoutendijk's theorem

↑Index

**2.2** Proof *ch2-v2, Theorem 3.1, pp. 4–5; corollary p. 6*

Let $f : \mathbb{R}^n \to \mathbb{R}$ be of class $C^1$, bounded below, with $\nabla f$ Lipschitz with constant $L > 0$. Let $x_{k+1} = x_k + \alpha_k p_k$ with $p_k$ a descent direction and $\alpha_k$ satisfying the weak Wolfe conditions, $f(x_k + \alpha_k p_k) \le f(x_k) + c_1\alpha_k \nabla f(x_k)^\mathsf{T} p_k$ and $\nabla f(x_k + \alpha_k p_k)^\mathsf{T} p_k \ge c_2 \nabla f(x_k)^\mathsf{T} p_k$, with $0 < c_1 < c_2 < 1$. Prove that

$$\sum_{k=0}^{\infty} \cos^2\theta_k \, \|\nabla f(x_k)\|^2 < \infty, \qquad \cos\theta_k = \frac{-\nabla f(x_k)^\mathsf{T} p_k}{\|\nabla f(x_k)\| \, \|p_k\|}.$$

Deduce that, for $p_k = -\nabla f(x_k)$, $\lim_{k\to\infty} \|\nabla f(x_k)\| = 0$.

**2.11** Counterexample / observation *ch2-v2, p. 5*

Let $f(x) = |x|^{4/3}$.

a) Prove that $f$ is $C^1$ and that $f'$ is not Lipschitz on any neighbourhood of 0.

b) Identify the step of Zoutendijk's proof that is no longer valid.

c) Discuss whether the conclusion of the theorem actually fails for this $f$.

**Note.** The notes only show that the proof breaks, not that the conclusion is false.

## U5. Strict and strong convexity

↑Index

**2.3** Proof *ch2-v2, Theorem 4.1, p. 6*

Let $f:\mathbb{R}^n\to\mathbb{R}$ be of class $C^1$ and strictly convex, i.e. $f(y)>f(x)+\nabla f(x)^{\mathsf T}(y-x)$ for all $x\neq y$. Prove that if $f$ has a global minimum, it is unique.

**2.4** Proof *ch2-v2, Theorem 4.2, pp. 6–7*

Let $f:\mathbb{R}^n\to\mathbb{R}$ be of class $C^2$ and $m>0$. Prove that

$$f(y)\geq f(x)+\nabla f(x)^{\mathsf T}(y-x)+\tfrac{m}{2}\|y-x\|^2\ \ \forall x,y \iff y^{\mathsf T}\nabla^2 f(x)\,y\geq m\|y\|^2\ \ \forall x,y.$$

**2.5** Proof *ch2-v2, Theorem 4.3, pp. 7–8*

Let $f:\mathbb{R}^n\to\mathbb{R}$ be of class $C^1$ and $m$-strongly convex. Prove that $f$ is coercive and has a unique global minimum.

## U6. Rate of convergence of steepest descent

↑Index

**P4** Practice *added*

Let $f(x_1,x_2)=10x_1^2+x_2^2$, with Hessian $Q=\operatorname{diag}(20,2)$ and $\kappa=\lambda_{\max}(Q)/\lambda_{\min}(Q)=10$, and steepest descent $x_{k+1}=x_k-\alpha_k\nabla f(x_k)$ with exact line search. The minimizer is $x^*=0$.

a) From $x_0=(1,1)^{\mathsf T}$, compute $x_1$ and $x_2$, verify that $\nabla f(x_1)\perp\nabla f(x_0)$, and compare the ratio $\big(f(x_{k+1})-f(x^*)\big)/\big(f(x_k)-f(x^*)\big)$ with the bound $\left(\frac{\kappa-1}{\kappa+1}\right)^2$.

b) Repeat from $x_0=(1,10)^{\mathsf T}$ and verify that the bound is attained with equality.

c) Describe the geometric pattern of the iterates in both cases.

**A3** Pseudocode *steepest descent*

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**SteepestDescent**$(f,\ \nabla f,\ x_0;\ \varepsilon, k_{\max})$

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1: **for** $k=0,1,\dots,k_{\max}$ **do**
2: $\quad g_k\leftarrow\nabla f(x_k)$; **if** $\|g_k\|<\varepsilon$ **then return** $x_k$
3: $\quad p_k\leftarrow -g_k$
4: $\quad \alpha_k\leftarrow$ Backtracking$(f,x_k,p_k,g_k;1,\frac12,10^{-4})$ or WeakWolfe$(f,\nabla f,x_k,p_k,g_k;1,10^{-4},0.9)$ or the exact step $\alpha_k=g_k^{\mathsf T}g_k/(g_k^{\mathsf T}Qg_k)$ if $f$ is quadratic with Hessian $Q$
5: $\quad x_{k+1}\leftarrow x_k+\alpha_k p_k$

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**V3** Validation table *A3 with exact step; solves P4*

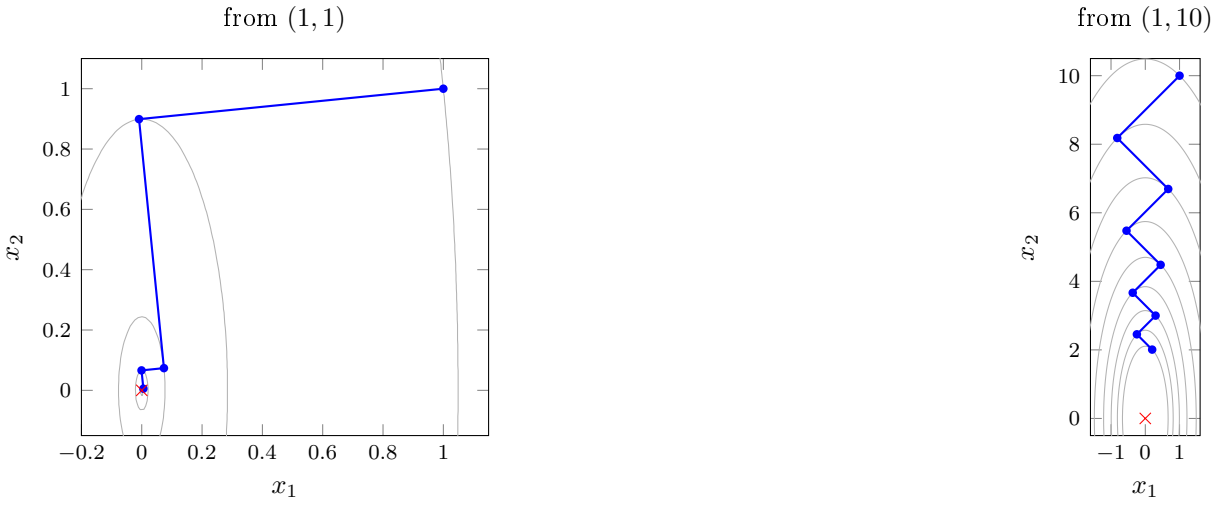
$f(x)=10x_1^2+x_2^2=\frac12 x^{\mathsf T}Qx$ with $Q=\operatorname{diag}(20,2)$, $\kappa=10$, $x^*=0$; exact step $\alpha_k=g_k^{\mathsf T}g_k/(g_k^{\mathsf T}Qg_k)$, $g_k=Qx_k$. Theoretical bound: $\left(\frac{\kappa-1}{\kappa+1}\right)^2=\frac{81}{121}\approx 0.6694$.

| $k$ | $x_k$ | $g_k$ | $\alpha_k$ | $f(x_k)$ | $f(x_{k+1})/f(x_k)$ |
|---|---|---|---|---|---|
| | | From $x_0 = (1,1)^\mathsf{T}$ | | | |
| 0 | $(1,\ 1)$ | $(20,\ 2)$ | $\frac{101}{2002} \approx 0.050450$ | 11 | $\frac{810}{11011} \approx 0.07356$ |
| 1 | $\left(-\frac{9}{1001},\ \frac{900}{1001}\right)$ | $\left(-\frac{180}{1001},\ \frac{1800}{1001}\right)$ | $\frac{101}{220} \approx 0.459091$ | $\frac{810}{1001} \approx 0.809191$ | 0.07356 |
| 2 | $\left(\frac{810}{11011},\ \frac{810}{11011}\right)$ | $\left(\frac{16200}{11011},\ \frac{1620}{11011}\right)$ | $\frac{101}{2002}$ | 0.059526 | 0.07356 |
| 3 | $(-0.000661,\ 0.066140)$ | $(-0.013228,\ 0.132281)$ | $\frac{101}{220}$ | 0.004379 | 0.07356 |
| | | From $x_0 = (1,10)^\mathsf{T}$ (worst case: the bound is attained) | | | |
| 0 | $(1,\ 10)$ | $(20,\ 20)$ | $\frac{1}{11}$ | 110 | $\frac{81}{121}$ |
| 1 | $\left(-\frac{9}{11},\ \frac{90}{11}\right)$ | $\left(-\frac{180}{11},\ \frac{180}{11}\right)$ | $\frac{1}{11}$ | $\frac{810}{11}$ | $\frac{81}{121}$ |
| 2 | $\left(\frac{81}{121},\ \frac{810}{121}\right)$ | $\left(\frac{1620}{121},\ \frac{1620}{121}\right)$ | $\frac{1}{11}$ | $\frac{65610}{1331}$ | $\frac{81}{121}$ |

In both cases $g_{k+1}^\mathsf{T} g_k = 0$ (orthogonal zigzag).

**What it shows.** The Kantorovich bound is attained in the worst case, and the orthogonal zigzag explains the slowness when $\kappa \gg 1$.

**Figure.**




Level curves of $f = 10x_1^2 + x_2^2$ through each iterate. Each step is tangent to the level curve at the next point, so $g_{k+1} \perp g_k$. From $(1,10)$ the pattern repeats at scale $\frac{9}{11}$ and the reduction per step is exactly $(\frac{\kappa-1}{\kappa+1})^2$.

## V4 Validation table

*A3 with Backtracking (A1)*

$f(x) = 10x_1^2 + x_2^2$, from $x_0 = (1,1)^\mathsf{T}$, Backtracking with $\alpha_0 = 1$, $\rho = \frac{1}{2}$, $c_1 = 10^{-4}$. Column "trials" counts the evaluations of $f$ inside A1.

| $k$ | $x_k$ | $f(x_k)$ | $\|g_k\|$ | $\alpha_k$ | trials |
|---|---|---|---|---|---|
| 0 | $(1,\ 1)$ | 11 | 20.0998 | 0.0625 | 5 |
| 1 | $(-0.25,\ 0.875)$ | 1.390625 | 5.2974 | 0.0625 | 5 |
| 2 | $(0.0625,\ 0.765625)$ | 0.625244 | 1.9767 | 0.125 | 4 |
| 3 | $(-0.09375,\ 0.574219)$ | 0.417618 | 2.1988 | 0.125 | 4 |
| 4 | $(0.140625,\ 0.430664)$ | 0.383225 | 2.9414 | 0.0625 | 5 |
| 5 | $(-0.035156,\ 0.376831)$ | 0.154361 | 1.0307 | 0.125 | 4 |

Note that $f$ always decreases, but $\|g_k\|$ is not monotone.

**2.6** Proof *ch2-v2, equation (11), p. 8*

Let $f(x) = \frac{1}{2}x^{\mathsf{T}}Qx - b^{\mathsf{T}}x$ with $Q \succ 0$, $x^* = Q^{-1}b$ and $\|v\|_Q = \sqrt{v^{\mathsf{T}}Qv}$. Prove that $\frac{1}{2}\|x - x^*\|_Q^2 = f(x) - f(x^*)$ for all $x \in \mathbb{R}^n$.

**2.7** Proof *ch2-v2, Lemma 4.1 (Kantorovich), pp. 8–10*

Let $Q \succ 0$ with eigenvalues $0 < m = \lambda_1 \leq \cdots \leq \lambda_n = M$. Prove that for every $g \neq 0$,

$$\frac{(g^{\mathsf{T}}g)^2}{(g^{\mathsf{T}}Qg)(g^{\mathsf{T}}Q^{-1}g)} \geq \frac{4Mm}{(M+m)^2}.$$

**2.8** Proof *ch2-v2, Theorem 4.4, pp. 10–11*

Let $f(x) = \frac{1}{2}x^{\mathsf{T}}Qx - b^{\mathsf{T}}x$ with $Q \succ 0$ with eigenvalues $0 < \lambda_1 \leq \cdots \leq \lambda_n$, $x^* = Q^{-1}b$ and $\|v\|_Q = \sqrt{v^{\mathsf{T}}Qv}$. For $x_{k+1} = x_k - \alpha_k g_k$, $g_k = Qx_k - b$, with exact line search:

a) Prove that $\alpha_k = \dfrac{g_k^{\mathsf{T}}g_k}{g_k^{\mathsf{T}}Qg_k}$.

b) Prove that $\|x_{k+1} - x^*\|_Q^2 \leq \left(\dfrac{\lambda_n - \lambda_1}{\lambda_n + \lambda_1}\right)^2 \|x_k - x^*\|_Q^2$.

**2.12** Observation *ch2-v2, pp. 11 and 15*

For $f(x) = \frac{1}{2}x^{\mathsf{T}}Qx - b^{\mathsf{T}}x$, $Q \succ 0$ with eigenvalues $\lambda_1 \leq \cdots \leq \lambda_n$, and steepest descent with exact line search, the contraction factor of 2.8 is $\frac{\lambda_n - \lambda_1}{\lambda_n + \lambda_1}$ in the norm $\|v\|_Q = \sqrt{v^{\mathsf{T}}Qv}$.

a) Express the contraction factor in terms of $\kappa(Q) = \lambda_n/\lambda_1$ and explain the cases $\kappa = 1$ (convergence in one iteration) and $\kappa \gg 1$ (zigzag).

b) Explain the difference between Theorem 4.4 (global, exact bound from $k = 0$) and Theorem 4.5 (asymptotic bound, for every $r$ strictly larger than $\frac{\lambda_n - \lambda_1}{\lambda_n + \lambda_1}$).

**2.9** Proof *ch2-v2, Theorem 4.5, pp. 12–15*

Let $f : \mathbb{R}^n \to \mathbb{R}$ be of class $C^2$ and $\{x_k\}$ generated by steepest descent with exact line search. Suppose $x_k \to x^*$ with $\nabla^2 f(x^*) \succ 0$, with eigenvalues $\lambda_1 \leq \cdots \leq \lambda_n$. Prove that for every $r \in \left(\frac{\lambda_n - \lambda_1}{\lambda_n + \lambda_1}, 1\right)$ and every sufficiently large $k$,

$$f(x_{k+1}) - f(x^*) \leq r^2\,[f(x_k) - f(x^*)].$$

**2.15*** Proposed exercise *ch2-v2, p. 16*

Let $f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$ (Rosenbrock), $\nabla f(x) = \big(-400x_1(x_2 - x_1^2) - 2(1 - x_1),\ 200(x_2 - x_1^2)\big)^{\mathsf{T}}$, $\nabla^2 f(x) = \begin{pmatrix} 1200x_1^2 - 400x_2 + 2 & -400x_1 \\ -400x_1 & 200 \end{pmatrix}$.

a) Verify that $x^* = (1, 1)^{\mathsf{T}}$ is the unique global minimum and compute $\nabla^2 f(x^*)$.

b) Find the eigenvalues $\lambda_1, \lambda_2$ of $\nabla^2 f(x^*)$ and $\kappa(\nabla^2 f(x^*))$.

c) Compute the asymptotic rate $r_0 = \dfrac{\lambda_2 - \lambda_1}{\lambda_2 + \lambda_1}$ of Theorem 4.5 (item 2.9) for steepest descent with exact line search.

## U7. Newton's method and modified Hessian

↑Index

**3.16** Numerical example *ch3, p. 4*

Let $f(x) = e^x - x - 1$, with minimum at $x^* = 0$. Apply Newton from $x_0 = 1$, compute $x_1, \dots, x_4$ and verify numerically that $|x_{k+1}|/|x_k|^2$ settles down.

**Note.** The values $x_2, x_3, x_4$ printed in the notes do not match the iteration; see table V5.

**A4** Pseudocode *Newton with modified Hessian*

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**ModifiedNewton**$(f,\ \nabla f,\ \nabla^2 f,\ x_0;\ \delta, \varepsilon, k_{\max})$

---

1: **for** $k = 0, 1, \dots, k_{\max}$ **do**
2: $\quad g_k \leftarrow \nabla f(x_k)$; **if** $\|g_k\| < \varepsilon$ **then return** $x_k$
3: $\quad \tau_k \leftarrow \max\{0,\ \delta - \lambda_{\min}(\nabla^2 f(x_k))\}$; $B_k \leftarrow \nabla^2 f(x_k) + \tau_k I$
4: $\quad$ solve $B_k p_k = -g_k$ (linear system; do not invert $B_k$)
5: $\quad \alpha_k \leftarrow \text{Backtracking}(f, x_k, p_k, g_k; 1, \frac{1}{2}, 10^{-4})$
6: $\quad x_{k+1} \leftarrow x_k + \alpha_k p_k$

---

Pure Newton is the case $\tau_k = 0$, $\alpha_k = 1$ (without lines 3 and 5).

**V5** Validation table *pure Newton on $f(x) = e^x - x - 1$; solves 3.16*

Iteration $x_{k+1} = x_k - (e^{x_k} - 1)/e^{x_k}$ from $x_0 = 1$; $x^* = 0$ and the ratio tends to $\frac{1}{2} f'''(0)/f''(0) = \frac{1}{2}$.

| $k$ | $x_k$ | $f'(x_k)$ | $f''(x_k)$ | $x_{k+1}/x_k^2$ |
|---|---|---|---|---|
| 0 | 1 | 1.71828183 | 2.71828183 | 0.3679 |
| 1 | 0.36787944 | 0.44466786 | 1.44466786 | 0.4439 |
| 2 | 0.06008007 | 0.06192157 | 1.06192157 | 0.4901 |
| 3 | 0.00176920 | 0.00177077 | 1.00177077 | 0.4997 |
| 4 | $1.5641 \times 10^{-6}$ | $1.5641 \times 10^{-6}$ | 1.00000156 | 0.5000 |
| 5 | $1.2233 \times 10^{-12}$ | | | |

**Note.** The notes (ch3, p. 4) give $x_2 \approx 0.05807$, $x_3 \approx 0.00159$, $x_4 \approx 1.26 \times 10^{-6}$; these values do not match the iteration. The correct ones are in this table.

**3.1** Proof *ch3, Theorem 2.1, p. 3*

Let $f : \mathbb{R}^n \to \mathbb{R}$ be of class $C^2$ on an open set $D$, and $x^* \in D$ with $\nabla f(x^*) = 0$, $\nabla^2 f(x^*)$ invertible and $\|\nabla^2 f(x^*)^{-1}\| \le M$. Suppose $\nabla^2 f$ is Lipschitz on $D$ with constant $L$. Prove that there exists $r > 0$ such that, if $x_0 \in B(x^*, r)$, the Newton iteration $x_{k+1} = x_k - \nabla^2 f(x_k)^{-1} \nabla f(x_k)$ is well defined, converges to $x^*$ and satisfies

$$\|x_{k+1} - x^*\| \le LM \, \|x_k - x^*\|^2.$$

**3.12** Counterexample *ch3, p. 4*

Let $f(x) = x^4$. Prove that the Newton iteration is $x_{k+1} = \frac{2}{3} x_k$, so convergence is only linear, and identify the hypothesis of Theorem 2.1 that fails.

**3.13** Counterexample *ch3, p. 4 (marked "verify")*

a) For $f(x) = |x|^{7/3}$, compute the Newton iteration explicitly and its order of convergence. Determine which hypotheses of Theorem 2.1 fail.

b) For $f(x) = \frac{1}{2}x^2 + |x|^{7/3}$, prove that $f''(0) \neq 0$, that $f''$ is not Lipschitz near 0, and that Newton converges with order exactly $4/3$.

**Note.** The notes claim superlinear convergence for $x^{7/3}$; part a) checks that claim and part b) isolates the Lipschitz hypothesis.

**3.20*** Proposed exercise *ch3, p. 18*

Let $f(x) = \frac{1}{2}x^{\mathsf{T}}Ax - b^{\mathsf{T}}x$ with $A \succ 0$.

a) Prove that Newton reaches $x^* = A^{-1}b$ in exactly one iteration from any $x_0$.

b) With $B_0 = A + \tau I$, $\tau > 0$, express $x_1 - x^*$ in terms of $x_0 - x^*$ and analyse the effect of $\tau$ on the speed of convergence.

**3.17** Numerical example *ch3, p. 6*

Let $f(x_1, x_2) = -2x_1 + 4x_2 - x_1^2 + x_1x_2 + \frac{3}{2}x_2^2$ and $x_k = (0,0)^{\mathsf{T}}$.

a) Compute $\nabla f(x_k)$, $\nabla^2 f(x_k)$ and its eigenvalues.

b) Compute the Newton direction and verify that it is not a descent direction.

c) With $B_k = \nabla^2 f(x_k) + 3.5\, I$, compute $p_k = -B_k^{-1}\nabla f(x_k)$ and verify that it is a descent direction.

**3.2** Proof *ch3, Proposition 3.1, p. 5*

Let $B_k$ be symmetric positive definite and $p_k = -B_k^{-1}\nabla f(x_k)$. Prove that

$$\nabla f(x_k)^{\mathsf{T}} p_k \leq -\frac{1}{\|B_k\|}\|\nabla f(x_k)\|^2.$$

**3.21*** Proposed exercise *ch3, p. 18*

Let $f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$ (Rosenbrock).

a) Show that the unique global minimum is $x^* = (1,1)^{\mathsf{T}}$.

b) Compute $\nabla^2 f(x)$. Prove that Newton is applicable at $x_0 = (0,0)^{\mathsf{T}}$ and find $x_1$.

c) Decide whether the Hessian is positive definite at $x_0 = (0,0)^{\mathsf{T}}$ and at $\tilde{x} = (0,1)^{\mathsf{T}}$. If it is not at $\tilde{x}$, find the smallest $\tau > 0$ such that $\nabla^2 f(\tilde{x}) + \tau I$ is positive semidefinite (boundary of positive definiteness).

**P6** Practice *added*

Let $f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$ (Rosenbrock), $\nabla f(x) = \big(-400x_1(x_2 - x_1^2) - 2(1 - x_1),\ 200(x_2 - x_1^2)\big)^{\mathsf{T}}$, $\nabla^2 f(x) = \begin{pmatrix} 1200x_1^2 - 400x_2 + 2 & -400x_1 \\ -400x_1 & 200 \end{pmatrix}$, at the point $\tilde{x} = (0,1)^{\mathsf{T}}$, and let $m(p) = f(\tilde{x}) + \nabla f(\tilde{x})^{\mathsf{T}}p + \frac{1}{2}p^{\mathsf{T}}\nabla^2 f(\tilde{x})p$ be the quadratic model.

a) Compute the pure Newton direction and decide whether it is a descent direction.

b) Show that $p^N$ is a saddle point of the model $m$, not its minimum.

c) With $\tau = 400$ and $B = \nabla^2 f(\tilde{x}) + \tau I$, compute $p = -B^{-1}\nabla f(\tilde{x})$, compare both directions and verify the bound $\nabla f(\tilde{x})^{\mathsf{T}}p \leq -\|\nabla f(\tilde{x})\|^2/\|B\|$.

**V6** Validation table *A4 on Rosenbrock from $(0,1)^{\mathsf T}$; solves P6*

$f(x)=100(x_2-x_1^2)^2+(1-x_1)^2$, $x_0=(0,1)^{\mathsf T}$, $\delta=1$, Backtracking with $\alpha_0=1$, $\rho=\tfrac12$, $c_1=10^{-4}$; $x^*=(1,1)^{\mathsf T}$.

| $k$ | $x_k$ | $f(x_k)$ | $\|g_k\|$ | $\lambda_{\min}$ | $\tau_k$ | $p_k$ | $\alpha_k$ |
|---|---|---|---|---|---|---|---|
| 0 | $(0,\ 1)$ | 101 | 200.010 | $-398$ | 399 | $\left(2,\ -\frac{200}{599}\right)$ | 0.5 |
| 1 | $\left(1,\ \frac{499}{599}\right)$ | 2.787060 | 74.660 | 13.029 | 0 | $\left(0,\ \frac{100}{599}\right)$ | 1 |
| 2 | $(1,\ 1)$ | 0 | 0 | | | | |

Detail of $k=0$: $g_0=(-2,200)^{\mathsf T}$, $\nabla^2 f(x_0)=\operatorname{diag}(-398,200)$, $B_0=\operatorname{diag}(1,599)$, $g_0^{\mathsf T}p_0\approx -70.778$. In the backtracking, $\alpha=1$ gives $f=1112.48$ (reject) and $\alpha=0.5$ gives $f=2.787$ (accept). The pure Newton direction would be $p^N=\left(-\frac{1}{199},-1\right)^{\mathsf T}$, with $g_0^{\mathsf T}p^N\approx -199.99<0$ but pointing to the saddle point of the model. Detail of $k=1$: $g_1=\left(\frac{40000}{599},-\frac{20000}{599}\right)^{\mathsf T}$ and the Hessian is already positive definite.

## U8. Quasi-Newton: DFP and BFGS formulas

↑Index

**3.14** Observation *ch3, p. 7*

With $s_k=x_{k+1}-x_k$, $y_k=\nabla f(x_{k+1})-\nabla f(x_k)$, prove that $y_k=\left(\int_0^1 \nabla^2 f(x_k+ts_k)\,dt\right)s_k$. Give an example where $\nabla^2 f(x_{k+1})s_k\neq y_k$, and prove that for quadratic $f$ the exact Hessian does satisfy the secant equation.

**3.15** Counterexample *ch3, p. 12*

Let $f(x)=-x^2$, $x_0=1$, $s_0=1$. Compute $y_0$ and $s_0^{\mathsf T}y_0$, apply the inverse BFGS update $H_{k+1}^{\mathrm{BFGS}}=(I-\rho_k s_k y_k^{\mathsf T})H_k(I-\rho_k y_k s_k^{\mathsf T})+\rho_k s_k s_k^{\mathsf T}$, $\rho_k=1/(y_k^{\mathsf T}s_k)$ (here $k=0$) with $H_0=1$, and show that $H_1<0$, so that $p_1=-H_1\nabla f(x_1)$ is an ascent direction.

**3.3** Proof *ch3, Proposition 4.1, pp. 7–8*

Let $p_k$ be a descent direction, $\alpha_k>0$ such that $\nabla f(x_k+\alpha_k p_k)^{\mathsf T}p_k\geq c_2\nabla f(x_k)^{\mathsf T}p_k$ with $c_2<1$, $s_k=\alpha_k p_k$ and $y_k=\nabla f(x_k+s_k)-\nabla f(x_k)$. Prove that $s_k^{\mathsf T}y_k>0$.

**3.5** Derivation *ch3, p. 9 (DFP) and p. 11 (BFGS, "verify the details")*

Let $s_k=x_{k+1}-x_k$, $y_k=\nabla f(x_{k+1})-\nabla f(x_k)$, with $s_k^{\mathsf T}y_k>0$.

a) Starting from $H_{k+1}=H_k+\alpha uu^{\mathsf T}+\beta vv^{\mathsf T}$ and $H_{k+1}y_k=s_k$, with $u=s_k$, $v=H_ky_k$, derive the DFP formula $H_{k+1}^{\mathrm{DFP}}=H_k+\dfrac{s_ks_k^{\mathsf T}}{s_k^{\mathsf T}y_k}-\dfrac{H_ky_ky_k^{\mathsf T}H_k}{y_k^{\mathsf T}H_ky_k}$.

b) Starting from $B_{k+1}=B_k+\alpha uu^{\mathsf T}+\beta vv^{\mathsf T}$ and $B_{k+1}s_k=y_k$, with $u=y_k$, $v=B_ks_k$, derive the BFGS formula $B_{k+1}^{\mathrm{BFGS}}=B_k-\dfrac{B_ks_ks_k^{\mathsf T}B_k}{s_k^{\mathsf T}B_ks_k}+\dfrac{y_ky_k^{\mathsf T}}{y_k^{\mathsf T}s_k}$.

c) Justify the "equate coefficients" step (it requires $s_k$ and $H_ky_k$ to be linearly independent; see 3.4).

**3.7** Proof *ch3, formula (15), p. 10 ("verify")*

Let $s_k = x_{k+1} - x_k$, $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$, $\rho_k = 1/(y_k^{\mathsf{T}} s_k)$, $B_k \succ 0$, $H_k = B_k^{-1}$ and $s_k^{\mathsf{T}} y_k > 0$. Prove that the inverse of $B_{k+1}^{\mathrm{BFGS}} = B_k - \dfrac{B_k s_k s_k^{\mathsf{T}} B_k}{s_k^{\mathsf{T}} B_k s_k} + \dfrac{y_k y_k^{\mathsf{T}}}{y_k^{\mathsf{T}} s_k}$ is

$$\begin{aligned} H_{k+1}^{\mathrm{BFGS}} &= (I - \rho_k s_k y_k^{\mathsf{T}}) H_k (I - \rho_k y_k s_k^{\mathsf{T}}) + \rho_k s_k s_k^{\mathsf{T}} \\ &= H_k - \rho_k (s_k y_k^{\mathsf{T}} H_k + H_k y_k s_k^{\mathsf{T}}) + \rho_k (1 + \rho_k y_k^{\mathsf{T}} H_k y_k) s_k s_k^{\mathsf{T}}. \end{aligned}$$

**3.18** Numerical example *ch3, p. 10 (DFP) and p. 12 (BFGS)*

Let $f(x_1, x_2) = \frac{5}{8}x_1^2 + \frac{3}{2}x_2^2 - 2x_1 - x_2$, $x_0 = (0,0)^{\mathsf{T}}$, $H_0 = I_2$ and step $x_1 = x_0 - \alpha_0 H_0 \nabla f(x_0)$ with $\alpha_0 = 1$. Notation: $s_k = x_{k+1} - x_k$, $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$, and the formulas $H_{k+1}^{\mathrm{DFP}} = H_k + \dfrac{s_k s_k^{\mathsf{T}}}{s_k^{\mathsf{T}} y_k} - \dfrac{H_k y_k y_k^{\mathsf{T}} H_k}{y_k^{\mathsf{T}} H_k y_k}$ and $H_{k+1}^{\mathrm{BFGS}} = (I - \rho_k s_k y_k^{\mathsf{T}}) H_k (I - \rho_k y_k s_k^{\mathsf{T}}) + \rho_k s_k s_k^{\mathsf{T}}$, $\rho_k = 1/(y_k^{\mathsf{T}} s_k)$.

a) Compute $x_1$, $s_0$, $y_0$ and verify $s_0^{\mathsf{T}} y_0 > 0$.

b) Compute $H_1^{\mathrm{DFP}}$ and $H_1^{\mathrm{BFGS}}$.

c) Verify in both cases the secant equation $H_1 y_0 = s_0$ and positive definiteness.

**3.4** Proof *ch3, Lemma 4.1, pp. 8–9*

Let $f$ be of class $C^1$, $H_k \succ 0$, $s_k = -\alpha_k H_k \nabla f(x_k)$ with $\alpha_k > 0$ obtained by exact line search, $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$ with $x_{k+1} = x_k + s_k$, $s_k^{\mathsf{T}} y_k > 0$ and $\nabla f(x_{k+1}) \neq 0$. Prove that $s_k$ and $H_k y_k$ are linearly independent.

**3.6** Proof *ch3, Theorem 4.1, pp. 9–10*

Let $H_k$ be symmetric positive definite, $s_k, y_k \in \mathbb{R}^n$ with $s_k^{\mathsf{T}} y_k > 0$, and $H_{k+1}^{\mathrm{DFP}} = H_k + \dfrac{s_k s_k^{\mathsf{T}}}{s_k^{\mathsf{T}} y_k} - \dfrac{H_k y_k y_k^{\mathsf{T}} H_k}{y_k^{\mathsf{T}} H_k y_k}$. Prove that $H_{k+1}^{\mathrm{DFP}}$ is symmetric positive definite.

**Note.** The notes close with "they cannot vanish simultaneously" without justification; the equality case of Cauchy–Schwarz must be completed.

**3.8** Proof *ch3, Theorem 4.2, p. 11*

Let $H_k \succ 0$ be symmetric, $s_k, y_k \in \mathbb{R}^n$ with $s_k^{\mathsf{T}} y_k > 0$, and $H_{k+1}^{\mathrm{BFGS}} = (I - \rho_k s_k y_k^{\mathsf{T}}) H_k (I - \rho_k y_k s_k^{\mathsf{T}}) + \rho_k s_k s_k^{\mathsf{T}}$, $\rho_k = 1/(y_k^{\mathsf{T}} s_k)$. Prove that $H_{k+1}^{\mathrm{BFGS}}$ is symmetric positive definite.

**3.22*** Proposed exercise *ch3, p. 19*

With $H_0 = I_2$, $x_0 = (1,2)^{\mathsf{T}}$, $\nabla f(x_0) = (3,-1)^{\mathsf{T}}$, $x_1 = (2,0)^{\mathsf{T}}$, $\nabla f(x_1) = (1,1)^{\mathsf{T}}$, $s_0 = x_1 - x_0$, $y_0 = \nabla f(x_1) - \nabla f(x_0)$, and the formulas $H_{k+1}^{\mathrm{DFP}} = H_k + \dfrac{s_k s_k^{\mathsf{T}}}{s_k^{\mathsf{T}} y_k} - \dfrac{H_k y_k y_k^{\mathsf{T}} H_k}{y_k^{\mathsf{T}} H_k y_k}$ and $H_{k+1}^{\mathrm{BFGS}} = (I - \rho_k s_k y_k^{\mathsf{T}}) H_k (I - \rho_k y_k s_k^{\mathsf{T}}) + \rho_k s_k s_k^{\mathsf{T}}$, $\rho_k = 1/(y_k^{\mathsf{T}} s_k)$ (with $k = 0$):

a) Check whether $s_0^{\mathsf{T}} y_0 > 0$ holds.

b) Obtain $H_1^{\mathrm{BFGS}}$ with the inverse formula.

c) Prove that, for any $s, y \in \mathbb{R}^n$ with $s^{\mathsf{T}} y > 0$, both $H_1^{\mathrm{DFP}}$ and $H_1^{\mathrm{BFGS}}$ satisfy $H_1 y = s$.

**A5** Pseudocode *BFGS and DFP (inverse form)*

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**QuasiNewton**($f$, $\nabla f$, $x_0$, $H_0 \succ 0$; $\varepsilon, k_{\max}$)

---

1: $g_0 \leftarrow \nabla f(x_0)$
2: **for** $k = 0, 1, \dots, k_{\max}$ **do**
3: **if** $\|g_k\| < \varepsilon$ **then return** $x_k$
4: $p_k \leftarrow -H_k g_k$
5: $\alpha_k \leftarrow \text{WeakWolfe}(f, \nabla f, x_k, p_k, g_k; 1, 10^{-4}, 0.9)$ (or exact step if $f$ is quadratic)
6: $s_k \leftarrow \alpha_k p_k$; $x_{k+1} \leftarrow x_k + s_k$; $g_{k+1} \leftarrow \nabla f(x_{k+1})$; $y_k \leftarrow g_{k+1} - g_k$
7: **if** $s_k^\mathsf{T} y_k \le 0$ **then** $H_{k+1} \leftarrow H_k$ (skip the update)
8: **else** (BFGS): $\rho_k \leftarrow 1/(y_k^\mathsf{T} s_k)$; $H_{k+1} \leftarrow (I - \rho_k s_k y_k^\mathsf{T}) H_k (I - \rho_k y_k s_k^\mathsf{T}) + \rho_k s_k s_k^\mathsf{T}$
9: (DFP): $H_{k+1} \leftarrow H_k + \dfrac{s_k s_k^\mathsf{T}}{s_k^\mathsf{T} y_k} - \dfrac{H_k y_k y_k^\mathsf{T} H_k}{y_k^\mathsf{T} H_k y_k}$

---

With WeakWolfe, line 7 is never triggered, because the curvature condition implies $s_k^\mathsf{T} y_k > 0$ (see 3.3); it is kept as a safeguard.

## U9. Convergence of quasi-Newton methods

↑Index

**P5** Practice *added*

Let $f(x_1, x_2) = \frac{5}{8}x_1^2 + \frac{3}{2}x_2^2 - 2x_1 - x_2$, i.e. $A = \operatorname{diag}(\frac{5}{4}, 3)$, $b = (2,1)^\mathsf{T}$. From $x_0 = (0,0)^\mathsf{T}$ and $H_0 = I_2$, perform two DFP iterations, $x_{k+1} = x_k + \alpha_k p_k$ with $p_k = -H_k g_k$, $g_k = \nabla f(x_k)$, exact step and $H_{k+1}^{\text{DFP}} = H_k + \dfrac{s_k s_k^\mathsf{T}}{s_k^\mathsf{T} y_k} - \dfrac{H_k y_k y_k^\mathsf{T} H_k}{y_k^\mathsf{T} H_k y_k}$ ($s_k = x_{k+1} - x_k$, $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$), and verify:

a) $g_1^\mathsf{T} s_0 = 0$ and $s_0^\mathsf{T} A s_1 = 0$.

b) $x_2 = x^* = A^{-1} b$.

c) $H_2 = A^{-1}$.

**V7** Validation table *A5 with exact step; solves P5 and illustrates 3.10*

$f(x) = \frac{5}{8}x_1^2 + \frac{3}{2}x_2^2 - 2x_1 - x_2 = \frac{1}{2}x^\mathsf{T} A x - b^\mathsf{T} x$, $A = \operatorname{diag}(\frac{5}{4}, 3)$, $b = (2,1)^\mathsf{T}$, $x^* = (\frac{8}{5}, \frac{1}{3})^\mathsf{T}$; $x_0 = (0,0)^\mathsf{T}$, $H_0 = I_2$, exact step $\alpha_k = -g_k^\mathsf{T} p_k / (p_k^\mathsf{T} A p_k)$. BFGS and DFP generate the same iterates and differ only in $p_1$, $\alpha_1$ and $H_1$.

| $k$ | $x_k$ | $g_k$ | $p_k$ | $\alpha_k$ | $s_k$ | $y_k$ |
|---|---|---|---|---|---|---|
| 0 | $(0,\ 0)$ | $(-2,\ -1)$ | $(2,\ 1)$ | $\frac{5}{8}$ | $(\frac{5}{4},\ \frac{5}{8})$ | $(\frac{25}{16},\ \frac{15}{8})$ |
| 1 (DFP) | $(\frac{5}{4},\ \frac{5}{8})$ | $(-\frac{7}{16},\ \frac{7}{8})$ | $(\frac{42}{61},\ -\frac{35}{61})$ | $\frac{61}{120}$ | $(\frac{7}{20},\ -\frac{7}{24})$ | $(\frac{7}{16},\ -\frac{7}{8})$ |
| 1 (BFGS) | $(\frac{5}{4},\ \frac{5}{8})$ | $(-\frac{7}{16},\ \frac{7}{8})$ | $(\frac{105}{128},\ -\frac{175}{256})$ | $\frac{32}{75}$ | $(\frac{7}{20},\ -\frac{7}{24})$ | $(\frac{7}{16},\ -\frac{7}{8})$ |
| 2 | $(\frac{8}{5},\ \frac{1}{3})$ | $(0,\ 0)$ | | | | |

$$H_1^{\text{DFP}} = \begin{pmatrix} \frac{133}{122} & -\frac{59}{244} \\ -\frac{59}{244} & \frac{261}{488} \end{pmatrix}, \quad H_1^{\text{BFGS}} = \begin{pmatrix} \frac{77}{64} & -\frac{43}{128} \\ -\frac{43}{128} & \frac{157}{256} \end{pmatrix}, \quad H_2^{\text{DFP}} = H_2^{\text{BFGS}} = \begin{pmatrix} \frac{4}{5} & 0 \\ 0 & \frac{1}{3} \end{pmatrix} = A^{-1}.$$

Checks: $s_0^\mathsf{T} y_0 = \frac{25}{8}$, $s_1^\mathsf{T} y_1 = \frac{49}{120}$, $g_1^\mathsf{T} s_0 = 0$, $s_0^\mathsf{T} A s_1 = 0$, $H_1 y_0 = s_0$.

**What it shows.** On a quadratic with exact line search, DFP and BFGS terminate in $n$ steps with $A$-conjugate directions and $H_n = A^{-1}$.

**3.10** Proof *ch3, Theorem 5.2, pp. 15–17*

Let $f(x) = \frac{1}{2}x^{\mathsf{T}}Ax - b^{\mathsf{T}}x$ with $A \succ 0$, $g_k = \nabla f(x_k)$, and $\{x_k\}$ generated by $x_{k+1} = x_k - \alpha_k H_k g_k$ with exact line search and $H_{k+1}$ given by $H_{k+1}^{\mathrm{DFP}} = H_k + \dfrac{s_k s_k^{\mathsf{T}}}{s_k^{\mathsf{T}} y_k} - \dfrac{H_k y_k y_k^{\mathsf{T}} H_k}{y_k^{\mathsf{T}} H_k y_k}$ or $H_{k+1}^{\mathrm{BFGS}} = (I - \rho_k s_k y_k^{\mathsf{T}}) H_k (I - \rho_k y_k s_k^{\mathsf{T}}) + \rho_k s_k s_k^{\mathsf{T}}$, $\rho_k = 1/(y_k^{\mathsf{T}} s_k)$ ($s_k = x_{k+1} - x_k$, $y_k = g_{k+1} - g_k$), from arbitrary $x_0$ and $H_0 \succ 0$. Prove by induction that, as long as $g_0, \dots, g_{k-1} \neq 0$:

(P1)$_k$ $s_i^{\mathsf{T}} A s_j = 0$ for $0 \le i < j \le k-1$;

(P2)$_k$ $g_k^{\mathsf{T}} s_i = 0$ for $0 \le i \le k-1$;

(P3)$_k$ $H_k y_i = s_i$ for $0 \le i \le k-1$.

Conclude that the method reaches $x^* = A^{-1}b$ in at most $n$ iterations and that, if $n$ nonzero steps are generated, $H_n = A^{-1}$. Complete the BFGS case of (P3), which the notes leave as "analogous".

**3.9** Proof *ch3_13-15, Theorem 5.1 (replaces ch3, pp. 13–14)*

Let $f : \mathbb{R}^n \to \mathbb{R}$ be of class $C^2$ with $m\|d\|^2 \le d^{\mathsf{T}} \nabla^2 f(x) d \le M\|d\|^2$ for all $x, d$, $0 < m \le M$. Let $x_{k+1} = x_k + \alpha_k p_k$, $p_k = -B_k^{-1} \nabla f(x_k)$, with arbitrary $B_0 \succ 0$, $B_{k+1}$ updated by $B_{k+1}^{\mathrm{BFGS}} = B_k - \dfrac{B_k s_k s_k^{\mathsf{T}} B_k}{s_k^{\mathsf{T}} B_k s_k} + \dfrac{y_k y_k^{\mathsf{T}}}{y_k^{\mathsf{T}} s_k}$ (or the corresponding DFP update) with $s_k = x_{k+1} - x_k$, $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$, and $\alpha_k$ satisfying the weak Wolfe conditions, $f(x_k + \alpha_k p_k) \le f(x_k) + c_1 \alpha_k \nabla f(x_k)^{\mathsf{T}} p_k$ and $\nabla f(x_k + \alpha_k p_k)^{\mathsf{T}} p_k \ge c_2 \nabla f(x_k)^{\mathsf{T}} p_k$, with $0 < c_1 < c_2 < 1$. Prove that for every $x_0$,

$$\lim_{k\to\infty} x_k = x^* \quad \text{and} \quad \lim_{k\to\infty} \|\nabla f(x_k)\| = 0,$$

where $x^*$ is the unique global minimizer. Structure: (1) bounds $m\|s_k\|^2 \le s_k^{\mathsf{T}} y_k \le M\|s_k\|^2$; (2) lower bound on $\alpha_k$ and decrease; (3) Zoutendijk series; (4) lower bound on $\cos\theta_k$; (5) $\|\nabla f(x_k)\| \to 0$ and $\|x_k - x^*\| \le \frac{1}{m}\|\nabla f(x_k)\|$.

**Note.** Step 4 of the notes bounds $\frac{s_k^{\mathsf{T}} B_{k+1} s_k}{\|B_{k+1} s_k\| \|s_k\|}$, which is the angle between $s_k$ and $y_k$; but $\cos\theta_{k+1}$ involves $s_{k+1}$. That step needs to be revised.

**3.11** Statement *ch3, Theorem 5.3, p. 17 (no proof in the notes)*

Let $f$ be of class $C^2$, $x^*$ a strict local minimizer with $\nabla^2 f(x^*) \succ 0$ and $\nabla^2 f$ Lipschitz on a neighbourhood of $x^*$. If the sequence generated by BFGS or DFP with unit step, $x_{k+1} = x_k - B_k^{-1} \nabla f(x_k)$ with $B_{k+1}$ given by $B_{k+1}^{\mathrm{BFGS}} = B_k - \dfrac{B_k s_k s_k^{\mathsf{T}} B_k}{s_k^{\mathsf{T}} B_k s_k} + \dfrac{y_k y_k^{\mathsf{T}}}{y_k^{\mathsf{T}} s_k}$ ($s_k = x_{k+1} - x_k$, $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$), converges to $x^*$, then

$$\lim_{k\to\infty} \frac{\|x_{k+1} - x^*\|}{\|x_k - x^*\|} = 0.$$

Be able to state it, and to distinguish linear, superlinear and quadratic convergence.

**3.19** Numerical example *ch3, p. 18*

Let $f(x_1, x_2) = x_1^4 + x_1 x_2 + (1 + x_2)^2$, $x_0 = (1, 1)^{\mathsf{T}}$, $B_0 = I_2$.

a) Find the unique stationary point $x^*$ of $f$ (solve $\nabla f(x) = 0$) and verify that $\nabla^2 f(x^*) \succ 0$.

b) From the errors $\|x_k - x^*\|$ tabulated in the notes, compute the ratios $\|x_{k+1} - x^*\| / \|x_k - x^*\|$ and $\|x_{k+1} - x^*\| / \|x_k - x^*\|^2$, and interpret the order of convergence.

**Note.** The point $x^* \approx (0.160738, -1.080369)^{\mathsf{T}}$ given in the notes does not make the gradient vanish ($4x_1^3 + x_2 \approx -1.06$), and the Hessian is indefinite there. The correct stationary point is $x^* \approx (0.695884, -1.347942)^{\mathsf{T}}$; see table V8.

**V8** Validation table *A5 (BFGS + WeakWolfe); complements 3.19*

$f(x) = x_1^4 + x_1x_2 + (1+x_2)^2$, $\nabla f(x) = (4x_1^3 + x_2,\ x_1 + 2(1+x_2))^{\mathsf T}$, $x^* \approx (0.695884, -1.347942)^{\mathsf T}$; $x_0 = (1,1)^{\mathsf T}$, $H_0 = I_2$, $c_1 = 10^{-4}$, $c_2 = 0.9$, $\alpha_{\text{ini}} = 1$, $\varepsilon = 10^{-6}$. "eval." is the number of steps tried in A2 and $e_k = \|x_k - x^*\|$.

| $k$ | $x_k$ | $f(x_k)$ | $\|g_k\|$ | $\alpha_k$ | eval. | $e_k$ | $e_{k+1}/e_k$ |
|---|---|---|---|---|---|---|---|
| 0 | (1, 1) | 6 | 7.07107 | 0.25 | 3 | 2.36756 | 0.612 |
| 1 | (−0.25, −0.25) | 0.628906 | 1.28847 | 1 | 1 | 1.44920 | 0.251 |
| 2 | (0.334126, −1.390012) | −0.299866 | 1.31849 | 0.5 | 2 | 0.36420 | 0.853 |
| 3 | (0.708036, −1.658502) | −0.489338 | 0.65408 | 1 | 1 | 0.31080 | 0.202 |
| 4 | (0.674863, −1.407046) | −0.576451 | 0.22568 | 1 | 1 | 0.06273 | 0.327 |
| 5 | (0.716260, −1.350442) | −0.581260 | 0.12039 | 1 | 1 | 0.02053 | 0.294 |
| 6 | (0.692711, −1.342803) | −0.582406 | 0.01501 | 1 | 1 | 0.00604 | 0.090 |
| 7 | (0.695821, −1.348479) | −0.582445 | $1.45 \times 10^{-3}$ | 1 | 1 | $5.41 \times 10^{-4}$ | 0.038 |
| 8 | (0.695896, −1.347926) | −0.582445 | $9.52 \times 10^{-5}$ | 1 | 1 | $2.03 \times 10^{-5}$ | 0.009 |
| 9 | (0.695884, −1.347942) | −0.582445 | $7.28 \times 10^{-7}$ | | | $1.89 \times 10^{-7}$ | |

First step in detail: $g_0 = (5,5)^{\mathsf T}$, $p_0 = (-5,-5)^{\mathsf T}$; A2 rejects $\alpha = 1$ and $\alpha = 0.5$ by Armijo and accepts 0.25. Then $s_0 = (-1.25, -1.25)^{\mathsf T}$, $g_1 = (-0.3125, 1.25)^{\mathsf T}$, $y_0 = (-5.3125, -3.75)^{\mathsf T}$, $s_0^{\mathsf T} y_0 = 11.328125$ and

$$H_1 = \begin{pmatrix} 0.480380 & -0.347206 \\ -0.347206 & 0.825208 \end{pmatrix}, \qquad p_1 = -H_1 g_1 = (0.584126,\ -1.140012)^{\mathsf T}.$$

The ratio $e_{k+1}/e_k \to 0$ shows the superlinear convergence stated in 3.11.

## U10. Optimality over convex sets

↑Index

Propositions 1.1 and 1.2 of ch4 (pp. 1–2) repeat 1.3 and 1.4; they are not listed again.

**P7** Practice *added*

Let $f(x) = \frac{1}{2}\|x - z\|^2$ with $z = (2,-1)^{\mathsf T}$ and $C = [0,1]^2$.

a) Find the minimizer $\bar x$ of $f$ on $C$ and check that $\nabla f(\bar x) \neq 0$.

b) Verify the variational inequality $\langle \nabla f(\bar x), x - \bar x\rangle \geq 0$ for all $x \in C$ and draw $-\nabla f(\bar x)$ as a normal vector to $C$ at $\bar x$.

c) Verify that $\bar x = \operatorname{proj}_C(\bar x - \alpha \nabla f(\bar x))$ for every $\alpha > 0$.

d) Now let $\Omega = \{x \in \mathbb{R}^2 :\ 1 \leq \|x\|_\infty \leq 2\}$ and $f(x) = x_1$, which has a local minimum at $(1,0)$ on $\Omega$. Show that the variational inequality $\langle \nabla f(1,0), x - (1,0)\rangle \geq 0$ fails for some $x \in \Omega$, and explain which hypothesis of 4.1 is violated.

**4.1** Proof *ch4, Proposition 1.3, p. 2*

Let $C \subset \mathbb{R}^n$ be convex and nonempty and $f : \Omega \to \mathbb{R}$ of class $C^1$ on an open set $\Omega \supset C$.

a) If $f$ attains a local minimum on $C$ at $\bar x \in C$, then $\langle \nabla f(\bar x), x - \bar x\rangle \geq 0$ for all $x \in C$.

b) If $f$ is convex on $C$ and $\bar x \in C$ satisfies the inequality above, then $\bar x$ is a global minimum of $f$ on $C$.

**4.2** Proof *ch4, Proposition 1.4, pp. 2–3*

Let $C \subset \mathbb{R}^n$ be convex, closed and nonempty.

a) Prove that for every $y \in \mathbb{R}^n$ there is a unique $\operatorname{proj}_C(y) \in C$ minimizing $\|x - y\|$ over $C$, and that $p = \operatorname{proj}_C(y)$ if and only if $p \in C$ and $\langle y - p, x - p\rangle \le 0$ for all $x \in C$.

b) Let $f$ be convex of class $C^1$. Prove that, for any $\alpha > 0$, $\bar{x} \in C$ minimizes $f$ on $C$ if and only if $\bar{x} = \operatorname{proj}_C(\bar{x} - \alpha\nabla f(\bar{x}))$.

c) Prove that $\operatorname{proj}_C$ is nonexpansive: $\|\operatorname{proj}_C(y) - \operatorname{proj}_C(z)\| \le \|y - z\|$.

**Note.** The notes use characterization a) without proving it; c) is the ingredient for analysing projected gradient (A6).

**4.30*** Proposed exercise *ch4, p. 13*

Prove that if $f : \mathbb{R}^n \to \mathbb{R}$ is strongly convex with parameter $\alpha > 0$ on a closed convex set $C$, then $\min_{x\in C} f(x)$ has a unique optimal solution.

**P9** Practice *added*

*Projected gradient by hand.* Let $f(x_1, x_2) = 2x_1^2 + x_2^2 - 2x_1x_2 + 2x_1 - 4x_2$ (the one in 1.8*) on the box $C = [0, 2]^2$, with $\operatorname{proj}_C$ computed componentwise.

a) Verify with the variational inequality $\langle\nabla f(\bar{x}), x - \bar{x}\rangle \ge 0$ $(\forall x \in C)$ that $\bar{x} = (\frac{1}{2}, 2)^\mathsf{T}$ is the minimizer of $f$ on $C$, and find $f(\bar{x})$.

b) With $L = \lambda_{\max}(\nabla^2 f)$, check that the fixed step $\alpha = \frac{1}{5}$ satisfies $\alpha < 2/L$.

c) Run projected gradient $x_{k+1} = \operatorname{proj}_C(x_k - \alpha\nabla f(x_k))$ (A6) from $x_0 = (0, 0)^\mathsf{T}$ with $\alpha = \frac{1}{5}$: for each $k$ record $g_k$, $y_k = x_k - \alpha g_k$, $x_{k+1} = \operatorname{proj}_C(y_k)$ and the residual $\|x_{k+1} - x_k\|$ (table V9).

d) Identify the iteration from which the bound $x_2 \le 2$ stays active forever, and explain why from then on the residual is multiplied by $\frac{1}{5}$ at each step.

**A6** Pseudocode *projected gradient with fixed step (ch4, Prop. 1.4)*

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**ProjectedGradient**$(f,\ \nabla f,\ \operatorname{proj}_C,\ x_0;\ \alpha, \varepsilon, k_{\max})$

---

1: **for** $k = 0, 1, \ldots, k_{\max}$ **do**
2: $\quad g_k \leftarrow \nabla f(x_k)$; $\quad y_k \leftarrow x_k - \alpha g_k$; $\quad x_{k+1} \leftarrow \operatorname{proj}_C(y_k)$
3: $\quad$ **if** $\|x_{k+1} - x_k\| < \varepsilon$ **then return** $x_{k+1}$ (fixed-point residual, 4.2 b)

---

For a box $C = \prod_i [l_i, u_i]$: $[\operatorname{proj}_C(y)]_i = \min\{\max\{y_i, l_i\}, u_i\}$. With $f$ convex and $\nabla f$ Lipschitz with constant $L$, any $\alpha \in (0, 2/L)$ is admissible. If $C = \mathbb{R}^n$, A6 is steepest descent with fixed step.

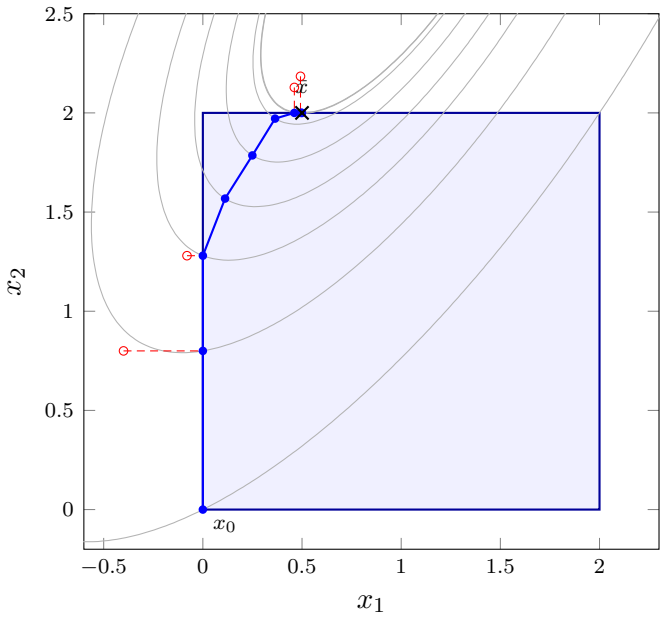
**V9** Validation table *A6 on $C = [0, 2]^2$, $\alpha = \frac{1}{5}$; solves P9*

$f(x) = 2x_1^2 + x_2^2 - 2x_1x_2 + 2x_1 - 4x_2$, $\nabla f(x) = (4x_1 - 2x_2 + 2,\ -2x_1 + 2x_2 - 4)^\mathsf{T}$, $C = [0, 2]^2$, $\bar{x} = (\frac{1}{2}, 2)^\mathsf{T}$, $f(\bar{x}) = -\frac{9}{2}$; from $x_0 = (0, 0)^\mathsf{T}$. Rows 0–3 in exact fractions; $r_k = \|x_{k+1} - x_k\|$.

| $k$ | $x_k$ | $g_k$ | $y_k = x_k - \alpha g_k$ | $f(x_k)$ | $r_k$ |
|---|---|---|---|---|---|
| 0 | $(0,\ 0)$ | $(2,\ -4)$ | $\left(-\frac{2}{5},\ \frac{4}{5}\right)$ | 0 | 0.8 |
| 1 | $\left(0,\ \frac{4}{5}\right)$ | $\left(\frac{2}{5},\ -\frac{12}{5}\right)$ | $\left(-\frac{2}{25},\ \frac{32}{25}\right)$ | $-2.56$ | 0.48 |
| 2 | $\left(0,\ \frac{32}{25}\right)$ | $\left(-\frac{14}{25},\ -\frac{36}{25}\right)$ | $\left(\frac{14}{125},\ \frac{196}{125}\right)$ | $-3.4816$ | 0.3090 |
| 3 | $\left(\frac{14}{125},\ \frac{196}{125}\right)$ | $\left(-\frac{86}{125},\ -\frac{136}{125}\right)$ | $\left(\frac{156}{625},\ \frac{1116}{625}\right)$ | $-3.915520$ | 0.2575 |
| 4 | $(0.249600,\ 1.785600)$ | $(-0.572800,\ -0.928000)$ | $(0.364160,\ 1.971200)$ | $-4.221604$ | 0.2181 |
| 5 | $(0.364160,\ 1.971200)$ | $(-0.485760,\ -0.785920)$ | $(0.461312,\ 2.128384)$ | $-4.441290$ | 0.1013 |
| 6 | $(0.461312,\ 2)$ | $(-0.154752,\ -0.922624)$ | $(0.492262,\ 2.184525)$ | $-4.497006$ | $3.095 \times 10^{-2}$ |
| 7 | $(0.492262,\ 2)$ | $(-0.030950,\ -0.984525)$ | $(0.498452,\ 2.196905)$ | $-4.499880$ | $6.190 \times 10^{-3}$ |
| 8 | $(0.498452,\ 2)$ | $(-0.006190,\ -0.996905)$ | $(0.499690,\ 2.199381)$ | $-4.499995$ | $1.238 \times 10^{-3}$ |
| 9 | $(0.499690,\ 2)$ | $(-0.001238,\ -0.999381)$ | $(0.499938,\ 2.199876)$ | $-4.500000$ | $2.476 \times 10^{-4}$ |

At $k = 0, 1$ the projection clips $x_1$ to 0; from $k = 5$ it clips $x_2$ to 2 and the bound $x_2 \le 2$ stays active forever. On that face, $x_1^{k+1} = \frac{1}{5}x_1^k + \frac{2}{5}$, so $r_{k+1}/r_k = \frac{1}{5}$. At $\bar{x} = (\frac{1}{2}, 2)$: $g = (0, -1)$, the outward normal of the face $x_2 = 2$, and the variational inequality holds.

**Figure.**




Box $C = [0, 2]^2$, level curves of $f$ through the iterates and through $f(\bar{x}) = -\frac{9}{2}$. In blue, the iterates $x_k$; in red, the points $y_k = x_k - \alpha g_k$ that fall outside $C$ and their projection (dashed line). First the bound $x_1 \ge 0$ becomes active, then the bound $x_2 \le 2$, and on that face the method becomes a one-variable gradient method with factor $\frac{1}{5}$.

## U11. Tangent cone and constraint qualifications

↑Index

### 4.15 Counterexample

*ch4, p. 4*

Let $C = \{x \in \mathbb{R}^2 :\ g_1(x) = x_1^2 + x_2 \le 0,\ g_2(x) = x_1^2 - x_2 \le 0\}$ and $\bar{x} = (0, 0)$. The tangent cone is $T(a, C) = \{d :\ \exists\, d_k \to d,\ t_k \to 0^+ \text{ with } a + t_k d_k \in C\}$. The linearized cone is $L(x, C) = \{d :\ \langle \nabla g_i(x), d\rangle \le 0\ (i \in I(x)),\ \langle \nabla h_j(x), d\rangle = 0\ (\forall j)\}$. KKT multipliers at $\bar{x}$: $\lambda_i \ge 0$, $\mu_j \in \mathbb{R}$ with $\nabla f(\bar{x}) + \sum_i \lambda_i \nabla g_i(\bar{x}) + \sum_j \mu_j \nabla h_j(\bar{x}) = 0$ and $\lambda_i g_i(\bar{x}) = 0$.

a) Prove that $C = \{(0, 0)\}$, that $T(\bar{x}, C) = \{0\}$ and that $L(\bar{x}, C) = \{d :\ d_2 = 0\}$.

b) With $f(x) = x_1$, check that $\bar{x}$ is a global minimum on $C$ and that there are no KKT multipliers. Relate this to the failure of CQ.

**4.3** Proof *ch4, Proposition 2.1, p. 3*

Let $C \subset \mathbb{R}^n$ be nonempty. The tangent cone is $T(a,C) = \{d :\ \exists\, d_k \to d,\ t_k \to 0^+ \text{ with } a + t_k d_k \in C\}$. Prove that, for every $a \in C$, $T(a,C)$ is a closed cone containing the origin.

**Note.** The notes close the closedness part with "selecting a suitable diagonal"; the diagonal subsequence must be constructed explicitly.

**4.4** Proof *ch4, Theorem 2.1, p. 3*

Let $C \subset \mathbb{R}^n$. The tangent cone is $T(a,C) = \{d :\ \exists\, d_k \to d,\ t_k \to 0^+ \text{ with } a + t_k d_k \in C\}$. If $f : \mathbb{R}^n \to \mathbb{R}$ of class $C^1$ attains a local minimum on $C$ at $a \in C$, prove that $\langle \nabla f(a), d\rangle \geq 0$ for all $d \in T(a,C)$.

**4.5** Proof *ch4, Proposition 2.2, p. 4*

Let $C = \{x \in \mathbb{R}^n :\ g_i(x) \leq 0\ (i = 1,\dots,p),\ h_j(x) = 0\ (j = 1,\dots,q)\}$ with $g_i, h_j$ of class $C^1$, and let $I(x) = \{i :\ g_i(x) = 0\}$ be the set of active indices at $x \in C$. The tangent cone is $T(a,C) = \{d :\ \exists\, d_k \to d,\ t_k \to 0^+ \text{ with } a + t_k d_k \in C\}$. The linearized cone is $L(x,C) = \{d :\ \langle \nabla g_i(x), d\rangle \leq 0\ (i \in I(x)),\ \langle \nabla h_j(x), d\rangle = 0\ (\forall j)\}$. Prove that $T(x,C) \subset L(x,C)$ for all $x \in C$.

**4.21** Numerical example *ch4, p. 11 (solved exercise)*

Let $C = \{x \in \mathbb{R}^2 :\ x_1 + 2x_2 \leq 4,\ x_1 \geq 0,\ x_2 \geq 0\}$ and $\bar{x} = (0,2)^{\mathsf{T}}$. Identify the active constraints, compute the linearized cone $L(\bar{x},C) = \{d :\ \langle \nabla g_i(\bar{x}), d\rangle \leq 0,\ i \text{ active}\}$ and conclude $T(\bar{x},C) = \{d :\ d_1 + 2d_2 \leq 0,\ d_1 \geq 0\}$ using that $T = L$ for affine constraints (4.6). Draw $C$ and the cone translated to $\bar{x}$.

**4.6** Proof *ch4, Proposition 2.3, p. 5*

Let $C = \{x \in \mathbb{R}^n :\ g_i(x) \leq 0\ (i = 1,\dots,p),\ h_j(x) = 0\ (j = 1,\dots,q)\}$ with $g_i, h_j$ of class $C^1$, and let $I(x) = \{i :\ g_i(x) = 0\}$ be the set of active indices at $x \in C$. Suppose $g_i(x) = a_i^{\mathsf{T}} x - b_i$ and $h_j(x) = a_j'^{\mathsf{T}} x - b_j'$ are affine. The tangent cone is $T(a,C) = \{d :\ \exists\, d_k \to d,\ t_k \to 0^+ \text{ with } a + t_k d_k \in C\}$. The linearized cone is $L(x,C) = \{d :\ \langle \nabla g_i(x), d\rangle \leq 0\ (i \in I(x)),\ \langle \nabla h_j(x), d\rangle = 0\ (\forall j)\}$. Prove that $T(x,C) = L(x,C)$ for all $x \in C$.

**4.26*** Proposed exercise *ch4, p. 12*

For $C = \{x \in \mathbb{R}^2 :\ x_2 \geq x_1^2,\ x_1^2 + x_2^2 \leq 2\}$, compute the tangent cone $T(\bar{x},C) = \{d :\ \exists\, d_k \to d,\ t_k \to 0^+,\ \bar{x} + t_k d_k \in C\}$ at the intersection point $\bar{x} = (1,1)^{\mathsf{T}}$.

**4.27*** Proposed exercise *ch4, p. 12*

The tangent cone is $T(a,C) = \{d :\ \exists\, d_k \to d,\ t_k \to 0^+ \text{ with } a + t_k d_k \in C\}$. Prove that if $C \subset \mathbb{R}^n$ is closed and convex and $\bar{x} \in C$, then $T(\bar{x},C) = \operatorname{cl}\big(\bigcup_{\gamma>0} \gamma(C - \bar{x})\big)$.

**4.7** Proof *ch4, Proposition 2.4, pp. 5–6*

Let $C = \{x :\ g_i(x) \leq 0,\ i = 1,\dots,p\}$ with $g_i$ convex of class $C^1$, and $I(x) = \{i :\ g_i(x) = 0\}$. Assume Slater's condition: there is $\hat{x}$ with $g_i(\hat{x}) < 0$ for all $i$. The tangent cone is $T(a,C) = \{d :\ \exists\, d_k \to d,\ t_k \to 0^+ \text{ with } a + t_k d_k \in C\}$. The linearized cone is $L(x,C) = \{d :\ \langle \nabla g_i(x), d\rangle \leq 0,\ i \in I(x)\}$. Prove that $T(x,C) = L(x,C)$ for all $x \in C$.

**Note.** The step "by Taylor's theorem there is $t_\epsilon$" must treat active constraints (sign of the directional derivative) and inactive constraints (continuity) separately.

**4.24** NUMERICAL EXAMPLE *ch4, p. 12 (solved exercise)*

Let $C = \{x \in \mathbb{R}^2 :\ x_2 \geq (1-x_1)^3,\ x_2 \geq 0\}$ and $\bar{x} = (1,0)^{\mathsf{T}}$. Write $C$ with $g_1(x) = (1-x_1)^3 - x_2 \leq 0$, $g_2(x) = -x_2 \leq 0$, check that both are active, compute their gradients and show that $d = (0,1)^{\mathsf{T}}$ satisfies MFCQ: $\langle \nabla g_i(\bar{x}), d\rangle < 0$ for every active $i$ (there are no equalities).

**4.8** PROOF *ch4, Theorem 2.2, p. 6*

Let $C = \{x \in \mathbb{R}^n :\ g_i(x) \leq 0\ (i = 1,\dots,p),\ h_j(x) = 0\ (j = 1,\dots,q)\}$ with $g_i, h_j$ of class $C^1$, and let $I(x) = \{i :\ g_i(x) = 0\}$ be the set of active indices at $x \in C$. The tangent cone is $T(a,C) = \{d :\ \exists\, d_k \to d,\ t_k \to 0^+$ with $a + t_k d_k \in C\}$. The linearized cone is $L(x,C) = \{d :\ \langle \nabla g_i(x), d\rangle \leq 0\ (i \in I(x)),\ \langle \nabla h_j(x), d\rangle = 0\ (\forall j)\}$. MFCQ at $\bar{x}$: $\{\nabla h_j(\bar{x})\}$ linearly independent and there is $d$ with $\langle \nabla g_i(\bar{x}), d\rangle < 0\ (i \in I(\bar{x}))$ and $\langle \nabla h_j(\bar{x}), d\rangle = 0\ (\forall j)$. Prove that if $\bar{x} \in C$ satisfies MFCQ, then $T(\bar{x},C) = L(\bar{x},C)$. The implicit function theorem may be cited in the form stated in the notes.

**Note.** One must also check that the curve $\gamma_k(t)$ respects the inactive constraints and that $(\gamma_k(t) - \bar{x})/t \to d^{(k)}$, which is what gives $d^{(k)} \in T(\bar{x},C)$.

**4.29*** PROPOSED EXERCISE *ch4, p. 12*

Let $C = \{x \in \mathbb{R}^2 :\ (x_1 - 1)^3 + x_2 \leq 0,\ x_1 \geq 0,\ x_2 \geq 0\}$. Analyse whether LICQ and MFCQ hold at $\bar{x} = (1,0)^{\mathsf{T}}$. LICQ at $\bar{x}$: the gradients $\{\nabla g_i(\bar{x})\}_{i \in I(\bar{x})} \cup \{\nabla h_j(\bar{x})\}_j$ are linearly independent. MFCQ at $\bar{x}$: $\{\nabla h_j(\bar{x})\}$ linearly independent and there is $d$ with $\langle \nabla g_i(\bar{x}), d\rangle < 0\ (i \in I(\bar{x}))$ and $\langle \nabla h_j(\bar{x}), d\rangle = 0\ (\forall j)$. The tangent cone is $T(a,C) = \{d :\ \exists\, d_k \to d,\ t_k \to 0^+$ with $a + t_k d_k \in C\}$. The linearized cone is $L(x,C) = \{d :\ \langle \nabla g_i(x), d\rangle \leq 0\ (i \in I(x)),\ \langle \nabla h_j(x), d\rangle = 0\ (\forall j)\}$. KKT multipliers at $\bar{x}$: $\lambda_i \geq 0$, $\mu_j \in \mathbb{R}$ with $\nabla f(\bar{x}) + \sum_i \lambda_i \nabla g_i(\bar{x}) + \sum_j \mu_j \nabla h_j(\bar{x}) = 0$ and $\lambda_i g_i(\bar{x}) = 0$. Also compute $T(\bar{x},C)$ and $L(\bar{x},C)$, and show that for $f(x) = -x_1$ the point $\bar{x}$ is a global minimum on $C$ without KKT multipliers.

## U12. Farkas' lemma and KKT conditions

↑Index

**4.16** COUNTEREXAMPLE *ch4, p. 7*

Consider $\min x_1$ subject to $g_1(x) = -x_2 \leq 0$ and $g_2(x) = x_2 - x_1^3 \leq 0$.

a) Prove that $\bar{x} = (0,0)^{\mathsf{T}}$ is the only optimum.

b) Show that there are no KKT multipliers at $\bar{x}$: $\lambda_1, \lambda_2 \geq 0$ with $\nabla f(\bar{x}) + \lambda_1 \nabla g_1(\bar{x}) + \lambda_2 \nabla g_2(\bar{x}) = 0$.

c) Compute $T(\bar{x},C)$ and $L(\bar{x},C)$, and verify that MFCQ and LICQ fail at $\bar{x}$.

The tangent cone is $T(a,C) = \{d :\ \exists\, d_k \to d,\ t_k \to 0^+$ with $a + t_k d_k \in C\}$. The linearized cone is $L(x,C) = \{d :\ \langle \nabla g_i(x), d\rangle \leq 0\ (i \in I(x)),\ \langle \nabla h_j(x), d\rangle = 0\ (\forall j)\}$. MFCQ at $\bar{x}$: $\{\nabla h_j(\bar{x})\}$ linearly independent and there is $d$ with $\langle \nabla g_i(\bar{x}), d\rangle < 0\ (i \in I(\bar{x}))$ and $\langle \nabla h_j(\bar{x}), d\rangle = 0\ (\forall j)$. LICQ at $\bar{x}$: the gradients $\{\nabla g_i(\bar{x})\}_{i \in I(\bar{x})} \cup \{\nabla h_j(\bar{x})\}_j$ are linearly independent.

**4.9** PROOF *ch4, Lemma 3.1 (Farkas), p. 6 (no proof in the notes)*

Let $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{q \times n}$ and $c \in \mathbb{R}^n$. Prove that the following are equivalent:

i) $Ad \leq 0$, $Bd = 0$ implies $c^{\mathsf{T}} d \geq 0$.

ii) There exist $u \in \mathbb{R}^m$, $v \in \mathbb{R}^q$ with $u \geq 0$ and $c + A^{\mathsf{T}} u + B^{\mathsf{T}} v = 0$.

The separation theorem for a point and a closed convex set may be cited; one must prove that the cone $\{A^{\mathsf{T}} u + B^{\mathsf{T}} v :\ u \geq 0\}$ is closed.

**4.10** Proof *ch4, Theorem 3.1 (KKT), p. 7*

Let $C = \{x \in \mathbb{R}^n : \ g_i(x) \le 0 \ (i = 1, \dots, p), \ h_j(x) = 0 \ (j = 1, \dots, q)\}$ with $g_i, h_j$ of class $C^1$, and let $I(x) = \{i : \ g_i(x) = 0\}$ be the set of active indices at $x \in C$. Let $\bar{x}$ be a local minimum of $f \in C^1$ on $C$, and suppose $\bar{x}$ satisfies a constraint qualification, i.e. $T(\bar{x}, C) = L(\bar{x}, C)$ (for instance under MFCQ, LICQ, Slater, or $C$ a polyhedron). The tangent cone is $T(a, C) = \{d : \ \exists\, d_k \to d, \ t_k \to 0^+ \text{ with } a + t_k d_k \in C\}$. The linearized cone is $L(x, C) = \{d : \ \langle \nabla g_i(x), d\rangle \le 0 \ (i \in I(x)), \ \langle \nabla h_j(x), d\rangle = 0 \ (\forall j)\}$. Farkas' lemma (4.9) may be used. Prove that there exist $\lambda_1, \dots, \lambda_p \ge 0$ and $\mu_1, \dots, \mu_q \in \mathbb{R}$ such that

$$\nabla f(\bar{x}) + \sum_{i=1}^{p} \lambda_i \nabla g_i(\bar{x}) + \sum_{j=1}^{q} \mu_j \nabla h_j(\bar{x}) = 0, \qquad \lambda_i g_i(\bar{x}) = 0 \ \ (i = 1, \dots, p).$$

**4.20** Numerical example *ch4, p. 11 (solved exercise)*

Let $f(x) = x_1^2 + x_2^2 - 4x_1 - 6x_2$, $g_1(x) = 2 - x_1 - x_2 \le 0$, $g_2(x) = x_1 + 2x_2 - 5 \le 0$. Write the KKT conditions, solve the case $g_2$ active, $g_1$ inactive, and justify that $(1.4,\ 1.8)^{\mathsf{T}}$ with $\lambda = (0, \frac{6}{5})^{\mathsf{T}}$ is the unique global minimum.

**P10** Practice *added*

*Active-set enumeration.* Let $\min f(x) = x_1^2 + x_2^2 - 4x_1 - 6x_2$ s.t. $g_1(x) = 2 - x_1 - x_2 \le 0$, $g_2(x) = x_1 + 2x_2 - 5 \le 0$. Run A7: for each $\mathcal{A} \subseteq \{1, 2\}$ solve the linear stationarity system with $g_i = 0$ for $i \in \mathcal{A}$, and discard the case if some $\lambda_i < 0$ $(i \in \mathcal{A})$ or some $g_i > 0$ $(i \notin \mathcal{A})$ (table V10). Explain why the procedure is exact here and why it does not scale ($2^p$ cases).

**A7** Pseudocode *KKT by active-set enumeration (small problems)*

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**KKTEnumeration**$(f,\, g_1, \dots, g_p)$ $\quad f$ convex quadratic, $g_i$ affine

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1: **for each** $\mathcal{A} \subseteq \{1, \dots, p\}$ **do**
2: solve the linear system $\nabla f(x) + \sum_{i \in \mathcal{A}} \lambda_i \nabla g_i = 0, \ \ g_i(x) = 0 \ (i \in \mathcal{A})$
3: **if** the system is singular **then** go to the next $\mathcal{A}$
4: **if** $\lambda_i \ge 0 \ \forall i \in \mathcal{A}$ **and** $g_i(x) \le 0 \ \forall i \notin \mathcal{A}$ **then return** $(x, \lambda)$ $\qquad (\lambda_i = 0 \text{ if } i \notin \mathcal{A})$

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With affine constraints no further qualification is needed, and by convexity the first KKT point found is a global minimum. The cost is $2^p$ linear systems: it is for validating by hand, not an algorithm.

**V10** Validation table *A7 on 4.20; solves P10*

$f = x_1^2 + x_2^2 - 4x_1 - 6x_2$, $g_1 = 2 - x_1 - x_2$, $g_2 = x_1 + 2x_2 - 5$.

| $\mathcal{A}$ | $x$ | $\lambda_{\mathcal{A}}$ | $(g_1(x),\, g_2(x))$ | $f(x)$ | decision |
|---|---|---|---|---|---|
| $\emptyset$ | $(2,\ 3)$ | — | $(-3,\ 3)$ | $-13$ | $g_2 > 0$: infeasible |
| $\{1\}$ | $(0.5,\ 1.5)$ | $\lambda_1 = -3$ | $(0,\ -1.5)$ | $-8.5$ | $\lambda_1 < 0$ |
| $\{2\}$ | $(1.4,\ 1.8)$ | $\lambda_2 = 1.2$ | $(-1.2,\ 0)$ | $-11.2$ | **KKT** |
| $\{1,2\}$ | $(-1,\ 3)$ | $(\lambda_1, \lambda_2) = (-12,\ -6)$ | $(0,\ 0)$ | $-4$ | $\lambda < 0$ |

**What it shows.** The sign of the multipliers is what rules out the wrong active sets; feasibility alone is not enough (case $\{1\}$ is feasible).

**4.25*** Proposed exercise *ch4, p. 12*

Minimize $f(x_1, x_2) = 2x_1^2 + x_2^2 - 12x_1 - 8x_2$ subject to $x_1 + x_2 \le 4$ and $x_1, x_2 \ge 0$. Find the global minimum with KKT and justify uniqueness using the Hessian.

**4.31*** Proposed exercise *ch4, p. 13*

Let $f(x) = \frac{1}{2}(x_1^2 + x_2^2 + x_3^2)$ subject to $x_1 + x_2 + x_3 = 10$, $x_1 \ge 1$, $x_2 \ge 2$, $x_3 \ge 0$. Compute the optimal Lagrange multipliers analytically.

**4.17** Example *ch4, p. 7*

Let $A \in \mathbb{R}^{m \times n}$ have rank $m < n$, $b \in \mathbb{R}^m$ and $y \in \mathbb{R}^n$. Derive by KKT that the projection of $y$ onto $\{x : \ Ax = b\}$ is $x = y - A^{\mathsf{T}}(AA^{\mathsf{T}})^{-1}(Ay - b)$. Justify that the qualification $T = L$ holds (affine constraints) and that the KKT point is the global minimum.

**4.18** Example *ch4, p. 8*

Let $M \in \mathbb{R}^{n \times n}$ be symmetric. Show that the KKT points of $\min / \max \ x^{\mathsf{T}} M x$ on $\|x\|^2 = 1$ are the unit eigenvectors of $M$, with multiplier equal to the eigenvalue, and conclude that the optimal values are $\lambda_{\min}(M)$ and $\lambda_{\max}(M)$.

**Note.** The notes justify neither the existence of optima (compactness of the sphere) nor the qualification (LICQ on the sphere); both are needed for a complete argument.

**4.19** Example *ch4, p. 8*

Maximize the entropy $H(p) = -\sum_{i=1}^{n} p_i \ln p_i$ over the simplex $\{p \ge 0, \ \sum p_i = 1\}$. Show that the optimum is $p_i = 1/n$ and deduce $H(p) \le \ln n$.

**Note.** The notes omit the constraints $p_i \ge 0$ from the Lagrangian, and ln is not differentiable at 0. One must argue with the concavity of $H$ (or with 4.1 on the simplex) that the interior stationary point is the global maximum.

**4.32*** Proposed exercise *ch4, p. 13*

Give a nonconvex problem in which the KKT conditions hold at a point that is a local maximum instead of a minimum.

**4.33*** Proposed exercise *ch4, p. 13*

Let $f(x, y) = x^2 - y^2$. Prove that $(0,0)^{\mathsf{T}}$ satisfies $\nabla f(0,0) = 0$ but is not a local extremum. Identify the directions of strict descent from the spectral analysis of $\nabla^2 f(0,0)$.

## U13. Lagrangian, saddle points and duality

↑Index

**P8** Practice *added*

*Inf-sup gap.* Let $A = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ and $\Phi(x, y) = x^{\mathsf{T}} A y$, minimized in $x$ and maximized in $y$.

a) With $x, y \in \{e_1, e_2\}$ (vertices), compute $\max_y \min_x \Phi$ and $\min_x \max_y \Phi$, and verify that the inf-sup inequality is strict.

b) With $x, y \in \Delta_2 = \{z \in \mathbb{R}^2 : \ z \ge 0, \ z_1 + z_2 = 1\}$, compute both values and verify equality, with saddle point $x = y = (\frac{1}{2}, \frac{1}{2})^{\mathsf{T}}$.

**4.11** Proof *ch4, Proposition 4.1, p. 8*

For any $L : X \times Y \to \mathbb{R}$, prove that $\sup_{y\in Y} \inf_{x\in X} L(x,y) \le \inf_{x\in X} \sup_{y\in Y} L(x,y)$.

**4.22** Numerical example *ch4, p. 11 (solved exercise)*

Let $L(x,\lambda) = x^2 + 4x + \lambda(1-x)$. Identify the primal problem ($\min x^2 + 4x$ s.t. $1 - x \le 0$), verify that $(\bar{x},\bar{\lambda}) = (1,6)$ is a saddle point, i.e. $L(\bar{x},\lambda) \le L(\bar{x},\bar{\lambda}) \le L(x,\bar{\lambda})$ for all $x \in \mathbb{R}$, $\lambda \ge 0$, and compute the dual function $q(\lambda) = \inf_x L(x,\lambda)$ to check $p^* = d^* = 5$.

**4.12** Proof *ch4, Theorem 4.1, p. 9*

For $\min f(x)$ s.t. $g_i(x) \le 0$ $(i = 1,\dots,p)$, $h_j(x) = 0$ $(j = 1,\dots,q)$, let $L(x,\lambda,\mu) = f(x) + \sum_i \lambda_i g_i(x) + \sum_j \mu_j h_j(x)$ with $\lambda \ge 0$, the dual function $q(\lambda,\mu) = \inf_{x\in\mathbb{R}^n} L(x,\lambda,\mu)$, $p^*$ the primal optimal value and $d^* = \sup_{\lambda\ge 0,\mu} q(\lambda,\mu)$. A saddle point is $(\bar{x},\bar{\lambda},\bar{\mu})$ with $\bar{\lambda} \ge 0$ and $L(\bar{x},\lambda,\mu) \le L(\bar{x},\bar{\lambda},\bar{\mu}) \le L(x,\bar{\lambda},\bar{\mu})$ for all $x$, $\lambda \ge 0$, $\mu$. Prove that $(\bar{x},\bar{\lambda},\bar{\mu})$ is a saddle point if and only if $\bar{x}$ is primal optimal, $(\bar{\lambda},\bar{\mu})$ is dual optimal and $p^* = d^*$.

**Note.** In ($\Leftarrow$) complementary slackness must be deduced from $f(\bar{x}) = q(\bar{\lambda},\bar{\mu}) \le L(\bar{x},\bar{\lambda},\bar{\mu}) \le f(\bar{x})$ before asserting the two saddle inequalities.

**4.13** Proof *ch4, Theorem 5.1, p. 10*

For $\min f(x)$ s.t. $g_i(x) \le 0$ $(i = 1,\dots,p)$, $h_j(x) = 0$ $(j = 1,\dots,q)$, let $L(x,\lambda,\mu) = f(x) + \sum_i \lambda_i g_i(x) + \sum_j \mu_j h_j(x)$ with $\lambda \ge 0$, the dual function $q(\lambda,\mu) = \inf_{x\in\mathbb{R}^n} L(x,\lambda,\mu)$, $p^*$ the primal optimal value and $d^* = \sup_{\lambda\ge 0,\mu} q(\lambda,\mu)$.

a) (Weak duality) Prove that $q(\lambda,\mu) \le f(x)$ for every feasible $x$ and $\lambda \ge 0$; in particular $d^* \le p^*$.

b) (Strong duality) If $f, g_i$ are convex, $h_j$ affine, $p^*$ finite and Slater holds (there is $\hat{x}$ with $g_i(\hat{x}) < 0$ for all $i$ and $h_j(\hat{x}) = 0$ for all $j$), prove that there is $(\lambda^*,\mu^*)$ with $\lambda^* \ge 0$ and $q(\lambda^*,\mu^*) = p^*$.

**Note.** The notes apply *strict* separation between the set $\mathcal{A}$ and the point $(0,0,p^*)$, but that point lies in the closure of $\mathcal{A}$. The correct argument separates $\mathcal{A}$ from $\mathcal{B} = \{(0,0,t) : t < p^*\}$ (disjoint convex sets) and uses Slater to show the hyperplane is not vertical. With equality constraints a relative-interior hypothesis is also required.

**4.28*** Proposed exercise *ch4, p. 12*

For $\min (x_1-3)^2 + (x_2-4)^2$ subject to $x_1 + x_2 = 5$, build the dual function $q(\mu) = \inf_x \big[(x_1-3)^2 + (x_2-4)^2 + \mu(x_1+x_2-5)\big]$, solve $\max_\mu q(\mu)$ and check that there is no duality gap.

**4.23** Numerical example *ch4, p. 12 (solved exercise)*

For $\min 3x_1 + 5x_2$ s.t. $x_1 + 2x_2 \ge 4$, $2x_1 + x_2 \ge 3$, $x \ge 0$, formulate the dual from the Lagrangian $L(x,\lambda) = 3x_1 + 5x_2 + \lambda_1(4 - x_1 - 2x_2) + \lambda_2(3 - 2x_1 - x_2)$, $\lambda \ge 0$, minimizing over $x \ge 0$; solve both problems by vertices and verify $x^* = (\frac{2}{3},\frac{5}{3})^\mathsf{T}$, $\lambda^* = (\frac{7}{3},\frac{1}{3})^\mathsf{T}$ and common value $\frac{31}{3}$. Verify complementary slackness too.

**4.14** Proof *ch4, Theorem 5.3, p. 10 (no proof in the notes)*

Let $A \in \mathbb{R}^{m\times n}$. Prove that exactly one of the following holds: (1) there is $x > 0$ with $Ax = 0$; (2) there is $y \in \mathbb{R}^m$ with $A^\mathsf{T} y \ge 0$ and $A^\mathsf{T} y \ne 0$.

**Note.** The notes call it Gordan's theorem; with this statement it is Stiemke's theorem. Gordan's is: there is $x \ge 0$, $x \ne 0$ with $Ax = 0$, or there is $y$ with $A^\mathsf{T} y > 0$. It can be derived from Farkas' lemma: $Ad \le 0$, $Bd = 0 \Rightarrow c^\mathsf{T} d \ge 0$ if and only if there are $u \ge 0$, $v$ with $c + A^\mathsf{T} u + B^\mathsf{T} v = 0$ (4.9).

**4.34*** PROPOSED EXERCISE *ch4, p. 13*

For $\min c^\mathsf{T}x$ s.t. $Ax = b$, $x \geq 0$, use Farkas' lemma (4.9) to prove the strong duality theorem of linear programming (Theorem 5.2, p. 10): if the primal or the dual $\max b^\mathsf{T}y$ s.t. $A^\mathsf{T}y \leq c$ has a finite optimal solution, both do and $c^\mathsf{T}x^* = b^\mathsf{T}y^*$. Do not use differentiability.

## U14. Trust region: model, ratio $\rho_k$ and radius

↑Index

**5.16** NUMERICAL EXAMPLE *ch5, p. 24 (solved exercise 4)*

Let $f(x_k) = 100$, $x_k = (0,0)^\mathsf{T}$, $g_k = (-2,0)^\mathsf{T}$, $B_k = 2I$, $p_k = (1,0)^\mathsf{T}$, $\Delta_k = 1$ and $f(x_k+p_k) = 98.8$. Compute the predicted reduction $m_k(0) - m_k(p_k) = -g_k^\mathsf{T}p_k - \frac{1}{2}p_k^\mathsf{T}B_kp_k$, the actual reduction and $\rho_k$; decide acceptance with $\eta_1 = 0.1$ and the new radius with $\eta_2 = 0.75$. Interpret $\rho_k > 1$. With $\rho_k = \dfrac{f(x_k) - f(x_k+p_k)}{m_k(0) - m_k(p_k)}$ and $0 < \eta_1 \leq \eta_2 < 1$, the rule in the notes is: if $\rho_k < \eta_1$ the step is rejected ($x_{k+1} = x_k$) and $\Delta_{k+1} = \frac{1}{2}\Delta_k$; if $\rho_k \geq \eta_1$ it is accepted ($x_{k+1} = x_k + p_k$), with $\Delta_{k+1} = \min(2\Delta_k, \Delta_{\max})$ if moreover $\rho_k > \eta_2$ and $\|p_k\| = \Delta_k$, and $\Delta_{k+1} = \Delta_k$ otherwise.

**P12** PRACTICE *added*

With $\rho_k = \dfrac{f(x_k) - f(x_k+p_k)}{m_k(0) - m_k(p_k)}$ and $0 < \eta_1 \leq \eta_2 < 1$, the rule in the notes is: if $\rho_k < \eta_1$ the step is rejected ($x_{k+1} = x_k$) and $\Delta_{k+1} = \frac{1}{2}\Delta_k$; if $\rho_k \geq \eta_1$ it is accepted ($x_{k+1} = x_k + p_k$), with $\Delta_{k+1} = \min(2\Delta_k, \Delta_{\max})$ if moreover $\rho_k > \eta_2$ and $\|p_k\| = \Delta_k$, and $\Delta_{k+1} = \Delta_k$ otherwise. With $\eta_1 = 0.1$, $\eta_2 = 0.75$, $\Delta_k = 1$, $\Delta_{\max} = 4$, decide $x_{k+1}$ and $\Delta_{k+1}$ in each case: (i) $\rho_k = -0.5$; (ii) $\rho_k = 0.05$; (iii) $\rho_k = 0.5$, $\|p_k\| = 1$; (iv) $\rho_k = 0.9$, $\|p_k\| = 0.6$; (v) $\rho_k = 0.9$, $\|p_k\| = 1$. Explain why in (iv) enlarging the radius would not change the next iteration if the model were the same, and why the rule of the notes does not shrink the radius when the step is accepted with small $\rho_k$ (compare with N&W, Alg. 4.1, which shrinks if $\rho_k < \frac{1}{4}$).

**5.25*** PROPOSED EXERCISE *ch5, p. 26 (exercise 8)*

With $f(x_k) = 25.0$, $m_k(0) - m_k(p_k) = 4.0$, $f(x_k + p_k) = 21.8$, $\eta_1 = 0.1$, $\eta_2 = 0.75$, $\Delta_k = 0.5$ and $\|p_k\| = 0.5$, decide whether the step is accepted and compute $\Delta_{k+1}$. With $\rho_k = \dfrac{f(x_k) - f(x_k+p_k)}{m_k(0) - m_k(p_k)}$ and $0 < \eta_1 \leq \eta_2 < 1$, the rule in the notes is: if $\rho_k < \eta_1$ the step is rejected ($x_{k+1} = x_k$) and $\Delta_{k+1} = \frac{1}{2}\Delta_k$; if $\rho_k \geq \eta_1$ it is accepted ($x_{k+1} = x_k + p_k$), with $\Delta_{k+1} = \min(2\Delta_k, \Delta_{\max})$ if moreover $\rho_k > \eta_2$ and $\|p_k\| = \Delta_k$, and $\Delta_{k+1} = \Delta_k$ otherwise.

## U15. Trust-region subproblem: exact solution

↑Index

**5.9** NUMERICAL EXAMPLE *ch5, p. 7*

For the subproblem $\min\{g^\mathsf{T}p + \frac{1}{2}p^\mathsf{T}Bp : \ \|p\|_2 \leq \Delta\}$ with $g = (-4,-6)^\mathsf{T}$, $B = \begin{pmatrix} 2 & 1 \\ 1 & 4 \end{pmatrix}$ and $\Delta = 1$:

a) Compute $p^N = -B^{-1}g = (\frac{10}{7}, \frac{8}{7})^\mathsf{T}$ and verify $\|p^N\| > \Delta$, so the solution lies on the boundary.

b) Set up $\|p(\lambda)\| = 1$ with $p(\lambda) = -(B + \lambda I)^{-1}g$ and find $\lambda^*$ (table V13).

c) Compare $g^\mathsf{T}p^* + \frac{1}{2}p^{*\mathsf{T}}Bp^*$ with the value at the Cauchy point $p^C = -\frac{\Delta}{\|g\|}g$ (truncated here; see V11).

**Note.** The notes leave the example at "the minimum must be on the boundary", without computing $\lambda^*$.

**5.1** Proof *ch5, Theorem 3.1, pp. 7–11*

Let $m_k(p) = f(x_k) + g_k^{\mathsf T} p + \tfrac12 p^{\mathsf T} B_k p$ with $g_k = \nabla f(x_k)$ and $B_k$ symmetric, and the subproblem $\min\{m_k(p) :\ \|p\|_2 \le \Delta_k\}$. Prove that $p^*$ is a global solution of the subproblem if and only if $\|p^*\| \le \Delta_k$ and there is $\lambda^* \ge 0$ with

$$(B_k + \lambda^* I)p^* = -g_k, \qquad \lambda^*(\|p^*\| - \Delta_k) = 0, \qquad B_k + \lambda^* I \succeq 0.$$

Include the derivation of the identity $m_k(p) - m_k(p^*) = \tfrac12 (p-p^*)^{\mathsf T}(B_k + \lambda^* I)(p-p^*) + \tfrac{\lambda^*}{2}\big(\|p^*\|^2 - \|p\|^2\big)$, which the notes obtain by "rearranging".

**Note.** In ($\Rightarrow$), the case $v^{\mathsf T} p^* = 0$ assumes $\|p^*\| = \Delta_k$; the case $\|p^*\| < \Delta_k$ ("follows directly") requires saying that then $\lambda^* = 0$, $p^*$ is an unconstrained local minimum of $m_k$, and SONC (1.4) gives $B_k \succeq 0$.

**5.18*** Proposed exercise *ch5, p. 25 (exercise 1)*

Let $m_k(p) = f(x_k) + g_k^{\mathsf T} p + \tfrac12 p^{\mathsf T} B_k p$ with $g_k = \nabla f(x_k)$ and $B_k$ symmetric, and the subproblem $\min\{m_k(p) :\ \|p\|_2 \le \Delta_k\}$. Prove that if $B_k \succ 0$ and $\|B_k^{-1} g_k\| \le \Delta_k$, then $p_k^N = -B_k^{-1} g_k$ is the unique global solution of the subproblem.

**5.13** Numerical example *ch5, p. 23 (solved exercise 1)*

Solve the subproblem $\min\{g^{\mathsf T} p + \tfrac12 p^{\mathsf T} B p :\ \|p\|_2 \le \Delta\}$ with $g = (-2,-4)^{\mathsf T}$, $B = \operatorname{diag}(4,2)$, $\Delta = 1$, looking for $p^* = p(\lambda^*) = -(B + \lambda^* I)^{-1} g$: verify $\|p^N\| = \sqrt{4.25} > 1$, set up $\frac{4}{(4+\lambda)^2} + \frac{16}{(2+\lambda)^2} = 1$ and find $\lambda^*$ and $p^*$.

**Note.** The notes give $\lambda^* \approx 2.21$ and $p^* \approx (0.322, 0.950)^{\mathsf T}$, but with $\lambda = 2.21$ the equation gives 1.0064, and $\|(0.322, 0.950)\| \approx 1.003$. The correct values are $\lambda^* \approx 2.224019$, $p^* \approx (0.321336, 0.946965)^{\mathsf T}$; see V13.

**5.2** Proof *ch5, Proposition 3.1, pp. 12–13*

Let $B_k = \sum_i \lambda_i q_i q_i^{\mathsf T}$ be symmetric (spectral decomposition, $\gamma_i = q_i^{\mathsf T} g_k$), $g_k \neq 0$ and $p(\lambda) = -(B_k + \lambda I)^{-1} g_k$. On every interval where $B_k + \lambda I \succ 0$, prove: $p'(\lambda) = -(B_k + \lambda I)^{-1} p(\lambda)$; $\lambda \mapsto \|p(\lambda)\|$ is strictly decreasing; $p(\lambda) \to 0$ as $\lambda \to +\infty$; and $p(0) = p_k^N$ if $B_k \succ 0$. Deduce, with the spectral representation $\|p(\lambda)\|^2 = \sum_i \gamma_i^2/(\lambda_i + \lambda)^2$, that if $B_k \succ 0$ and $\|p_k^N\| > \Delta_k$ there is a unique $\lambda^* > 0$ with $\|p(\lambda^*)\| = \Delta_k$.

**A9** Pseudocode *exact subproblem: Newton on the secular equation (ch5, p. 11, footnote)*

Solves $\min\{g^{\mathsf T} p + \tfrac12 p^{\mathsf T} B p :\ \|p\| \le \Delta\}$ with $p(\lambda) = -(B + \lambda I)^{-1} g$.

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**ExactSubproblem**$(g,\ B,\ \Delta;\ \lambda^{(0)},\ \text{tol},\ l_{\max})$

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1: $\lambda_1 \leftarrow \lambda_{\min}(B)$
2: **if** $\lambda_1 > 0$ **then** solve $Bp = -g$; **if** $\|p\| \le \Delta$ **then return** $(p, 0)$
3: **if** $q_i^{\mathsf T} g = 0$ for every eigenvector $q_i$ of $\lambda_1$ **and** $\|p(-\lambda_1)\| < \Delta$ **then** (hard case)
4: $\quad z \leftarrow$ unit eigenvector of $\lambda_1$; $\quad \tau \leftarrow \sqrt{\Delta^2 - \|p(-\lambda_1)\|^2}$; $\quad$ **return** $(p(-\lambda_1) + \tau z,\ -\lambda_1)$
5: $\lambda \leftarrow \lambda^{(0)}$ (with $\lambda^{(0)} > \max\{0, -\lambda_1\}$ and $\|p(\lambda^{(0)})\| > \Delta$)
6: **for** $l = 0, 1, \ldots, l_{\max}$ **do**
7: $\quad$ factor $B + \lambda I = R^{\mathsf T} R$ (Cholesky); $\quad$ solve $R^{\mathsf T} R p = -g$; $\quad$ solve $R^{\mathsf T} q = p$
8: $\quad$ **if** $\big|\|p\| - \Delta\big| \le \text{tol} \cdot \Delta$ **then return** $(p, \lambda)$
9: $\quad \lambda \leftarrow \lambda + \Big(\dfrac{\|p\|}{\|q\|}\Big)^2 \dfrac{\|p\| - \Delta}{\Delta}$

---

Since $\|q\|^2 = p^{\mathsf T}(B + \lambda I)^{-1} p$, line 9 is the update of the notes, $\lambda \leftarrow \lambda + \big(\frac{\|p\|}{\Delta} - 1\big)\frac{\|p\|^2}{p^{\mathsf T}(B+\lambda I)^{-1} p}$, i.e. Newton on $\phi(\lambda) = 1/\|p(\lambda)\| - 1/\Delta$. If $\|p(\lambda^{(0)})\| > \Delta$, the iterates increase monotonically to $\lambda^*$. For $2 \times 2$ problems line 7 can be replaced by the explicit formula for $p(\lambda)$.

**V13** VALIDATION TABLE *A9 on 5.13 and 5.9*

Subproblem $\min\{g^{\mathsf T}p+\tfrac12 p^{\mathsf T}Bp:\ \|p\|\le\Delta\}$, $p(\lambda)=-(B+\lambda I)^{-1}g$, Newton on $\phi(\lambda)=1/\|p(\lambda)\|-1/\Delta$ from $\lambda^{(0)}=0$ ($B\succ 0$ in both cases).

| $l$ | $\lambda^{(l)}$ | $p(\lambda^{(l)})$ | $\|p\|$ | $\phi(\lambda^{(l)})$ |
|---|---|---|---|---|
| | | 5.13: $g=(-2,-4)$, $B=\mathrm{diag}(4,2)$, $\Delta=1$ | | |
| 0 | 0 | (0.5, 2) | 2.0615528 | $-5.149\times10^{-1}$ |
| 1 | 2.1874422 | (0.323235, 0.955237) | 1.0084438 | $-8.373\times10^{-3}$ |
| 2 | 2.2240147 | (0.321336, 0.946966) | 1.0000011 | $-1.087\times10^{-6}$ |
| 3 | 2.2240195 | (0.321336, 0.946965) | 1.0000000 | $\approx 0$ |
| | | 5.9: $g=(-4,-6)$, $B=\left(\begin{smallmatrix}2&1\\1&4\end{smallmatrix}\right)$, $\Delta=1$ | | |
| 0 | 0 | (1.428571, 1.142857) | 1.8294641 | $-4.534\times10^{-1}$ |
| 1 | 2.5875673 | (0.696428, 0.805088) | 1.0645089 | $-6.060\times10^{-2}$ |
| 2 | 3.0107837 | (0.645867, 0.763700) | 1.0001907 | $-1.907\times10^{-4}$ |
| 3 | 3.0121233 | (0.645719, 0.763575) | 1.0000000 | $-1.553\times10^{-9}$ |

In 5.9: $g^{\mathsf T}p^*+\tfrac12 p^{*\mathsf T}Bp^*\approx -5.088225$, against $-5.057256$ with the Cauchy point and with Dogleg (V11), which coincide here because the minimizer along the ray $-g$ lies outside the region.

**5.10** NUMERICAL EXAMPLE *ch5, p. 11*

For the subproblem $\min\{g^{\mathsf T}p+\tfrac12 p^{\mathsf T}Bp:\ \|p\|_2\le\Delta\}$ with $g=(2,-4)^{\mathsf T}$, $B=\mathrm{diag}(-1,3)$ and $\Delta=2$, and $p(\lambda)=-(B+\lambda I)^{-1}g$:

a) Justify that one must look for $\lambda^*\ge 1$ (so that $B+\lambda^* I\succeq 0$) and that this is the easy case: the component of $g$ in the eigenspace of $\lambda_{\min}(B)=-1$ is nonzero.

b) Solve $\frac{4}{(\lambda-1)^2}+\frac{16}{(\lambda+3)^2}=4$ with Newton on $\phi(\lambda)=1/\|p(\lambda)\|-1/\Delta$ from $\lambda^{(0)}=2$ (table V12) and verify $(B+\lambda^* I)p^*=-g$, $\lambda^*(\|p^*\|-\Delta)=0$, $B+\lambda^* I\succeq 0$, $\lambda^*\ge 0$.

**Note.** The notes give $\lambda^*\approx 2.083$, $p^*\approx(-1.846,0.786)^{\mathsf T}$; that $p^*$ has norm $\approx 2.006$. The correct values are $\lambda^*\approx 2.087561$, $p^*\approx(-1.838978,0.786231)^{\mathsf T}$.

**V12** VALIDATION TABLE *A9 on 5.10*

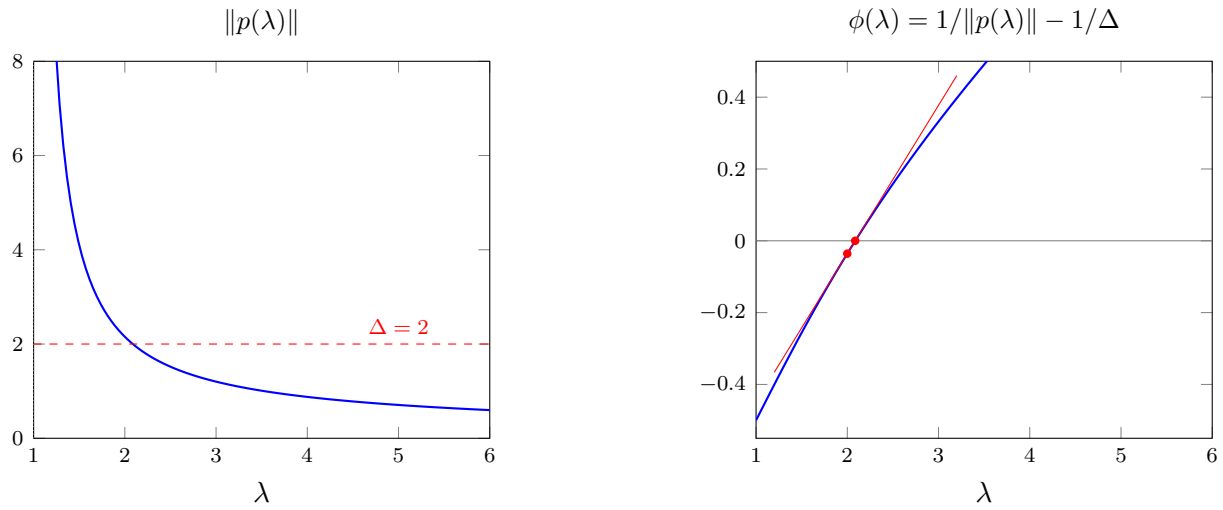
Subproblem $\min\{g^{\mathsf T}p+\tfrac12 p^{\mathsf T}Bp:\ \|p\|\le\Delta\}$ with $g=(2,-4)$, $B=\mathrm{diag}(-1,3)$, $\Delta=2$; Newton on $\phi(\lambda)=1/\|p(\lambda)\|-1/\Delta$ from $\lambda^{(0)}=2$. Easy case ($\gamma_1=2\neq 0$). Here $p(\lambda)=\left(-\frac{2}{\lambda-1},\ \frac{4}{\lambda+3}\right)$.

| $l$ | $\lambda^{(l)}$ | $p(\lambda^{(l)})$ | $\|p\|$ | $\phi(\lambda^{(l)})$ |
|---|---|---|---|---|
| 0 | 2 | (−2, 0.8) | 2.1540659 | $-3.576\times10^{-2}$ |
| 1 | 2.0865874 | (−1.840625, 0.786382) | 2.0015736 | $-3.931\times10^{-4}$ |
| 2 | 2.0875605 | (−1.838978, 0.786231) | 2.0000002 | $-4.850\times10^{-8}$ |
| 3 | 2.0875606 | (−1.838978, 0.786231) | 2.0000000 | $\approx 0$ |

$\lambda^*\approx 2.087561$, $B+\lambda^* I=\mathrm{diag}(1.087561,\ 5.087561)\succ 0$, $g^{\mathsf T}p^*+\tfrac12 p^{*\mathsf T}Bp^*\approx -7.586562$.

**What it shows.** Starting with $\|p(\lambda^{(0)})\|>\Delta$, Newton on $\phi$ increases monotonically and converges quadratically, even with indefinite $B$.

**Figure.**




$g = (2, -4)$, $B = \mathrm{diag}(-1, 3)$. Left: $\|p(\lambda)\|$ has a pole at $\lambda = -\lambda_{\min}(B) = 1$ and is strongly nonlinear near $\lambda^*$, so Newton on $\|p\| - \Delta$ would take poor steps. Right: $\phi$ is almost linear and concave; the tangent at $\lambda^{(0)} = 2$ (red) lands almost on the root, which explains the 3-step convergence in the table.

**5.23*** PROPOSED EXERCISE *ch5, p. 26 (exercise 6)*

With $g = (5, -12)^\mathsf{T}$, $B = \begin{pmatrix} 6 & 2 \\ 2 & 3 \end{pmatrix}$ and $\Delta = 1$, find the exact solution of the subproblem $\min\{g^\mathsf{T}p + \frac{1}{2}p^\mathsf{T}Bp : \|p\|_2 \le \Delta\}$ through the multiplier $\lambda^*$, with $(B + \lambda^* I)p^* = -g$, $\lambda^*(\|p^*\| - \Delta) = 0$, $B + \lambda^* I \succeq 0$, $\lambda^* \ge 0$.

**5.24*** PROPOSED EXERCISE *ch5, p. 26 (exercise 7)*

With $g = (1, 1)^\mathsf{T}$, $B = \mathrm{diag}(-2, 4)$ and $\Delta = 0.8$, find the exact solution of the subproblem $\min\{g^\mathsf{T}p + \frac{1}{2}p^\mathsf{T}Bp : \|p\|_2 \le \Delta\}$. Decide first whether it is the easy or the hard case (hard: $g$ orthogonal to the eigenspace of $\lambda_{\min}(B)$).

**5.3** PROOF *ch5, pp. 13–15 (easy case, hard case and Proposition 3.2)*

Let $m_k(p) = f(x_k) + g_k^\mathsf{T}p + \frac{1}{2}p^\mathsf{T}B_kp$ with $g_k = \nabla f(x_k)$ and $B_k$ symmetric, and the subproblem $\min\{m_k(p) : \|p\|_2 \le \Delta_k\}$. Let $B_k = \sum_i \lambda_i q_i q_i^\mathsf{T}$ with $\lambda_1 \le \cdots \le \lambda_n$, $\lambda_1 < 0$, $\gamma_i = q_i^\mathsf{T} g_k$, $E_1 = \{i : \lambda_i = \lambda_1\}$, $p(\lambda) = -(B_k + \lambda I)^{-1} g_k$ and $\psi(\lambda) = \|p(\lambda)\|^2 = \sum_i \gamma_i^2/(\lambda_i + \lambda)^2$. A solution is characterized (5.1) by $(B_k + \lambda^* I)p^* = -g_k$, $\lambda^*(\|p^*\| - \Delta_k) = 0$, $B_k + \lambda^* I \succeq 0$, $\lambda^* \ge 0$.

a) (Easy case) If $\gamma_i \neq 0$ for some $i \in E_1$, prove that there is a unique $\lambda^* \in (-\lambda_1, \infty)$ with $\|p(\lambda^*)\| = \Delta_k$, and that $p(\lambda^*)$ is the solution.

b) (Hard case) If $\gamma_i = 0$ for all $i \in E_1$ and $\psi(-\lambda_1) < \Delta_k^2$, prove that $p^* = p(-\lambda_1) + \tau z$, with $z \in \mathrm{Ker}(B_k - \lambda_1 I)$ of unit norm and $\tau = \sqrt{\Delta_k^2 - \psi(-\lambda_1)}$, satisfies those three conditions.

c) In the hard case with $\psi(-\lambda_1) \ge \Delta_k^2$, show that the solution is obtained as in the easy case.

**5.17** NUMERICAL EXAMPLE *ch5, p. 25 (solved exercise 5)*

For the subproblem $\min\{g^\mathsf{T}p + \frac{1}{2}p^\mathsf{T}Bp : \|p\|_2 \le \Delta\}$ with $g = (0, -3)^\mathsf{T}$, $B = \mathrm{diag}(-4, 2)$, $\Delta = 1$ and $p(\lambda) = -(B + \lambda I)^{-1} g$: show that the equation $\|p(\lambda)\| = 1$ with $\lambda > 4$ has no solution, identify the hard case, and obtain $p^* = (\pm\frac{\sqrt{3}}{2}, \frac{1}{2})^\mathsf{T}$ with $\lambda^* = 4$. Verify $(B + \lambda^* I)p^* = -g$, $\lambda^*(\|p^*\| - \Delta) = 0$, $B + \lambda^* I \succeq 0$, $\lambda^* \ge 0$ and that both give $g^\mathsf{T}p^* + \frac{1}{2}p^{*\mathsf{T}}Bp^* = -\frac{11}{4}$.

**P13** PRACTICE *added*

*Near the hard case.* For the subproblem $\min\{g^{\mathsf{T}}p + \frac{1}{2}p^{\mathsf{T}}Bp : \ \|p\|_2 \le \Delta\}$ with $B = \mathrm{diag}(-4, 2)$, $\Delta = 1$ and $g = (\varepsilon, -3)^{\mathsf{T}}$, $\varepsilon > 0$; $p(\lambda) = -(B + \lambda I)^{-1}g$. With $\varepsilon = 0$ the solutions are $(\pm\frac{\sqrt{3}}{2}, \frac{1}{2})^{\mathsf{T}}$ with $\lambda^* = 4$ (5.17).

a) Show that for $\varepsilon > 0$ it is the easy case and compute $\lambda^*(\varepsilon)$ and $p^*(\varepsilon)$ for $\varepsilon = 1,\ 0.1,\ 0.01,\ 0.001$ (table V14).

b) Verify that $p^*(\varepsilon) \to (-\frac{\sqrt{3}}{2}, \frac{1}{2})^{\mathsf{T}}$ and $\lambda^*(\varepsilon) \to 4^+$ as $\varepsilon \to 0^+$: the sign of $\varepsilon$ selects one of the two solutions of the case $\varepsilon = 0$.

c) Explain why Newton on $\phi(\lambda) = 1/\|p(\lambda)\| - 1/\Delta$ becomes difficult as $\varepsilon \to 0$ (pole of $\|p(\lambda)\|$ at $\lambda = 4$ with weight $\varepsilon^2$).

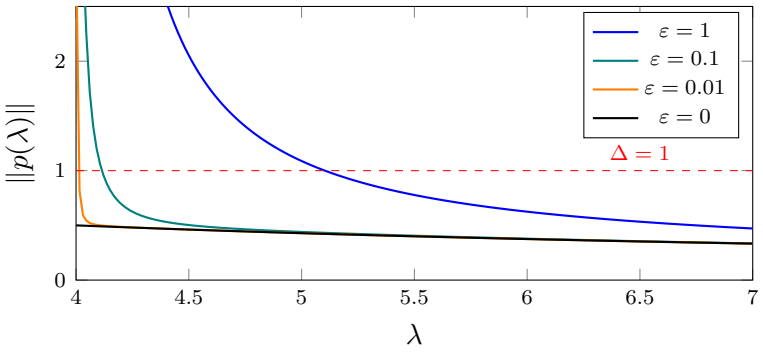
**V14** VALIDATION TABLE *near-hard case; solves P13*

Subproblem $\min\{g^{\mathsf{T}}p + \frac{1}{2}p^{\mathsf{T}}Bp : \ \|p\| \le \Delta\}$ with $B = \mathrm{diag}(-4, 2)$, $\Delta = 1$, $g = (\varepsilon, -3)$; for $\varepsilon > 0$ it is the easy case and $\lambda^* > 4$.

| $\varepsilon$ | $\lambda^*(\varepsilon)$ | $p^*(\varepsilon)$ | $g^{\mathsf{T}}p^* + \frac{1}{2}p^{*\mathsf{T}}Bp^*$ |
|---|---|---|---|
| 1 | 5.103222 | $(-0.906436,\ 0.422344)$ | $-3.638344$ |
| 0.1 | 4.114761 | $(-0.871376,\ 0.490616)$ | $-2.836873$ |
| 0.01 | 4.011540 | $(-0.866579,\ 0.499040)$ | $-2.758663$ |
| 0.001 | 4.001155 | $(-0.866081,\ 0.499904)$ | $-2.750866$ |
| 0 (5.17) | 4 | $(\pm 0.866025,\ 0.5)$ | $-2.75$ |

**What it shows.** As $\varepsilon \to 0$ the solution tends to one of the hard-case solutions and $\lambda^* \to -\lambda_1$, where the secular equation becomes ill-conditioned.

**Figure.**




$B = \mathrm{diag}(-4, 2)$, $g = (\varepsilon, -3)$, for $\lambda > -\lambda_{\min}(B) = 4$. The pole at $\lambda = 4$ has weight $\varepsilon^2$: as $\varepsilon$ decreases the curve drops almost vertically and the crossing with $\Delta = 1$ approaches $\lambda = 4$. With $\varepsilon = 0$ the pole disappears, $\|p(\lambda)\| \le \frac{1}{2} < \Delta$ and there is no crossing: this is the hard case, and the missing component is added along the eigenvector of $\lambda_{\min}$.

**5.20*** PROPOSED EXERCISE *ch5, p. 25 (exercise 3)*

Consider the trust-region subproblem in the $\ell_\infty$ norm: $\min g_k^{\mathsf{T}}p + \frac{1}{2}p^{\mathsf{T}}B_k p$ s.t. $\|p\|_\infty \le \Delta_k$. If $B_k = \mathrm{diag}(d_1, \ldots, d_n)$ with $d_i > 0$, find the exact analytic expression of the optimal step $p^*$.

# U16. Approximate methods: Cauchy point and Dogleg

↑Index

**5.11** Numerical example *ch5, p. 18*

With $g = (3,4)^\mathsf{T}$, $B = 5I$, $\Delta = 0.5$: The Cauchy point is $p^C = -\alpha^C g$, where $\alpha^C$ minimizes $g^\mathsf{T}(-\alpha g) + \frac{1}{2}\alpha^2 g^\mathsf{T} B g$ over $0 \le \alpha \le \Delta/\|g\|$; and $p^U = -\frac{g^\mathsf{T} g}{g^\mathsf{T} B g}\, g$ is the minimizer along the ray with no radius constraint (if $g^\mathsf{T} B g > 0$). Compute $\alpha^* = \|g\|^2/g^\mathsf{T} B g$, check that the unconstrained step leaves the region and obtain $p^C = (-0.3, -0.4)^\mathsf{T}$. Observe that here $p^C$ is also the exact solution of the subproblem $\min\{g^\mathsf{T} p + \frac{1}{2} p^\mathsf{T} B p : \ \|p\|_2 \le \Delta\}$ (why?).

**5.14** Numerical example *ch5, p. 23 (solved exercise 2)*

With $g = (3,-4)^\mathsf{T}$, $B = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $\Delta = 0.5$: The Cauchy point is $p^C = -\alpha^C g$, where $\alpha^C$ minimizes $g^\mathsf{T}(-\alpha g) + \frac{1}{2}\alpha^2 g^\mathsf{T} B g$ over $0 \le \alpha \le \Delta/\|g\|$; and $p^U = -\frac{g^\mathsf{T} g}{g^\mathsf{T} B g}\, g$ is the minimizer along the ray with no radius constraint (if $g^\mathsf{T} B g > 0$). Compute $g^\mathsf{T} B g = 26$, $\alpha^* = \frac{25}{26}$ and $p^C = (-0.3, 0.4)^\mathsf{T}$.

**5.4** Derivation *ch5, Definition 4.1 and eq. (12), p. 16*

Let $m_k(p) = f(x_k) + g_k^\mathsf{T} p + \frac{1}{2} p^\mathsf{T} B_k p$ with $g_k = \nabla f(x_k)$ and $B_k$ symmetric, and the subproblem $\min\{m_k(p) : \ \|p\|_2 \le \Delta_k\}$. The Cauchy point $p_k^C = -\alpha_k^C g_k$ minimizes $m_k$ over the segment $\{-\alpha g_k : \ 0 \le \alpha \le \Delta_k/\|g_k\|\}$. Derive its explicit formula

$$p_k^C = -\alpha_k^C g_k, \qquad \alpha_k^C = \begin{cases} \dfrac{\|g_k\|^2}{g_k^\mathsf{T} B_k g_k}, & g_k^\mathsf{T} B_k g_k > 0 \text{ and } \dfrac{\|g_k\|^3}{g_k^\mathsf{T} B_k g_k} \le \Delta_k, \\ \dfrac{\Delta_k}{\|g_k\|}, & \text{otherwise,} \end{cases}$$

by minimizing $m_k(-\alpha g_k)$ over $0 \le \alpha \le \Delta_k/\|g_k\|$.

**5.5** Proof *ch5, Proposition 4.1, p. 17*

Let $m_k(p) = f(x_k) + g_k^\mathsf{T} p + \frac{1}{2} p^\mathsf{T} B_k p$ with $g_k = \nabla f(x_k)$ and $B_k$ symmetric, and the subproblem $\min\{m_k(p) : \ \|p\|_2 \le \Delta_k\}$. Let $p_k^C$ be the Cauchy point (minimizer of $m_k$ over $\{-\alpha g_k : \ 0 \le \alpha \le \Delta_k/\|g_k\|\}$). Prove that $m_k(0) - m_k(p_k^C) \ge \frac{1}{2}\|g_k\| \min\big(\Delta_k, \|g_k\|/\|B_k\|\big)$.

**Note.** In subcase 2b the notes appeal to "the slope at the origin dominates". It must be made precise: if $\alpha^C \le \alpha^*$ then $\alpha^C g^\mathsf{T} B g \le \|g\|^2$, hence $m_k(0) - m_k(p^C) \ge \frac{1}{2}\alpha^C \|g\|^2 = \frac{1}{2}\Delta_k \|g\|$.

**5.21*** Proposed exercise *ch5, p. 26 (exercise 4)*

Let $f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$ (Rosenbrock), $\nabla f(x) = \big(-400x_1(x_2 - x_1^2) - 2(1 - x_1),\ 200(x_2 - x_1^2)\big)^\mathsf{T}$, $\nabla^2 f(x) = \begin{pmatrix} 1200x_1^2 - 400x_2 + 2 & -400x_1 \\ -400x_1 & 200 \end{pmatrix}$. At $x_0 = (-1.2, 1.0)^\mathsf{T}$, compute $g_0$, $B_0 = \nabla^2 f(x_0)$ and the Cauchy step $p_0^C$ for $\Delta_0 = 0.25$. The Cauchy point is $p^C = -\alpha^C g$, where $\alpha^C$ minimizes $g^\mathsf{T}(-\alpha g) + \frac{1}{2}\alpha^2 g^\mathsf{T} B g$ over $0 \le \alpha \le \Delta/\|g\|$; and $p^U = -\frac{g^\mathsf{T} g}{g^\mathsf{T} B g}\, g$ is the minimizer along the ray with no radius constraint (if $g^\mathsf{T} B g > 0$).

**5.12** Numerical example *ch5, p. 20*

With $g = (2,2)^\mathsf{T}$, $B = \mathrm{diag}(2, 8)$, $\Delta = 0.5$: With $B \succ 0$, $p^N = -B^{-1} g$ and $p^U = -\frac{g^\mathsf{T} g}{g^\mathsf{T} B g} g$, the Dogleg path is $\tilde{p}(\tau) = \tau p^U$ on $[0,1]$ and $\tilde{p}(\tau) = p^U + (\tau - 1)(p^N - p^U)$ on $[1,2]$; the Dogleg step is $\tilde{p}(\tau^*)$ with $\tau^* = \max\{\tau \in [0,2] : \ \|\tilde{p}(\tau)\| \le \Delta\}$. Compute $p^N = (-1, -0.25)^\mathsf{T}$ and $p^U = (-0.4, -0.4)^\mathsf{T}$, and conclude that the Dogleg step is on the first segment: $p^{DL} \approx (-0.3536, -0.3536)^\mathsf{T}$.

**5.15** NUMERICAL EXAMPLE *ch5, p. 24 (solved exercise 3)*

With $g = (1,2)^{\mathsf{T}}$, $B = \operatorname{diag}(3,1)$, $\Delta = 1.5$: With $B \succ 0$, $p^N = -B^{-1}g$ and $p^U = -\frac{g^{\mathsf{T}}g}{g^{\mathsf{T}}Bg}g$, the Dogleg path is $\tilde{p}(\tau) = \tau p^U$ on $[0,1]$ and $\tilde{p}(\tau) = p^U + (\tau-1)(p^N - p^U)$ on $[1,2]$; the Dogleg step is $\tilde{p}(\tau^*)$ with $\tau^* = \max\{\tau \in [0,2] : \ \|\tilde{p}(\tau)\| \le \Delta\}$. Compute $p^N = (-\frac{1}{3}, -2)^{\mathsf{T}}$, $p^U = -\frac{5}{7}(1,2)^{\mathsf{T}}$ with $\|p^U\| = \frac{5}{7}\sqrt{5} \approx 1.597$ and the Dogleg step $p^{DL} \approx (-0.6708, -1.3416)^{\mathsf{T}}$.

**5.6** PROOF *ch5, Lemma 4.1, p. 19*

Let $g \neq 0$. With $B \succ 0$, $p^N = -B^{-1}g$ and $p^U = -\frac{g^{\mathsf{T}}g}{g^{\mathsf{T}}Bg}g$, the Dogleg path is $\tilde{p}(\tau) = \tau p^U$ on $[0,1]$ and $\tilde{p}(\tau) = p^U + (\tau-1)(p^N - p^U)$ on $[1,2]$; the Dogleg step is $\tilde{p}(\tau^*)$ with $\tau^* = \max\{\tau \in [0,2] : \|\tilde{p}(\tau)\| \le \Delta\}$. Prove that $\|\tilde{p}(\tau)\|$ is strictly increasing and that $m(\tilde{p}(\tau))$, with $m(p) = g^{\mathsf{T}}p + \frac{1}{2}p^{\mathsf{T}}Bp$, is strictly decreasing on $[0,2]$.

**Note.** The notes state "one shows that $p_k^{C\,\mathsf{T}}(p_k^N - p_k^C) \ge 0$" without doing it, and part 2 only invokes convexity. Both need a computation: write $p^C = -\frac{g^{\mathsf{T}}g}{g^{\mathsf{T}}Bg}g$ and use Cauchy–Schwarz in the $B$ and $B^{-1}$ inner products.

**5.19*** PROPOSED EXERCISE *ch5, p. 25 (exercise 2)*

With $B \succ 0$, $p^N = -B^{-1}g$ and $p^U = -\frac{g^{\mathsf{T}}g}{g^{\mathsf{T}}Bg}g$, the Dogleg path is $\tilde{p}(\tau) = \tau p^U$ on $[0,1]$ and $\tilde{p}(\tau) = p^U + (\tau-1)(p^N - p^U)$ on $[1,2]$; the Dogleg step is $\tilde{p}(\tau^*)$ with $\tau^* = \max\{\tau \in [0,2] : \ \|\tilde{p}(\tau)\| \le \Delta\}$. Prove formally that $h(\tau) = \|\tilde{p}(\tau)\|^2$ is strictly increasing on $[1,2]$.

**P11** PRACTICE *added*

*Dogleg on the second segment.* With $g = (2,2)^{\mathsf{T}}$, $B = \operatorname{diag}(2,8)$ (the data of 5.12). With $B \succ 0$, $p^N = -B^{-1}g$ and $p^U = -\frac{g^{\mathsf{T}}g}{g^{\mathsf{T}}Bg}g$, the Dogleg path is $\tilde{p}(\tau) = \tau p^U$ on $[0,1]$ and $\tilde{p}(\tau) = p^U + (\tau-1)(p^N - p^U)$ on $[1,2]$; the Dogleg step is $\tilde{p}(\tau^*)$ with $\tau^* = \max\{\tau \in [0,2] : \ \|\tilde{p}(\tau)\| \le \Delta\}$. The Cauchy point is $p^C = -\alpha^C g$, where $\alpha^C$ minimizes $g^{\mathsf{T}}(-\alpha g) + \frac{1}{2}\alpha^2 g^{\mathsf{T}}Bg$ over $0 \le \alpha \le \Delta/\|g\|$; and $p^U = -\frac{g^{\mathsf{T}}g}{g^{\mathsf{T}}Bg}\,g$ is the minimizer along the ray with no radius constraint (if $g^{\mathsf{T}}Bg > 0$).

a) For $\Delta = 0.8$, verify $\|p^U\| < \Delta < \|p^N\|$, set up the quadratic $\|p^U + s(p^N - p^U)\|^2 = \Delta^2$ in $s = \tau - 1 \in [0,1]$ and find $\tau^*$ and $p^{DL}$.

b) For $\Delta = 1.2$, verify that $p^{DL} = p^N$.

c) For the three radii 0.5, 0.8, 1.2, compare the reduction $m(0) - m(p) = -g^{\mathsf{T}}p - \frac{1}{2}p^{\mathsf{T}}Bp$ of Dogleg with that of the Cauchy point and with the bound $m(0) - m(p) \ge \frac{1}{2}\|g\| \min\big(\Delta, \|g\|/\|B\|\big)$ (table V11).

**A8** PSEUDOCODE *Cauchy point and Dogleg (ch5, eq. 12 and Def. 4.2)*

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**CauchyPoint**$(g,\ B,\ \Delta)$

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1: $\gamma \leftarrow g^{\mathsf{T}}Bg$
2: **if** $\gamma > 0$ **and** $\|g\|^3 \le \Delta\,\gamma$ **then** $\alpha \leftarrow \|g\|^2/\gamma$ **else** $\alpha \leftarrow \Delta/\|g\|$
3: **return** $-\alpha g$

---

**Dogleg**$(g,\ B,\ \Delta)$ with $B \succ 0$

1: solve $Bp^N = -g$; **if** $\|p^N\| \le \Delta$ **then return** $p^N$ ($\tau^* = 2$)
2: $p^U \leftarrow -\dfrac{g^{\mathsf T} g}{g^{\mathsf T} B g}\, g$; **if** $\|p^U\| \ge \Delta$ **then return** $(\Delta/\|p^U\|)\, p^U$ (first segment)
3: $d \leftarrow p^N - p^U$; $s \leftarrow$ positive root of $\|d\|^2 s^2 + 2\,(p^U)^{\mathsf T} d\, s + \|p^U\|^2 - \Delta^2 = 0$
4: **return** $p^U + s\, d$ (second segment, $\tau^* = 1 + s$)

Here $m(p) = g^{\mathsf T} p + \frac{1}{2} p^{\mathsf T} B p$, $p^U$ is the minimizer of $m$ along the ray $-g$ with no radius constraint (the notation $p^C$ in Definition 4.2 of the notes), and the Cauchy point of line 2 of CauchyPoint is its version truncated to the ball $\|p\| \le \Delta$.

**V11** Validation table *A8; solves P11 and checks 5.11, 5.12, 5.14, 5.15, 5.9*

Model $m(p) = g^{\mathsf T} p + \frac{1}{2} p^{\mathsf T} B p$; $p^U = -\frac{g^{\mathsf T} g}{g^{\mathsf T} B g} g$, $p^N = -B^{-1} g$, $p^C$ the Cauchy point and $p^{DL}$ the Dogleg step. Data: 5.11: $g = (3, 4)$, $B = 5I$; 5.14: $g = (3, -4)$, $B = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$; 5.15: $g = (1, 2)$, $B = \mathrm{diag}(3, 1)$; 5.12 and P11: $g = (2, 2)$, $B = \mathrm{diag}(2, 8)$; 5.9: $g = (-4, -6)$, $B = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 4 \end{smallmatrix}\right)$. Model reduction $\mathrm{red}(p) = m(0) - m(p) = -g^{\mathsf T} p - \frac{1}{2} p^{\mathsf T} B p$ and Cauchy bound $\kappa = \frac{1}{2}\|g\| \min(\Delta, \|g\|/\|B\|_2)$.

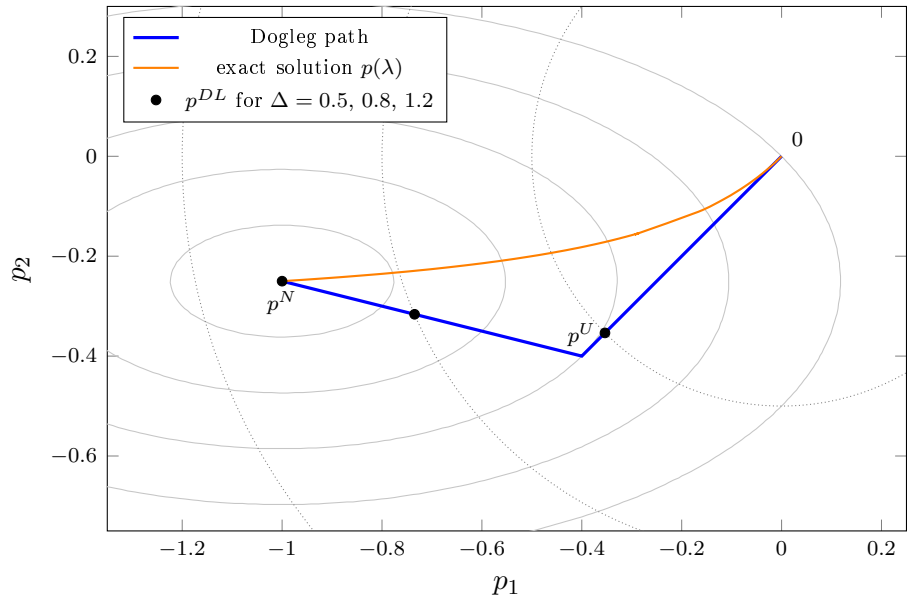
| case | $\Delta$ | $\|p^U\|$ | $\|p^N\|$ | $p^C$ | $\mathrm{red}(p^C)$ | $\kappa$ |
|---|---|---|---|---|---|---|
| 5.11 | 0.5 | 1 | 1 | (−0.3, −0.4) | 1.875 | 1.25 |
| 5.14 | 0.5 | 4.807692 | 4.955356 | (−0.3, 0.4) | 2.37 | 1.25 |
| 5.15 | 1.5 | 1.597191 | 2.027588 | (−0.670820, −1.341641) | 1.779102 | 0.833333 |
| 5.12 | 0.5 | 0.565685 | 1.030776 | (−0.353553, −0.353553) | 0.789214 | 0.5 |
| P11 a) | 0.8 | 0.565685 | 1.030776 | (−0.4, −0.4) | 0.8 | 0.5 |
| P11 b) | 1.2 | 0.565685 | 1.030776 | (−0.4, −0.4) | 0.8 | 0.5 |
| 5.9 | 1 | 1.674006 | 1.829464 | (0.554700, 0.832050) | 5.057256 | 3.605551 |

| case | $\Delta$ | segment | $p^{DL}$ | $\mathrm{red}(p^{DL})$ |
|---|---|---|---|---|
| 5.11 | 0.5 | 1 | (−0.3, −0.4) | 1.875 |
| 5.14 | 0.5 | 1 | (−0.3, 0.4) | 2.37 |
| 5.15 | 1.5 | 1 | (−0.670820, −1.341641) | 1.779102 |
| 5.12 | 0.5 | 1 | (−0.353553, −0.353553) | 0.789214 |
| P11 a) | 0.8 | 2, $s = 0.558030$, $\tau^* = 1.558030$ | (−0.734818, −0.316296) | 1.162098 |
| P11 b) | 1.2 | Newton, $\tau^* = 2$ | (−1, −0.25) | 1.25 |
| 5.9 | 1 | 1 | (0.554700, 0.832050) | 5.057256 |

Detail of P11 a): $p^U = (-0.4, -0.4)$, $d = p^N - p^U = (-0.6,\ 0.15)$, the equation is $0.3825 s^2 + 0.36 s - 0.32 = 0$. In 5.11, $\|p^U\| = \|p^N\| = 1$ because $B = 5I$: the gradient ray passes through $p^N$.

**What it shows.** In every case $\mathrm{red}(p^{DL}) \ge \mathrm{red}(p^C) \ge \kappa$, which is the inequality used by global convergence.

**Figure.**




$g = (2, 2)$, $B = \mathrm{diag}(2, 8)$. Level curves of the model (centred at $p^N$), dotted circles $\|p\| = \Delta$. The Dogleg path goes from 0 to $p^U$ along the ray $-g$ and then straight to $p^N$; the exact solution follows the curve $p(\lambda)$, tangent to $-g$ at the origin ($\lambda \to \infty$) and ending at $p^N$ ($\lambda = 0$). Dogleg approximates that curve with two segments; the norm increases and the model decreases along both (5.6).

### 5.7 Proof

*ch5, Theorem 4.1, p. 20*

Let $m(p) = g^\mathsf{T} p + \frac{1}{2} p^\mathsf{T} B p$ with $B \succ 0$. With $B \succ 0$, $p^N = -B^{-1} g$ and $p^U = -\frac{g^\mathsf{T} g}{g^\mathsf{T} B g} g$, the Dogleg path is $\tilde{p}(\tau) = \tau p^U$ on $[0, 1]$ and $\tilde{p}(\tau) = p^U + (\tau - 1)(p^N - p^U)$ on $[1, 2]$; the Dogleg step is $\tilde{p}(\tau^*)$ with $\tau^* = \max\{\tau \in [0, 2] : \ \|\tilde{p}(\tau)\| \le \Delta\}$. The Cauchy point is $p^C = -\alpha^C g$, where $\alpha^C$ minimizes $g^\mathsf{T}(-\alpha g) + \frac{1}{2}\alpha^2 g^\mathsf{T} B g$ over $0 \le \alpha \le \Delta / \|g\|$; and $p^U = -\frac{g^\mathsf{T} g}{g^\mathsf{T} B g}\, g$ is the minimizer along the ray with no radius constraint (if $g^\mathsf{T} B g > 0$). Prove that $m(0) - m(p^{DL}) \ge m(0) - m(p^C)$ and hence that the Dogleg step satisfies the bound $m(0) - m(p) \ge \frac{1}{2}\|g\| \min\big(\Delta, \|g\|/\|B\|\big)$ (5.5).

**Note.** The chain "$m_k(p^{DL}) \le m_k(p^C) \le m_k(p^C)$" in the notes has a typo: in the case $\|p^U\| < \Delta_k$ one must observe that the constrained Cauchy point coincides with $p^U$ (minimum along the ray inside the region) and then apply 5.6.

### 5.22* Proposed exercise

*ch5, p. 26 (exercise 5)*

Let $f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$ (Rosenbrock), $\nabla f(x) = \big(-400x_1(x_2 - x_1^2) - 2(1 - x_1),\ 200(x_2 - x_1^2)\big)^\mathsf{T}$, $\nabla^2 f(x) = \begin{pmatrix} 1200x_1^2 - 400x_2 + 2 & -400x_1 \\ -400x_1 & 200 \end{pmatrix}$. At $x_0 = (-1.2, 1.0)^\mathsf{T}$, check whether the Hessian is positive definite and build the full Dogleg step $p_0^{DL}$ for $\Delta_0 = 0.25$. With $B \succ 0$, $p^N = -B^{-1} g$ and $p^U = -\frac{g^\mathsf{T} g}{g^\mathsf{T} B g} g$, the Dogleg path is $\tilde{p}(\tau) = \tau p^U$ on $[0, 1]$ and $\tilde{p}(\tau) = p^U + (\tau - 1)(p^N - p^U)$ on $[1, 2]$; the Dogleg step is $\tilde{p}(\tau^*)$ with $\tau^* = \max\{\tau \in [0, 2] : \ \|\tilde{p}(\tau)\| \le \Delta\}$.

## U17. Global convergence of trust region

↑Index

**A10** Pseudocode *trust region (ch5, Def. 2.2–2.4)*

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**TrustRegion**$(f,\ \nabla f,\ \nabla^2 f,\ x_0;\ \Delta_0, \Delta_{\max}, \eta_1, \eta_2, \varepsilon, k_{\max})$

---

1: $\Delta \leftarrow \Delta_0$
2: **for** $k = 0, 1, \dots, k_{\max}$ **do**
3: $\quad g_k \leftarrow \nabla f(x_k)$; $\quad B_k \leftarrow \nabla^2 f(x_k)$; $\quad$ **if** $\|g_k\| < \varepsilon$ **then return** $x_k$
4: $\quad$ **if** $B_k \succ 0$ **then** $p_k \leftarrow \text{Dogleg}(g_k, B_k, \Delta)$ **else** $p_k \leftarrow \text{CauchyPoint}(g_k, B_k, \Delta)$
$\quad\quad$ (or $p_k \leftarrow \text{ExactSubproblem}(g_k, B_k, \Delta)$)
5: $\quad \text{pred} \leftarrow -g_k^{\mathsf T} p_k - \tfrac{1}{2} p_k^{\mathsf T} B_k p_k$; $\quad \text{ared} \leftarrow f(x_k) - f(x_k + p_k)$; $\quad \rho_k \leftarrow \text{ared}/\text{pred}$
6: $\quad$ **if** $\rho_k < \eta_1$ **then**
7: $\quad\quad x_{k+1} \leftarrow x_k$; $\quad \Delta \leftarrow \Delta/2$ (reject)
8: $\quad$ **else**
9: $\quad\quad x_{k+1} \leftarrow x_k + p_k$
10: $\quad\quad$ **if** $\rho_k > \eta_2$ **and** $\|p_k\| = \Delta$ **then** $\Delta \leftarrow \min(2\Delta, \Delta_{\max})$

---

Dogleg requires $B_k \succ 0$; using the Cauchy point when $B_k \not\succ 0$ is a practical choice (not in the notes) that keeps the bound $m_k(0) - m_k(p_k) \geq \tfrac{1}{2}\|g_k\| \min(\Delta_k, \|g_k\|/\|B_k\|)$ and hence the hypotheses of 5.8. The equality $\|p_k\| = \Delta$ is tested with a tolerance ($10^{-10}$).

**P14** Practice *added*

Let $f(x) = x_1^4 + x_1 x_2 + (1+x_2)^2$, with $\nabla f(x) = (4x_1^3 + x_2,\ x_1 + 2(1+x_2))^{\mathsf T}$ and $\nabla^2 f(x) = \begin{pmatrix} 12x_1^2 & 1 \\ 1 & 2 \end{pmatrix}$ (indefinite if $|x_1| < 1/\sqrt{24} \approx 0.204$). With $\rho_k = \dfrac{f(x_k) - f(x_k + p_k)}{m_k(0) - m_k(p_k)}$ and $0 < \eta_1 \leq \eta_2 < 1$, the rule in the notes is: if $\rho_k < \eta_1$ the step is rejected ($x_{k+1} = x_k$) and $\Delta_{k+1} = \tfrac{1}{2}\Delta_k$; if $\rho_k \geq \eta_1$ it is accepted ($x_{k+1} = x_k + p_k$), with $\Delta_{k+1} = \min(2\Delta_k, \Delta_{\max})$ if moreover $\rho_k > \eta_2$ and $\|p_k\| = \Delta_k$, and $\Delta_{k+1} = \Delta_k$ otherwise. Run by hand iterations $k = 0, \dots, 4$ of A10 from $x_0 = (0,0)^{\mathsf T}$, with $\Delta_0 = 1$, $\Delta_{\max} = 4$, $\eta_1 = 0.1$, $\eta_2 = 0.75$, using the Cauchy point if $B_k \not\succ 0$ and Dogleg if $B_k \succ 0$ (table V15). For each $k$: decide whether $B_k \succ 0$, compute $p_k$, the predicted and actual reductions, $\rho_k$, and the update rule applied. Point out the iteration in which the model predicts a reduction but $f$ increases.

**V15** Validation table *A10 from $(0,0)^{\mathsf T}$; solves P14*

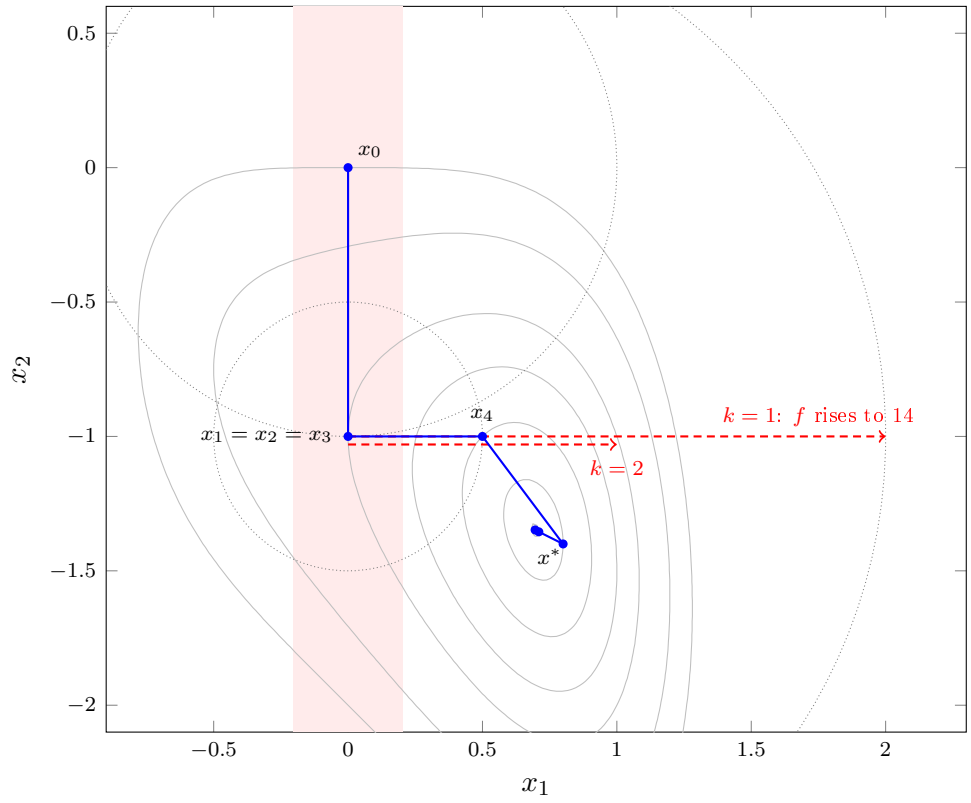
$f(x) = x_1^4 + x_1 x_2 + (1+x_2)^2$, $B_k = \nabla^2 f(x_k)$, $x^* \approx (0.695884, -1.347942)^{\mathsf T}$; $\Delta_0 = 1$, $\Delta_{\max} = 4$, $\eta_1 = 0.1$, $\eta_2 = 0.75$, $\varepsilon = 10^{-6}$. Column "step": C = Cauchy ($B_k \not\succ 0$), N = Newton inside the region (Dogleg with $\tau^* = 2$).

| $k$ | $x_k$ | $f(x_k)$ | $\|g_k\|$ | $\lambda_{\min}(B_k)$ | $\Delta$ |
|---|---|---|---|---|---|
| 0 | (0, 0) | 1 | 2 | −0.4142 | 1 |
| 1 | (0, −1) | 0 | 1 | −0.4142 | 2 |
| 2 | (0, −1) | 0 | 1 | −0.4142 | 1 |
| 3 | (0, −1) | 0 | 1 | −0.4142 | 0.5 |
| 4 | (0.5, −1) | −0.4375 | 0.707107 | 1.3820 | 1 |
| 5 | (0.8, −1.4) | −0.5504 | 0.648000 | 1.8291 | 1 |
| 6 | (0.709749, −1.354875) | −0.581927 | $7.5253 \times 10^{-2}$ | 1.7663 | 1 |
| 7 | (0.696178, −1.348089) | −0.582445 | $1.5587 \times 10^{-3}$ | 1.7538 | 1 |
| 8 | (0.695885, −1.347942) | −0.582445 | $7.18 \times 10^{-7}$ | | |

| $k$ | step | $p_k$ | pred | ared | $\rho_k$ | decision |
|---|---|---|---|---|---|---|
| 0 | C | $(0,\ -1)$ | 1 | 1 | 1.0000 | accept, $\Delta \leftarrow 2$ |
| 1 | C | $(2,\ 0)$ | 2 | $-14$ | $-7.0000$ | reject, $\Delta \leftarrow 1$ |
| 2 | C | $(1,\ 0)$ | 1 | 0 | 0.0000 | reject, $\Delta \leftarrow 0.5$ |
| 3 | C | $(0.5,\ 0)$ | 0.5 | 0.4375 | 0.8750 | accept, $\Delta \leftarrow 1$ |
| 4 | N | $(0.3,\ -0.4)$ | 0.175 | 0.1129 | 0.6451 | accept |
| 5 | N | $(-0.090251,\ 0.045125)$ | $2.9241 \times 10^{-2}$ | $3.1527 \times 10^{-2}$ | 1.0782 | accept |
| 6 | N | $(-0.013572,\ 0.006786)$ | $5.1065 \times 10^{-4}$ | $5.1772 \times 10^{-4}$ | 1.0138 | accept |
| 7 | N | $(-0.000293,\ 0.000147)$ | $2.2852 \times 10^{-7}$ | $2.2859 \times 10^{-7}$ | 1.0003 | accept |

Detail by hand. $k = 0$: $g_0 = (0, 2)$, $B_0 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, $g_0^{\mathsf{T}} B_0 g_0 = 8$, $\|g_0\|^3 = 8 \le \Delta \cdot 8$, so $\alpha = \frac{4}{8} = \frac{1}{2}$ and $p_0 = (0, -1)$, which lands exactly on the boundary. $k = 1$: $g_1 = (-1, 0)$, $g_1^{\mathsf{T}} B_1 g_1 = 0$, so $\alpha = \Delta / \|g_1\| = 2$, $p_1 = (2, 0)$ and $f(2, -1) = 14$: the model predicts a decrease of 2 and $f$ increases by 14. $k = 2$: $f(1, -1) = 0$, $\rho = 0$. $k = 3$: $f(0.5, -1) = -0.4375$. $k = 4$: $g_4 = (-\frac{1}{2}, \frac{1}{2})$, $B_4 = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 2 \end{smallmatrix}\right) \succ 0$, $p^N = (0.3, -0.4)$ with $\|p^N\| = 0.5 < \Delta$; it is accepted without enlarging ($\rho_4 < \eta_2$). From $k = 5$ the steps are Newton steps and convergence is quadratic ($\rho_k \to 1$).

**Figure.**




Level curves of $f = x_1^4 + x_1 x_2 + (1 + x_2)^2$; the red band $|x_1| < 1/\sqrt{24}$ is where $\nabla^2 f \not\succ 0$ and A10 uses the Cauchy point. Dotted circles: trust regions of radius 1 (at $x_0$) and 2, 1, 0.5 (at $x_1$). Red arrows: rejected steps. The Cauchy step at $k = 1$ follows $-g$ to the edge of a region that is too large, where the $x_1^4$ term the model does not see dominates; after shrinking $\Delta$ the model becomes reliable again, and from $x_4$ (positive definite Hessian) the steps are Newton steps.

**5.8** Proof *ch5, Theorem 5.1, pp. 21–22*

Let $m_k(p) = f(x_k) + g_k^{\mathsf{T}} p + \frac{1}{2} p^{\mathsf{T}} B_k p$ with $g_k = \nabla f(x_k)$ and $B_k$ symmetric, and the subproblem $\min\{m_k(p) : \|p\|_2 \le \Delta_k\}$. Let $f : \mathbb{R}^n \to \mathbb{R}$ be of class $C^2$, bounded below on $S = \{x : f(x) \le f(x_0)\}$, with $\|B_k\| \le M$ for all $k$, iterates generated with the acceptance and radius rule of the notes, and steps satisfying the Cauchy bound $m_k(0) - m_k(p_k) \ge \frac{1}{2}\|g_k\| \min(\Delta_k, \|g_k\|/\|B_k\|)$. With $\rho_k = \dfrac{f(x_k) - f(x_k + p_k)}{m_k(0) - m_k(p_k)}$ and $0 < \eta_1 \le \eta_2 < 1$, the rule in the notes is: if $\rho_k < \eta_1$ the step is rejected ($x_{k+1} = x_k$) and $\Delta_{k+1} = \frac{1}{2}\Delta_k$; if $\rho_k \ge \eta_1$ it is accepted ($x_{k+1} = x_k + p_k$), with $\Delta_{k+1} = \min(2\Delta_k, \Delta_{\max})$ if moreover $\rho_k > \eta_2$ and $\|p_k\| = \Delta_k$, and $\Delta_{k+1} = \Delta_k$ otherwise. Prove that $\lim_{k\to\infty} \|\nabla f(x_k)\| = 0$.

**Note.** (1) Step 1 uses a bound on $\nabla^2 f$ on a compact set, but $S$ is not assumed bounded; one must assume $\nabla^2 f$ bounded on a neighbourhood of $S$ (or $\nabla f$ Lipschitz). (2) The proof uses $\eta_1 < \frac{1}{4}$, which is not in the statement. (3) Assuming $\|g_k\| \ge \epsilon$ for infinitely many indices and concluding $\Delta_k \ge \bar{\Delta}$ for all $k$ is not valid; the argument of the notes proves $\liminf_k \|g_k\| = 0$ (N&W, Theorem 4.5). For lim the additional argument of N&W, Theorem 4.6, with $\nabla f$ Lipschitz and $\eta_1 > 0$, is needed.

**5.26*** Proposed exercise *ch5, p. 26 (exercise 9)*

Let $m_k(p) = f(x_k) + g_k^{\mathsf{T}} p + \frac{1}{2} p^{\mathsf{T}} B_k p$ with $g_k = \nabla f(x_k)$ and $B_k$ symmetric, and the subproblem $\min\{m_k(p) : \|p\|_2 \le \Delta_k\}$. Let $p_k^C$ be the Cauchy point. Prove that the conclusion $\lim_k \|\nabla f(x_k)\| = 0$ of 5.8 still holds if the step only satisfies $m_k(0) - m_k(p_k) \ge c_1\big(m_k(0) - m_k(p_k^C)\big)$ for a fixed $c_1 \in (0, 1]$.

**5.27*** Proposed exercise *ch5, p. 26 (exercise 10)*

Let $m_k(p) = f(x_k) + g_k^{\mathsf{T}} p + \frac{1}{2} p^{\mathsf{T}} B_k p$ with $g_k = \nabla f(x_k)$ and $B_k$ symmetric, and the subproblem $\min\{m_k(p) : \|p\|_2 \le \Delta_k\}$. Explain how truncated conjugate gradient (Steihaug–CG) solves the subproblem approximately at large scale without computing $B_k^{-1}$, and under which condition the algorithm stops because of negative curvature.

**Note.** ch5 says Steihaug–CG is covered in the next chapter, but ch6 only develops linear CG (U18–U19), on which Steihaug–CG is built; Steihaug–CG itself is not in ch6.

## U18. Conjugate directions and linear conjugate gradient

↑Index

**6.9** Numerical example *ch6, p. 10 (solved exercise 3)*

Nonzero vectors $\{p_0, \dots, p_k\}$ are $A$-conjugate if $p_i^{\mathsf{T}} A p_j = 0$ for all $i \ne j$. Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 1.5 \end{pmatrix}$, $p_0 = (1, 1)^{\mathsf{T}}$ and $p_1 = (1, -2)^{\mathsf{T}}$.

a) Decide whether $p_0$ and $p_1$ are $A$-conjugate.

b) Replace $p_1$ by $\tilde{p}_1 = p_1 - \dfrac{p_1^{\mathsf{T}} A p_0}{p_0^{\mathsf{T}} A p_0}\, p_0$ (one Gram–Schmidt step in the inner product $\langle u, v\rangle_A = u^{\mathsf{T}} A v$) and verify that $\tilde{p}_1 = (\frac{15}{11}, -\frac{18}{11})^{\mathsf{T}}$ is $A$-conjugate to $p_0$.

**P15** Practice *added*

*Why conjugacy matters.* Let $\varphi(x) = \frac{1}{2} x^{\mathsf{T}} A x - b^{\mathsf{T}} x$ with $A = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}$, $b = (1, 2)^{\mathsf{T}}$, $x^* = A^{-1} b = (\frac{1}{11}, \frac{7}{11})^{\mathsf{T}}$ and $\varphi(x^*) = -\frac{15}{22}$. In every case below, $x_{k+1} = x_k + \alpha_k p_k$ with the exact step $\alpha_k = -\dfrac{(A x_k - b)^{\mathsf{T}} p_k}{p_k^{\mathsf{T}} A p_k}$, from $x_0 = (0, 0)^{\mathsf{T}}$.

a) Cyclic coordinate directions $p_0 = e_1$, $p_1 = e_2$, $p_2 = e_1$, $p_3 = e_2$: compute $x_1, \dots, x_4$ and the gap $\varphi(x_k) - \varphi(x^*)$. Show that $x^*$ is not reached in finitely many steps.

b) Directions $p_0 = e_1$ and $p_1 = e_2 - \dfrac{e_2^{\mathsf{T}} A e_1}{e_1^{\mathsf{T}} A e_1} e_1 = (-\frac{1}{4}, 1)^{\mathsf{T}}$: verify $p_0^{\mathsf{T}} A p_1 = 0$ and that $x_2 = x^*$.

c) Steepest descent, $p_k = -(Ax_k - b)$: compute $x_1, \dots, x_4$ and the gap, and verify $p_{k+1} \perp p_k$.

d) Explain the difference between a) and b) in terms of the expansion $x^* - x_0 = \sum_i \sigma_i p_i$.

**V16** Validation table *solves P15*

$A = \left(\begin{smallmatrix} 4 & 1 \\ 1 & 3 \end{smallmatrix}\right)$, $b = (1, 2)$, $x_0 = 0$, exact step along each direction; gap $= \varphi(x_k) - \varphi(x^*)$ with $\varphi(x^*) = -\frac{15}{22}$.

| $k$ | $x_k$ | $p_k$ | $\alpha_k$ | gap |
|---|---|---|---|---|
| | a) cyclic coordinate directions | | | |
| 0 | $(0,\ 0)$ | $e_1$ | $\frac{1}{4}$ | $\frac{15}{22}$ |
| 1 | $(\frac{1}{4},\ 0)$ | $e_2$ | $\frac{7}{12}$ | $\frac{49}{88}$ |
| 2 | $(\frac{1}{4},\ \frac{7}{12})$ | $e_1$ | $-\frac{7}{48}$ | $\frac{49}{1056}$ |
| 3 | $(\frac{5}{48},\ \frac{7}{12})$ | $e_2$ | $\frac{7}{144}$ | $\frac{49}{12672}$ |
| 4 | $(\frac{5}{48},\ \frac{91}{144})$ | | | $\frac{49}{152064}$ |
| | b) $A$-conjugate directions | | | |
| 0 | $(0,\ 0)$ | $e_1$ | $\frac{1}{4}$ | $\frac{15}{22}$ |
| 1 | $(\frac{1}{4},\ 0)$ | $(-\frac{1}{4},\ 1)$ | $\frac{7}{11}$ | $\frac{49}{88}$ |
| 2 | $(\frac{1}{11},\ \frac{7}{11})$ | | | 0 |
| | c) steepest descent | | | |
| 0 | $(0,\ 0)$ | $(1,\ 2)$ | $\frac{1}{4}$ | $\frac{15}{22}$ |
| 1 | $(\frac{1}{4},\ \frac{1}{2})$ | $(-\frac{1}{2},\ \frac{1}{4})$ | $\frac{1}{3}$ | $\frac{5}{88}$ |
| 2 | $(\frac{1}{12},\ \frac{7}{12})$ | $(\frac{1}{12},\ \frac{1}{6})$ | $\frac{1}{4}$ | $\frac{5}{1056}$ |
| 3 | $(\frac{5}{48},\ \frac{5}{8})$ | $(-\frac{1}{24},\ \frac{1}{48})$ | $\frac{1}{3}$ | $\frac{5}{12672}$ |
| 4 | $(\frac{13}{144},\ \frac{91}{144})$ | | | $\frac{5}{152064}$ |

**What it shows.** An exact line search along directions that are not conjugate (coordinates, gradients) only shrinks the gap by a constant factor ($\frac{1}{12}$ here) per step; along $A$-conjugate directions the same exact step reaches $x^*$ in $n = 2$ steps, because each step removes one term of $x^* - x_0 = \sum_i \sigma_i p_i$ for good.

**6.1** Proof *ch6, Theorem 2.1, pp. 2–3*

Let $\varphi(x) = \frac{1}{2} x^{\mathsf{T}} A x - b^{\mathsf{T}} x$ with $A \in \mathbb{R}^{n \times n}$ symmetric positive definite, $x^* = A^{-1} b$, and residual $r_k = Ax_k - b = \nabla \varphi(x_k)$. Let $p_0, \dots, p_{n-1}$ be nonzero and $A$-conjugate ($p_i^{\mathsf{T}} A p_j = 0$ for $i \neq j$), $x_0 \in \mathbb{R}^n$ arbitrary, and $x_{k+1} = x_k + \alpha_k p_k$ with $\alpha_k = -\dfrac{r_k^{\mathsf{T}} p_k}{p_k^{\mathsf{T}} A p_k}$.

a) Prove that $p_0, \dots, p_{n-1}$ are linearly independent.

b) Prove that $\alpha_k$ is the exact minimizer of $\alpha \mapsto \varphi(x_k + \alpha p_k)$.

c) Prove that $r_k^{\mathsf{T}} p_i = 0$ for all $i < k$, and that $x_n = x^*$.

**Note.** The notes use $\{p_i\}$ as a basis without proving linear independence (part a), which is what makes the expansion $x^* - x_0 = \sum_i \sigma_i p_i$ possible.

**6.15*** PROPOSED EXERCISE *ch6, p. 12 (exercise 2)*

Let $\varphi(x) = \frac{1}{2}x^\mathsf{T}Ax - b^\mathsf{T}x$ with $A \in \mathbb{R}^{n\times n}$ symmetric positive definite, $x^* = A^{-1}b$, and residual $r_k = Ax_k - b = \nabla\varphi(x_k)$. The linear CG method is: $r_0 = Ax_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^\mathsf{T}r_k}{p_k^\mathsf{T}Ap_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k Ap_k$, $\beta_{k+1} = \dfrac{r_{k+1}^\mathsf{T}r_{k+1}}{r_k^\mathsf{T}r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1}p_k$.

Prove that the step $\alpha_k$ minimizing $\varphi(x_k + \alpha p_k)$ is exactly $\alpha_k = \dfrac{r_k^\mathsf{T}r_k}{p_k^\mathsf{T}Ap_k}$. You may use $r_k^\mathsf{T}p_{k-1} = 0$ (6.2).

**6.2** PROOF *ch6, Lemma 2.1, p. 3*

Let $\varphi(x) = \frac{1}{2}x^\mathsf{T}Ax - b^\mathsf{T}x$ with $A \in \mathbb{R}^{n\times n}$ symmetric positive definite, $x^* = A^{-1}b$, and residual $r_k = Ax_k - b = \nabla\varphi(x_k)$. Let $x_k \in \mathbb{R}^n$, $p_k \neq 0$, and $\alpha_k$ the minimizer of $\alpha \mapsto \varphi(x_k + \alpha p_k)$, $x_{k+1} = x_k + \alpha_k p_k$. Prove that $\alpha_k = -\dfrac{r_k^\mathsf{T}p_k}{p_k^\mathsf{T}Ap_k}$, that $r_{k+1} = r_k + \alpha_k Ap_k$ and that $r_{k+1}^\mathsf{T}p_k = 0$.

**6.4** PROOF *ch6, Lemma 2.3, pp. 5–6*

Let $\varphi(x) = \frac{1}{2}x^\mathsf{T}Ax - b^\mathsf{T}x$ with $A \in \mathbb{R}^{n\times n}$ symmetric positive definite, $x^* = A^{-1}b$, and residual $r_k = Ax_k - b = \nabla\varphi(x_k)$. The linear CG method is: $r_0 = Ax_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^\mathsf{T}r_k}{p_k^\mathsf{T}Ap_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k Ap_k$, $\beta_{k+1} = \dfrac{r_{k+1}^\mathsf{T}r_{k+1}}{r_k^\mathsf{T}r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1}p_k$.
Assume $r_0, r_1 \neq 0$. Prove that $p_0^\mathsf{T}Ap_1 = 0$.

**Note.** The notes write $r_0^\mathsf{T}Ap_0 = \frac{r_0^\mathsf{T}r_0}{\alpha_0}$; the correct value is $r_0^\mathsf{T}Ap_0 = \frac{r_0^\mathsf{T}r_1 - r_0^\mathsf{T}r_0}{\alpha_0} = -\frac{r_0^\mathsf{T}r_0}{\alpha_0}$. With the right sign, $p_0^\mathsf{T}Ap_0 = -r_0^\mathsf{T}Ap_0 = \frac{\|r_0\|^2}{\alpha_0}$ and $p_0^\mathsf{T}Ap_1 = -\frac{\|r_1\|^2}{\alpha_0} + \beta_1\frac{\|r_0\|^2}{\alpha_0} = 0$.

**6.3** PROOF *ch6, Lemma 2.2 and Theorem 2.2, pp. 3–6*

Let $\varphi(x) = \frac{1}{2}x^\mathsf{T}Ax - b^\mathsf{T}x$ with $A \in \mathbb{R}^{n\times n}$ symmetric positive definite, $x^* = A^{-1}b$, and residual $r_k = Ax_k - b = \nabla\varphi(x_k)$. The linear CG method is: $r_0 = Ax_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^\mathsf{T}r_k}{p_k^\mathsf{T}Ap_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k Ap_k$, $\beta_{k+1} = \dfrac{r_{k+1}^\mathsf{T}r_{k+1}}{r_k^\mathsf{T}r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1}p_k$.
Assume $r_0, \ldots, r_{k-1} \neq 0$. Prove by a single induction on $k \geq 1$ that

(i) $r_k^\mathsf{T}p_i = 0$ for $i < k$;

(ii) $r_k^\mathsf{T}r_i = 0$ for $i < k$;

(iii) $p_k^\mathsf{T}Ap_i = 0$ for $i < k$;

(iv) $\operatorname{span}\{p_0, \ldots, p_{k-1}\} = \operatorname{span}\{r_0, \ldots, r_{k-1}\}$.

Derive along the way the formulas $\alpha_{k-1} = \dfrac{\|r_{k-1}\|^2}{p_{k-1}^\mathsf{T}Ap_{k-1}}$ and $\beta_k = \dfrac{\|r_k\|^2}{\|r_{k-1}\|^2}$ from the conditions $r_k^\mathsf{T}p_{k-1} = 0$ and $p_k^\mathsf{T}Ap_{k-1} = 0$.

**Note.** In the notes, Lemma 2.2 uses the $A$-conjugacy of the previous directions as an induction hypothesis, but conjugacy is proved afterwards in Theorem 2.2 using Lemma 2.2. The two inductions must be merged into one, as above, to avoid circularity. The divisions by $\alpha_i$ need $\alpha_i \neq 0$, which holds because $r_i \neq 0$; and the formula for $\alpha_{k-1}$ uses $p_{k-1}^\mathsf{T}r_{k-1} = -\|r_{k-1}\|^2$, which itself follows from (i).

**6.14*** PROPOSED EXERCISE *ch6, p. 11 (exercise 1)*

Let $\varphi(x) = \frac{1}{2}x^\mathsf{T}Ax - b^\mathsf{T}x$ with $A \in \mathbb{R}^{n\times n}$ symmetric positive definite, $x^* = A^{-1}b$, and residual $r_k = Ax_k - b = \nabla\varphi(x_k)$. The linear CG method is: $r_0 = Ax_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^\mathsf{T}r_k}{p_k^\mathsf{T}Ap_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k Ap_k$, $\beta_{k+1} = \dfrac{r_{k+1}^\mathsf{T}r_{k+1}}{r_k^\mathsf{T}r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1}p_k$.

Prove formally by induction that the residuals satisfy $r_i^{\mathsf{T}} r_j = 0$ for all $i \neq j$ (as long as they are nonzero).

**6.16*** Proposed exercise *ch6, p. 12 (exercise 3)*

Let $\varphi(x) = \frac{1}{2} x^{\mathsf{T}} A x - b^{\mathsf{T}} x$ with $A \in \mathbb{R}^{n \times n}$ symmetric positive definite, $x^* = A^{-1} b$, and residual $r_k = A x_k - b = \nabla \varphi(x_k)$. The linear CG method is: $r_0 = A x_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^{\mathsf{T}} r_k}{p_k^{\mathsf{T}} A p_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k A p_k$, $\beta_{k+1} = \dfrac{r_{k+1}^{\mathsf{T}} r_{k+1}}{r_k^{\mathsf{T}} r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1} p_k$. Prove that linear CG reaches the exact solution $x^*$ of $Ax = b$ in at most $n$ iterations (exact arithmetic).

**6.8** Numerical example *ch6, p. 9 (solved exercise 1)*

The linear CG method is: $r_0 = A x_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^{\mathsf{T}} r_k}{p_k^{\mathsf{T}} A p_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k A p_k$, $\beta_{k+1} = \dfrac{r_{k+1}^{\mathsf{T}} r_{k+1}}{r_k^{\mathsf{T}} r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1} p_k$. With $A = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}$, $b = (1, 2)^{\mathsf{T}}$ and $x_0 = (0, 0)^{\mathsf{T}}$, compute $r_0$, $p_0$, $\alpha_0$, $x_1$ and $r_1$, and verify $r_1^{\mathsf{T}} r_0 = 0$.

**P16** Practice *added*

*Linear CG by hand.* The linear CG method is: $r_0 = A x_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^{\mathsf{T}} r_k}{p_k^{\mathsf{T}} A p_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k A p_k$, $\beta_{k+1} = \dfrac{r_{k+1}^{\mathsf{T}} r_{k+1}}{r_k^{\mathsf{T}} r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1} p_k$. Same data as 6.8: $A = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}$, $b = (1, 2)^{\mathsf{T}}$, $x_0 = (0, 0)^{\mathsf{T}}$.

a) Complete the second iteration: $\beta_1$, $p_1$, $A p_1$, $\alpha_1$, $x_2$, $r_2$.

b) Verify $r_1^{\mathsf{T}} p_0 = 0$, $p_1^{\mathsf{T}} A p_0 = 0$, $r_2 = 0$ and $x_2 = A^{-1} b$.

c) Steepest descent with exact step from the same $x_0$ takes the same first step. Explain which ingredient of CG makes the second step reach $x^*$ (compare with V16 c).

**A11** Pseudocode *linear conjugate gradient (ch6, p. 7)*

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**LinearCG**$(A,\ b,\ x_0;\ \varepsilon, k_{\max})$ $\quad A$ symmetric positive definite

---

1: $x \leftarrow x_0$; $\quad r \leftarrow A x_0 - b$; $\quad p \leftarrow -r$; $\quad \delta \leftarrow r^{\mathsf{T}} r$
2: **for** $k = 0, 1, \ldots, k_{\max}$ **do**
3: $\quad$ **if** $\sqrt{\delta} < \varepsilon$ **then return** $x$
4: $\quad w \leftarrow A p$ $\qquad$ (the only matrix–vector product of the iteration)
5: $\quad \alpha \leftarrow \delta / (p^{\mathsf{T}} w)$
6: $\quad x \leftarrow x + \alpha p$; $\quad r \leftarrow r + \alpha w$
7: $\quad \delta_{\text{new}} \leftarrow r^{\mathsf{T}} r$; $\quad \beta \leftarrow \delta_{\text{new}} / \delta$; $\quad \delta \leftarrow \delta_{\text{new}}$
8: $\quad p \leftarrow -r + \beta p$

---

Storage: four vectors ($x$, $r$, $p$, $w$); $A$ is only needed through products $Ap$. In floating point the recursively updated $r$ drifts away from $Ax - b$; recomputing $r \leftarrow Ax - b$ every so often is a common safeguard. Exact arithmetic: at most $n$ iterations (6.16*).

**V17** VALIDATION TABLE *A11; solves P16 and 6.8*

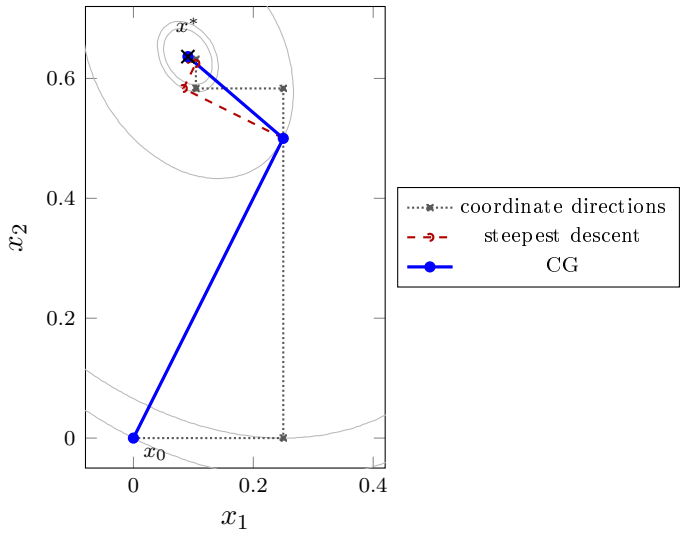
$A = \left(\begin{smallmatrix} 4 & 1 \\ 1 & 3 \end{smallmatrix}\right)$, $b = (1, 2)$, $x_0 = (0, 0)$, $x^* = (\frac{1}{11}, \frac{7}{11})$.

| $k$ | $x_k$ | $r_k$ | $p_k$ | $Ap_k$ | $\alpha_k$ | $\beta_{k+1}$ | $\varphi(x_k)$ |
|---|---|---|---|---|---|---|---|
| 0 | $(0,\ 0)$ | $(-1,\ -2)$ | $(1,\ 2)$ | $(6,\ 7)$ | $\frac{1}{4}$ | $\frac{1}{16}$ | $0$ |
| 1 | $(\frac{1}{4},\ \frac{1}{2})$ | $(\frac{1}{2},\ -\frac{1}{4})$ | $(-\frac{7}{16},\ \frac{3}{8})$ | $(-\frac{11}{8},\ \frac{11}{16})$ | $\frac{4}{11}$ | $0$ | $-\frac{5}{8}$ |
| 2 | $(\frac{1}{11},\ \frac{7}{11})$ | $(0,\ 0)$ | | | | | $-\frac{15}{22}$ |

Checks: $r_0^\mathsf{T} r_0 = 5$, $p_0^\mathsf{T} A p_0 = 20$, $r_1^\mathsf{T} r_1 = \frac{5}{16}$, $p_1^\mathsf{T} A p_1 = \frac{55}{64}$; $r_1^\mathsf{T} r_0 = 0$, $r_1^\mathsf{T} p_0 = 0$, $p_1^\mathsf{T} A p_0 = 0$.

**What it shows.** CG builds the conjugate direction $p_1 = -r_1 + \beta_1 p_0$ from $r_1$ and $p_0$ alone ($\beta_1 = \frac{1}{16}$), without a Gram–Schmidt sweep over all previous directions. The first step coincides with steepest descent; the correction $\beta_1 p_0$ is what turns the second step into one that reaches $x^*$.

**Figure.**




Level curves of $\varphi(x) = \frac{1}{2} x^\mathsf{T} A x - b^\mathsf{T} x$, $A = \left(\begin{smallmatrix} 4 & 1 \\ 1 & 3 \end{smallmatrix}\right)$, $b = (1, 2)$, through the iterates. All three methods use the exact step along their direction. Steepest descent and CG take the same first step; CG then moves along $p_1$, which is $A$-conjugate to $p_0$ and points at the centre of the ellipse, so it stops at $x^*$. Steepest descent zigzags with $g_{k+1} \perp g_k$, and cyclic coordinate directions (not conjugate since $A$ is not diagonal) also never reach $x^*$ in finitely many steps.

## U19. Krylov subspaces and rate of convergence of CG

↑Index

**P17** PRACTICE *added*

*Few distinct eigenvalues.* The linear CG method is: $r_0 = Ax_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^\mathsf{T} r_k}{p_k^\mathsf{T} A p_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k A p_k$, $\beta_{k+1} = \dfrac{r_{k+1}^\mathsf{T} r_{k+1}}{r_k^\mathsf{T} r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1} p_k$.
Let $A = \operatorname{diag}(1, 1, 10, 10)$, $b = (1, 1, 1, 1)^\mathsf{T}$, $x_0 = 0$, so $x^* = (1, 1, \frac{1}{10}, \frac{1}{10})^\mathsf{T}$, and $\|v\|_A = \sqrt{v^\mathsf{T} A v}$.

a) Run CG by hand and show that $r_2 = 0$, although $n = 4$.

b) Run steepest descent with exact step $\alpha_k = \frac{g_k^\mathsf{T} g_k}{g_k^\mathsf{T} A g_k}$, $g_k = Ax_k - b$, and show that $\|x_{k+1} - x^*\|_A / \|x_k - x^*\|_A = \frac{9}{11} = \frac{\kappa - 1}{\kappa + 1}$ at every step ($\kappa = 10$).

c) Explain a) through the dimension of $\mathcal{K}_k(A, r_0) = \operatorname{span}\{r_0, Ar_0, \ldots, A^{k-1} r_0\}$.

**V18** VALIDATION TABLE *solves P17*

$A = \mathrm{diag}(1,1,10,10)$, $b = (1,1,1,1)$, $x_0 = 0$; $\|e_k\|_A^2 = (x_k - x^*)^{\mathsf{T}} A (x_k - x^*)$.

| CG | | | | | | |
|---|---|---|---|---|---|---|
| $k$ | $x_k$ | $r_k$ | $p_k$ | $\alpha_k$ | $\beta_{k+1}$ | $\|e_k\|_A^2$ |
| 0 | $(0,0,0,0)$ | $(-1,-1,-1,-1)$ | $(1,1,1,1)$ | $\frac{2}{11}$ | $\frac{81}{121}$ | $\frac{11}{5}$ |
| 1 | $\frac{2}{11}(1,1,1,1)$ | $\frac{9}{11}(-1,-1,1,1)$ | $\frac{1}{121}(180,180,-18,-18)$ | $\frac{11}{20}$ | 0 | $\frac{81}{55}$ |
| 2 | $\left(1,1,\frac{1}{10},\frac{1}{10}\right)$ | 0 | | | | 0 |
| Steepest descent (exact step $\alpha_k = \frac{2}{11}$ at every $k$) | | | | | | |
| $\|e_k\|_A^2$ for $k = 0,1,2,3$: $\frac{11}{5}$, $\frac{81}{55}$, $\frac{6561}{6655}$, $\frac{531441}{805255}$ (ratio $\frac{81}{121} = (\frac{9}{11})^2$ each step) | | | | | | |

**What it shows.** CG terminates after as many steps as $A$ has distinct eigenvalues, because $\mathcal{K}_k(A, r_0)$ cannot grow beyond dimension 2; steepest descent, from this $x_0$, contracts by exactly the worst-case factor at every step.

**6.5** PROOF *ch6, Theorem 2.3, pp. 7–8*

Let $\varphi(x) = \frac{1}{2} x^{\mathsf{T}} A x - b^{\mathsf{T}} x$ with $A \in \mathbb{R}^{n \times n}$ symmetric positive definite, $x^* = A^{-1} b$, and residual $r_k = A x_k - b = \nabla \varphi(x_k)$. The linear CG method is: $r_0 = A x_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^{\mathsf{T}} r_k}{p_k^{\mathsf{T}} A p_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k A p_k$, $\beta_{k+1} = \dfrac{r_{k+1}^{\mathsf{T}} r_{k+1}}{r_k^{\mathsf{T}} r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1} p_k$.
Let $\mathcal{K}_k(A, r_0) = \mathrm{span}\{r_0, A r_0, \ldots, A^{k-1} r_0\}$. Assume $r_0, \ldots, r_{k-1} \neq 0$. Prove:

a) $r_k^{\mathsf{T}} r_i = 0$ for all $i < k$;

b) $\mathrm{span}\{r_0, \ldots, r_{k-1}\} = \mathrm{span}\{p_0, \ldots, p_{k-1}\} = \mathcal{K}_k(A, r_0)$;

c) $x_k$ minimizes $\varphi$ over the affine set $x_0 + \mathcal{K}_k(A, r_0)$.

You may use 6.3.

**Note.** The induction in the notes only shows $\mathrm{span}\{r_0, \ldots, r_k\} \subset \mathcal{K}_{k+1}(A, r_0)$. Equality needs a dimension count: $r_0, \ldots, r_k$ are nonzero and mutually orthogonal, so they span a space of dimension $k+1$, while $\dim \mathcal{K}_{k+1} \leq k+1$. Item 3 also needs $\{p_0, \ldots, p_{k-1}\}$ to be a basis of $\mathcal{K}_k$ (linear independence, 6.1 a).

**6.7** PROOF *ch6, Observation, p. 8 (no proof in the notes)*

Let $\varphi(x) = \frac{1}{2} x^{\mathsf{T}} A x - b^{\mathsf{T}} x$ with $A \in \mathbb{R}^{n \times n}$ symmetric positive definite, $x^* = A^{-1} b$, and residual $r_k = A x_k - b = \nabla \varphi(x_k)$. The linear CG method is: $r_0 = A x_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^{\mathsf{T}} r_k}{p_k^{\mathsf{T}} A p_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k A p_k$, $\beta_{k+1} = \dfrac{r_{k+1}^{\mathsf{T}} r_{k+1}}{r_k^{\mathsf{T}} r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1} p_k$.
Let $\|v\|_A = \sqrt{v^{\mathsf{T}} A v}$ and $\mathcal{P}_k^1$ the polynomials $P$ of degree $\leq k$ with $P(0) = 1$.

a) Using 6.5 (3), prove that $\|x_k - x^*\|_A = \min_{P \in \mathcal{P}_k^1} \|P(A)(x_0 - x^*)\|_A$, and hence $\|x_k - x^*\|_A \leq \min_{P \in \mathcal{P}_k^1} \max_{\lambda \in \sigma(A)} |P(\lambda)| \; \|x_0 - x^*\|_A$.

b) Deduce that if $A$ has only $m$ distinct eigenvalues, CG terminates in at most $m$ iterations.

**6.6** STATEMENT *ch6, Theorem 2.4, p. 8 (no proof in the notes)*

Let $\varphi(x) = \frac{1}{2} x^{\mathsf{T}} A x - b^{\mathsf{T}} x$ with $A \in \mathbb{R}^{n \times n}$ symmetric positive definite, $x^* = A^{-1} b$, and residual $r_k = A x_k - b = \nabla \varphi(x_k)$. The linear CG method is: $r_0 = A x_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^{\mathsf{T}} r_k}{p_k^{\mathsf{T}} A p_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k A p_k$, $\beta_{k+1} = \dfrac{r_{k+1}^{\mathsf{T}} r_{k+1}}{r_k^{\mathsf{T}} r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1} p_k$.

Let $\kappa = \lambda_{\max}(A)/\lambda_{\min}(A)$ and $\|v\|_A = \sqrt{v^{\mathsf T} A v}$. Then

$$\|x_k - x^*\|_A \le 2\Big(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\Big)^k \|x_0 - x^*\|_A .$$

Be able to state it and compare it with the steepest-descent factor $\frac{\kappa-1}{\kappa+1}$ per step in the same norm (2.8). The proof is 6.20*.

**6.20*** Proposed exercise *ch6, p. 13 (exercise 7)*

Let $\varphi(x) = \frac12 x^{\mathsf T} A x - b^{\mathsf T} x$ with $A \in \mathbb{R}^{n\times n}$ symmetric positive definite, $x^* = A^{-1}b$, and residual $r_k = Ax_k - b = \nabla\varphi(x_k)$. The linear CG method is: $r_0 = Ax_0 - b$, $p_0 = -r_0$ and, for $k \ge 0$, $\alpha_k = \dfrac{r_k^{\mathsf T} r_k}{p_k^{\mathsf T} A p_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k A p_k$, $\beta_{k+1} = \dfrac{r_{k+1}^{\mathsf T} r_{k+1}}{r_k^{\mathsf T} r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1} p_k$. Let $\kappa = \lambda_{\max}(A)/\lambda_{\min}(A)$. The Chebyshev polynomials satisfy $T_k(t) = \cos(k \arccos t)$, so $|T_k(t)| \le 1$, for $|t| \le 1$, and $T_k(t) = \cosh(k\,\mathrm{arccosh}\, t) = \frac12\big[(t+\sqrt{t^2-1})^k + (t-\sqrt{t^2-1})^k\big]$ for $|t| > 1$. Using the scaled polynomial

$$P_k^*(\lambda) = \frac{T_k\big(\frac{\kappa+1-2\lambda/\lambda_{\min}}{\kappa-1}\big)}{T_k\big(\frac{\kappa+1}{\kappa-1}\big)}, \qquad \max_{\lambda\in[\lambda_{\min},\lambda_{\max}]} |P_k^*(\lambda)| = \frac{1}{T_k\big(\frac{\kappa+1}{\kappa-1}\big)},$$

prove that $\|x_k - x^*\|_A \le 2\Big(\dfrac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\Big)^k \|x_0 - x^*\|_A$. You may use 6.7 a).

**6.22*** Proposed exercise *ch6, p. 13 (exercise 9)*

For $\varphi(x) = \frac12 x^{\mathsf T} A x - b^{\mathsf T} x$ with $\kappa(A) = 1000$, compare analytically steepest descent with exact line search, whose error satisfies $\|x_{k+1} - x^*\|_A \le \frac{\kappa-1}{\kappa+1}\|x_k - x^*\|_A$, with linear CG, whose error satisfies the bound of 6.6. In particular, estimate how many iterations each bound needs to reduce $\|x_k - x^*\|_A$ by a factor $10^{-6}$.

**P18** Practice *added*

*Bound against actual behaviour.* Let $A = \mathrm{diag}(1, 12, 23, \dots, 100)$ ($n = 10$, $\lambda_i = 1 + 11(i-1)$, $\kappa = 100$), $b = (1, \dots, 1)^{\mathsf T}$, $x_0 = 0$, $\|v\|_A = \sqrt{v^{\mathsf T} A v}$. Run linear CG (A11) and steepest descent with exact step $\alpha_k = \dfrac{g_k^{\mathsf T} g_k}{g_k^{\mathsf T} A g_k}$ (A3), and tabulate $\|x_k - x^*\|_A / \|x_0 - x^*\|_A$ for both, together with $2\big(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\big)^k = 2(\frac{9}{11})^k$ and $\big(\frac{\kappa-1}{\kappa+1}\big)^k = (\frac{99}{101})^k$ (table V19). Which bound is tight, and why does CG stop at $k = 10$?

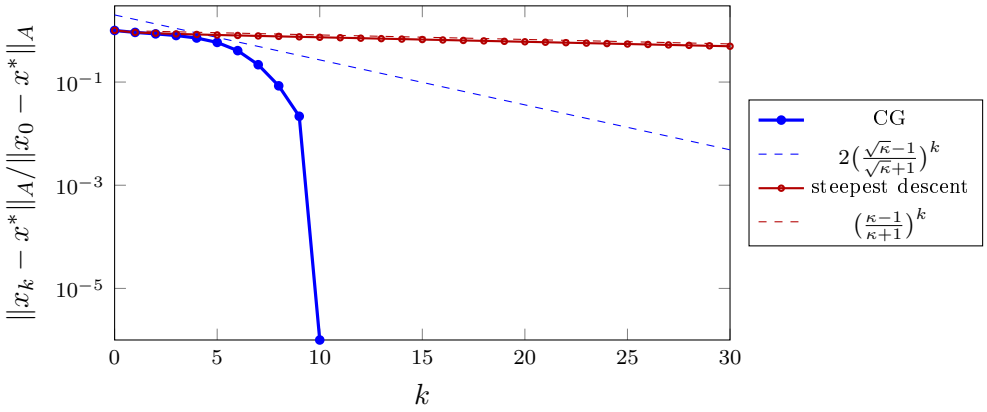
**V19** Validation table *solves P18*

$A = \mathrm{diag}(1, 12, \dots, 100)$, $b = (1, \dots, 1)$, $x_0 = 0$. Columns are $\|x_k - x^*\|_A / \|x_0 - x^*\|_A$ and the corresponding bounds.

| $k$ | CG | $2(\frac{9}{11})^k$ | steepest descent | $(\frac{99}{101})^k$ |
|---|---|---|---|---|
| 0 | 1 | 2 | 1 | 1 |
| 1 | 0.9171 | 1.6364 | 0.9171 | 0.9802 |
| 2 | 0.8595 | 1.3388 | 0.8825 | 0.9608 |
| 3 | 0.7980 | 1.0954 | 0.8576 | 0.9418 |
| 4 | 0.7144 | 0.8963 | 0.8368 | 0.9231 |
| 5 | 0.5869 | 0.7333 | 0.8180 | 0.9048 |
| 6 | 0.4065 | 0.6000 | 0.8006 | 0.8869 |
| 7 | 0.2169 | 0.4909 | 0.7840 | 0.8694 |
| 8 | $8.454 \times 10^{-2}$ | 0.4016 | 0.7681 | 0.8521 |
| 9 | $2.171 \times 10^{-2}$ | 0.3286 | 0.7526 | 0.8353 |
| 10 | $\approx 5 \times 10^{-16}$ | 0.2689 | 0.7376 | 0.8187 |
| 20 | | | 0.6037 | 0.6703 |
| 40 | | | 0.4046 | 0.4493 |

**What it shows.** The first CG step is a steepest-descent step (same value at $k = 1$); after that CG pulls away. The $\sqrt{\kappa}$ bound is pessimistic here because it ignores the finite spectrum, and CG ends at $k = n = 10$; steepest descent stays close to its own bound.

**Figure.**




$A = \mathrm{diag}(1, 12, 23, \ldots, 100)$ ($n = 10$, $\kappa = 100$), $b = (1, \ldots, 1)$, $x_0 = 0$. Steepest descent stays close to its bound, which only gives a factor $\approx 0.98$ per step. The CG bound gives $9/11 \approx 0.82$ per step, and the actual CG error falls faster than the bound and reaches 0 (rounding level) at $k = 10 = n$, as it must with 10 distinct eigenvalues.

## U20. Nonlinear conjugate gradient and L-BFGS

↑Index

ch6 develops only linear CG; nonlinear CG and L-BFGS appear in its solved and proposed exercises (they are developed in N&W, Sections 5.2 and 7.2).

**6.10** Numerical example *ch6, p. 10 (solved exercise 2)*

Nonlinear CG uses $p_0 = -g_0$, $p_k = -g_k + \beta_k p_{k-1}$ with $g_k = \nabla f(x_k)$ and either $\beta_k^{FR} = \dfrac{g_k^\mathsf{T} g_k}{g_{k-1}^\mathsf{T} g_{k-1}}$ (Fletcher–Reeves) or $\beta_k^{PR} = \dfrac{g_k^\mathsf{T}(g_k - g_{k-1})}{g_{k-1}^\mathsf{T} g_{k-1}}$ (Polak–Ribière). Let $g_{k-1} = (1, 2)^\mathsf{T}$ and $g_k = (0.5, 1)^\mathsf{T}$.

a) Compute $\beta_k^{FR}$ and $\beta_k^{PR}$.

b) With $p_{k-1} = -g_{k-1}$, compute $p_k$ for both choices and decide whether each is a descent direction ($g_k^\mathsf{T} p_k < 0$).

**Note.** Here $g_k = \frac{1}{2} g_{k-1}$, so $g_k^\mathsf{T} p_{k-1} \neq 0$: these gradients cannot come from an exact line search along $p_{k-1} = -g_{k-1}$. The example only exercises the formulas.

**6.17*** Proposed exercise *ch6, p. 12 (exercise 4)*

Nonlinear CG uses $p_0 = -g_0$, $p_k = -g_k + \beta_k p_{k-1}$ with $g_k = \nabla f(x_k)$ and either $\beta_k^{FR} = \dfrac{g_k^\mathsf{T} g_k}{g_{k-1}^\mathsf{T} g_{k-1}}$ (Fletcher–Reeves) or $\beta_k^{PR} = \dfrac{g_k^\mathsf{T}(g_k - g_{k-1})}{g_{k-1}^\mathsf{T} g_{k-1}}$ (Polak–Ribière). Let $f(x) = \frac{1}{2} x^\mathsf{T} A x - b^\mathsf{T} x$ with $A$ symmetric positive definite, and let $\alpha_k$ be the exact line-search step. Prove that $\beta_k^{FR} = \beta_k^{PR} = \dfrac{r_k^\mathsf{T} r_k}{r_{k-1}^\mathsf{T} r_{k-1}}$ with $r_k = g_k$, so that both methods generate the same iterates as linear CG.

**P19** Practice *added*

Nonlinear CG uses $p_0 = -g_0$, $p_k = -g_k + \beta_k p_{k-1}$ with $g_k = \nabla f(x_k)$ and either $\beta_k^{FR} = \dfrac{g_k^{\mathsf T} g_k}{g_{k-1}^{\mathsf T} g_{k-1}}$ (Fletcher–Reeves) or $\beta_k^{PR} = \dfrac{g_k^{\mathsf T}(g_k - g_{k-1})}{g_{k-1}^{\mathsf T} g_{k-1}}$ (Polak–Ribière). For $f(x) = \tfrac12 x^{\mathsf T} A x - b^{\mathsf T} x$ with $A = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}$, $b = (1,2)^{\mathsf T}$, $x_0 = 0$ and exact line search, compute $g_0$, $x_1$, $g_1$, $\beta_1^{FR}$ and $\beta_1^{PR}$, and check that both equal the linear CG value $\beta_1 = \frac{1}{16}$ of V17. Which term of $\beta^{PR}$ vanishes, and why?

**6.18*** Proposed exercise *ch6, p. 12 (exercise 5)*

Nonlinear CG uses $p_0 = -g_0$, $p_k = -g_k + \beta_k p_{k-1}$ with $g_k = \nabla f(x_k)$ and either $\beta_k^{FR} = \dfrac{g_k^{\mathsf T} g_k}{g_{k-1}^{\mathsf T} g_{k-1}}$ (Fletcher–Reeves) or $\beta_k^{PR} = \dfrac{g_k^{\mathsf T}(g_k - g_{k-1})}{g_{k-1}^{\mathsf T} g_{k-1}}$ (Polak–Ribière). Explain the theoretical reason why nonlinear CG should be restarted with $p_k = -g_k$ every $n$ iterations. Relate it to 6.17*: near a minimizer with $\nabla^2 f \succ 0$, $f$ is close to a quadratic, and conjugacy holds only for directions generated from a steepest-descent start.

**6.23*** Proposed exercise *ch6, p. 13 (exercise 10)*

Nonlinear CG uses $p_0 = -g_0$, $p_k = -g_k + \beta_k p_{k-1}$ with $g_k = \nabla f(x_k)$ and either $\beta_k^{FR} = \dfrac{g_k^{\mathsf T} g_k}{g_{k-1}^{\mathsf T} g_{k-1}}$ (Fletcher–Reeves) or $\beta_k^{PR} = \dfrac{g_k^{\mathsf T}(g_k - g_{k-1})}{g_{k-1}^{\mathsf T} g_{k-1}}$ (Polak–Ribière). Build a counterexample where $\beta_k^{PR} < 0$ and interpret the resulting direction $p_k$ geometrically (compare with 6.10). Explain why the variant $\beta_k^{PR+} = \max\{\beta_k^{PR}, 0\}$ is used in practice.

**6.21*** Proposed exercise *ch6, p. 13 (exercise 8)*

Nonlinear CG uses $p_0 = -g_0$, $p_k = -g_k + \beta_k p_{k-1}$ with $g_k = \nabla f(x_k)$ and either $\beta_k^{FR} = \dfrac{g_k^{\mathsf T} g_k}{g_{k-1}^{\mathsf T} g_{k-1}}$ (Fletcher–Reeves) or $\beta_k^{PR} = \dfrac{g_k^{\mathsf T}(g_k - g_{k-1})}{g_{k-1}^{\mathsf T} g_{k-1}}$ (Polak–Ribière). A problem has $n = 50\,000$ variables and is solved with Fletcher–Reeves. Write the one-dimensional line-search subproblem that determines $\alpha_0$, and give the first direction $p_0$ if $g_0 = (2, -1, 3, \dots)^{\mathsf T}$.

**Note.** The first direction does not depend on the choice of $\beta$: it is $p_0 = -g_0$.

**6.25*** Proposed exercise *ch6, p. 14 (exercise 12)*

L-BFGS with memory $m$ applies the inverse BFGS update $H_{k+1}^{\mathrm{BFGS}} = (I - \rho_k s_k y_k^{\mathsf T}) H_k (I - \rho_k y_k s_k^{\mathsf T}) + \rho_k s_k s_k^{\mathsf T}$, $\rho_k = 1/(y_k^{\mathsf T} s_k)$ $m$ times to an initial matrix $H_k^{(0)} = \gamma_k I$, $\gamma_k = \dfrac{s_{k-1}^{\mathsf T} y_{k-1}}{y_{k-1}^{\mathsf T} y_{k-1}}$, using only the last $m$ pairs $(s_i, y_i)$, $s_i = x_{i+1} - x_i$, $y_i = \nabla f(x_{i+1}) - \nabla f(x_i)$; it never forms $H_k$. Show why the scaling $\gamma_k = \dfrac{s_{k-1}^{\mathsf T} y_{k-1}}{y_{k-1}^{\mathsf T} y_{k-1}}$ makes the size of the step $-H_k g_k$ close to that of Newton's step. (For quadratic $f$, $y_{k-1} = A s_{k-1}$; relate $\gamma_k$ to $A^{-1}$ along the recent direction, and show $\gamma_k \in [1/\lambda_{\max}(A), 1/\lambda_{\min}(A)]$.)

**A12** PSEUDOCODE *L-BFGS: two-loop recursion with weak Wolfe search (N&W, Alg. 7.4–7.5)*

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**LBFGS**$(f,\ \nabla f,\ x_0;\ m, \varepsilon, k_{\max})$

---

1: $x \leftarrow x_0$; $g \leftarrow \nabla f(x)$; pair list $\mathcal{S} \leftarrow \emptyset$ (at most $m$ pairs $(s_i, y_i)$)
2: **for** $k = 0, 1, \ldots, k_{\max}$ **do**
3: **if** $\|g\| < \varepsilon$ **then return** $x$
4: $q \leftarrow g$; **for** $(s_i, y_i) \in \mathcal{S}$ from newest to oldest **do** $\rho_i \leftarrow 1/(y_i^{\mathsf{T}} s_i)$; $a_i \leftarrow \rho_i s_i^{\mathsf{T}} q$; $q \leftarrow q - a_i y_i$
5: **if** $\mathcal{S} \neq \emptyset$ **then** $\gamma \leftarrow s^{\mathsf{T}} y / (y^{\mathsf{T}} y)$ for the newest pair **else** $\gamma \leftarrow 1$; $r \leftarrow \gamma q$
6: **for** $(s_i, y_i) \in \mathcal{S}$ from oldest to newest **do** $b \leftarrow \rho_i y_i^{\mathsf{T}} r$; $r \leftarrow r + s_i(a_i - b)$
7: $p \leftarrow -r$ $(r = H_k g)$
8: $\alpha \leftarrow \text{WeakWolfe}(f, \nabla f, x, p, g; 1, 10^{-4}, 0.9)$
9: $s \leftarrow \alpha p$; $x_{\text{new}} \leftarrow x + s$; $g_{\text{new}} \leftarrow \nabla f(x_{\text{new}})$; $y \leftarrow g_{\text{new}} - g$
10: **if** $s^{\mathsf{T}} y > 0$ **then** append $(s, y)$ to $\mathcal{S}$, and drop the oldest pair if $|\mathcal{S}| > m$
11: $x \leftarrow x_{\text{new}}$; $g \leftarrow g_{\text{new}}$

---

WeakWolfe is A2. Storage: $2m$ vectors of length $n$ plus a few work vectors; no $n \times n$ matrix is formed. With the weak Wolfe conditions $s^{\mathsf{T}} y > 0$ always holds (3.3), so line 10 never discards a pair.

**P20** PRACTICE *added*

*Two-loop recursion by hand.* L-BFGS with memory $m$ applies the inverse BFGS update $H_{k+1}^{\text{BFGS}} = (I - \rho_k s_k y_k^{\mathsf{T}}) H_k (I - \rho_k y_k s_k^{\mathsf{T}}) + \rho_k s_k s_k^{\mathsf{T}}$, $\rho_k = 1/(y_k^{\mathsf{T}} s_k)$ $m$ times to an initial matrix $H_k^{(0)} = \gamma_k I$, $\gamma_k = \dfrac{s_{k-1}^{\mathsf{T}} y_{k-1}}{y_{k-1}^{\mathsf{T}} y_{k-1}}$, using only the last $m$ pairs $(s_i, y_i)$, $s_i = x_{i+1} - x_i$, $y_i = \nabla f(x_{i+1}) - \nabla f(x_i)$; it never forms $H_k$. Take $m = 1$ and the single pair $s_0 = (\frac{5}{4}, \frac{5}{8})^{\mathsf{T}}$, $y_0 = (\frac{25}{16}, \frac{15}{8})^{\mathsf{T}}$, with current gradient $g_1 = (-\frac{7}{16}, \frac{7}{8})^{\mathsf{T}}$. (These come from one exact-step iteration on $f(x) = \frac{5}{8}x_1^2 + \frac{3}{2}x_2^2 - 2x_1 - x_2$ from $x_0 = 0$ with $H_0 = I$.)

a) Run the two loops of A12 and obtain $r = H_1 g_1$.

b) Form $H_1 = (I - \rho_0 s_0 y_0^{\mathsf{T}})\gamma_1 I (I - \rho_0 y_0 s_0^{\mathsf{T}}) + \rho_0 s_0 s_0^{\mathsf{T}}$ explicitly and check $H_1 g_1 = r$ and $H_1 y_0 = s_0$.

c) Compare $p_1 = -H_1 g_1$ with the full BFGS direction $(\frac{105}{128}, -\frac{175}{256})^{\mathsf{T}}$ obtained with $H_0 = I$. Explain why the first loop gives $a_0 = 0$.

**V20** VALIDATION TABLE *A12, two loops with $m = 1$; solves P20*

$s_0 = (\frac{5}{4}, \frac{5}{8})$, $y_0 = (\frac{25}{16}, \frac{15}{8})$, $g_1 = (-\frac{7}{16}, \frac{7}{8})$.

| quantity | value |
|---|---|
| $\rho_0 = 1/(y_0^{\mathsf{T}} s_0)$ | $\frac{8}{25}$ |
| first loop: $a_0 = \rho_0 s_0^{\mathsf{T}} g_1$, $q = g_1 - a_0 y_0$ | $a_0 = 0$, $q = \left(-\frac{7}{16},\ \frac{7}{8}\right)$ |
| $\gamma_1 = s_0^{\mathsf{T}} y_0 / (y_0^{\mathsf{T}} y_0)$, $r = \gamma_1 q$ | $\gamma_1 = \frac{32}{61}$, $r = \left(-\frac{14}{61},\ \frac{28}{61}\right)$ |
| second loop: $b = \rho_0 y_0^{\mathsf{T}} r$, $r \leftarrow r + s_0(a_0 - b)$ | $b = \frac{49}{305}$, $r = \left(-\frac{105}{244},\ \frac{175}{488}\right)$ |
| explicit $H_1$ | $\begin{pmatrix} \frac{53}{61} & -\frac{7}{122} \\ -\frac{7}{122} & \frac{93}{244} \end{pmatrix}$, $H_1 y_0 = \left(\frac{5}{4},\ \frac{5}{8}\right) = s_0$ |
| $p_1 = -H_1 g_1$ | $\left(\frac{105}{244},\ -\frac{175}{488}\right) = \frac{32}{61}\left(\frac{105}{128},\ -\frac{175}{256}\right)$ |

**What it shows.** The two loops reproduce the BFGS update of $\gamma_1 I$ using only inner products and

vector updates. Because the previous step was exact, $s_0^\mathsf{T} g_1 = 0$ and the first loop does nothing; the result is the full BFGS direction scaled by $\gamma_1$, which an exact line search then compensates.

**V21** VALIDATION TABLE *A12 with $m = 2$; compare with V8*

$f(x) = x_1^4 + x_1x_2 + (1+x_2)^2$, $\nabla f(x) = (4x_1^3 + x_2,\ x_1 + 2(1+x_2))^\mathsf{T}$, $x^* \approx (0.695884, -1.347942)^\mathsf{T}$; $x_0 = (1,1)^\mathsf{T}$, $m = 2$, WeakWolfe with $c_1 = 10^{-4}$, $c_2 = 0.9$, $\alpha_{\text{ini}} = 1$; $\varepsilon = 10^{-6}$; $\gamma_0 = 1$. "eval." is the number of steps tried in A2 and $e_k = \|x_k - x^*\|$.

| $k$ | $x_k$ | $f(x_k)$ | $\|g_k\|$ | $\gamma_k$ | $\alpha_k$ | eval. | $e_k$ |
|---|---|---|---|---|---|---|---|
| 0 | (1, 1) | 6 | 7.0711 | 1 | 0.25 | 3 | 2.3676 |
| 1 | (−0.25, −0.25) | 0.628906 | 1.2885 | 0.267898 | 1 | 1 | 1.4492 |
| 2 | (−0.188182, −0.650076) | 0.246034 | 0.84839 | 0.402586 | 1 | 1 | 1.1263 |
| 3 | (0.159432, −1.155473) | −0.159401 | 1.1493 | 0.266752 | 0.25 | 3 | 0.56994 |
| 4 | (0.611213, −1.590987) | −0.483603 | 0.88598 | 1.005852 | 1 | 1 | 0.25737 |
| 5 | (0.847744, −1.467863) | −0.508990 | 0.97311 | 0.152450 | 1 | 1 | 0.19350 |
| 6 | (0.636341, −1.199139) | −0.559437 | 0.29163 | 0.234295 | 1 | 1 | 0.16027 |
| 7 | (0.686236, −1.330481) | −0.582041 | $4.5497 \times 10^{-2}$ | 0.552865 | 1 | 1 | $1.9949 \times 10^{-2}$ |
| 8 | (0.699374, −1.346888) | −0.582405 | $2.2154 \times 10^{-2}$ | 0.282448 | 1 | 1 | $3.6455 \times 10^{-3}$ |
| 9 | (0.695264, −1.349625) | −0.582440 | $6.6215 \times 10^{-3}$ | 0.168842 | 1 | 1 | $1.7936 \times 10^{-3}$ |
| 10 | (0.695905, −1.348008) | −0.582445 | $1.2292 \times 10^{-4}$ | 0.222591 | 1 | 1 | $6.8534 \times 10^{-5}$ |
| 11 | (0.695898, −1.347962) | −0.582445 | $6.6807 \times 10^{-5}$ | 0.528218 | 1 | 1 | $2.3837 \times 10^{-5}$ |
| 12 | (0.695880, −1.347942) | −0.582445 | $2.6836 \times 10^{-5}$ | 0.248886 | 1 | 1 | $4.6924 \times 10^{-6}$ |
| 13 | (0.695885, −1.347942) | −0.582445 | $1.6583 \times 10^{-6}$ | 0.166758 | 1 | 1 | $5.3385 \times 10^{-7}$ |
| 14 | (0.695884, −1.347942) | −0.582445 | $2.17 \times 10^{-7}$ | | | | $1.47 \times 10^{-7}$ |

Detail of $k = 1$: $s_0 = (-1.25, -1.25)$, $y_0 = (-5.3125, -3.75)$ and $\gamma_1 = 0.267898$, which give the direction $p_1 = (0.061818, -0.400076)$. The BFGS direction of V8 at the same point is (0.584126, −1.140012); they differ because L-BFGS starts from $\gamma_1 I$ instead of $I$. With $m = 1$ the same run needs 16 iterations.

**What it shows.** L-BFGS restarts from $\gamma_k I$ at every iteration, so even with $m = n = 2$ it is not BFGS: it needs more iterations (14 with $m = 2$, 16 with $m = 1$, against 9 for BFGS in V8) but stores only $2m$ vectors. The unit step is accepted from $k = 4$ on, which is what the scaling $\gamma_k$ is for (6.25*).

**6.19*** PROPOSED EXERCISE *ch6, p. 12 (exercise 6)*

L-BFGS with memory $m$ applies the inverse BFGS update $H_{k+1}^{\text{BFGS}} = (I - \rho_k s_k y_k^\mathsf{T}) H_k (I - \rho_k y_k s_k^\mathsf{T}) + \rho_k s_k s_k^\mathsf{T}$, $\rho_k = 1/(y_k^\mathsf{T} s_k)$ $m$ times to an initial matrix $H_k^{(0)} = \gamma_k I$, $\gamma_k = \dfrac{s_{k-1}^\mathsf{T} y_{k-1}}{y_{k-1}^\mathsf{T} y_{k-1}}$, using only the last $m$ pairs $(s_i, y_i)$, $s_i = x_{i+1} - x_i$, $y_i = \nabla f(x_{i+1}) - \nabla f(x_i)$; it never forms $H_k$. Write detailed pseudocode for the complete L-BFGS method with a Wolfe line search. (A12 is one answer.)

**6.11** NUMERICAL EXAMPLE *ch6, p. 11 (solved exercise 4)*

L-BFGS with memory $m$ applies the inverse BFGS update $H_{k+1}^{\text{BFGS}} = (I - \rho_k s_k y_k^\mathsf{T}) H_k (I - \rho_k y_k s_k^\mathsf{T}) + \rho_k s_k s_k^\mathsf{T}$, $\rho_k = 1/(y_k^\mathsf{T} s_k)$ $m$ times to an initial matrix $H_k^{(0)} = \gamma_k I$, $\gamma_k = \dfrac{s_{k-1}^\mathsf{T} y_{k-1}}{y_{k-1}^\mathsf{T} y_{k-1}}$, using only the last $m$ pairs $(s_i, y_i)$, $s_i = x_{i+1} - x_i$, $y_i = \nabla f(x_{i+1}) - \nabla f(x_i)$; it never forms $H_k$. The two-loop recursion (A12, lines 4–6) computes $H_k g_k$ from $m$ pairs $(s_i, y_i) \in \mathbb{R}^n \times \mathbb{R}^n$. Count the floating-point operations per iteration and show that the cost is linear in $n$.

**Note.** The notes announce "at most $4mn + O(m)$" but their own count gives $8mn + O(n)$. Both are linear in $n$; the difference is the convention: counting each inner product and each vector update as $2n$ flops gives $8mn$, while N&W's $4mn$ counts multiplications only.

**6.12** NUMERICAL EXAMPLE *ch6, p. 11 (solved exercise 5)*

L-BFGS with memory $m$ applies the inverse BFGS update $H_{k+1}^{\mathrm{BFGS}} = (I-\rho_k s_k y_k^{\mathsf{T}})H_k(I-\rho_k y_k s_k^{\mathsf{T}})+\rho_k s_k s_k^{\mathsf{T}}$, $\rho_k = 1/(y_k^{\mathsf{T}} s_k)$ $m$ times to an initial matrix $H_k^{(0)} = \gamma_k I$, $\gamma_k = \dfrac{s_{k-1}^{\mathsf{T}} y_{k-1}}{y_{k-1}^{\mathsf{T}} y_{k-1}}$, using only the last $m$ pairs $(s_i, y_i)$, $s_i = x_{i+1} - x_i$, $y_i = \nabla f(x_{i+1}) - \nabla f(x_i)$; it never forms $H_k$. A problem has $n = 10^5$ variables and L-BFGS is used with $m = 5$. Compute the memory needed to store the pairs $(s_i, y_i)$ in 64-bit floating point, and compare with storing a dense $n \times n$ matrix $H_k$.

**6.24*** PROPOSED EXERCISE *ch6, p. 14 (exercise 11)*

L-BFGS with memory $m$ applies the inverse BFGS update $H_{k+1}^{\mathrm{BFGS}} = (I-\rho_k s_k y_k^{\mathsf{T}})H_k(I-\rho_k y_k s_k^{\mathsf{T}})+\rho_k s_k s_k^{\mathsf{T}}$, $\rho_k = 1/(y_k^{\mathsf{T}} s_k)$ $m$ times to an initial matrix $H_k^{(0)} = \gamma_k I$, $\gamma_k = \dfrac{s_{k-1}^{\mathsf{T}} y_{k-1}}{y_{k-1}^{\mathsf{T}} y_{k-1}}$, using only the last $m$ pairs $(s_i, y_i)$, $s_i = x_{i+1} - x_i$, $y_i = \nabla f(x_{i+1}) - \nabla f(x_i)$; it never forms $H_k$. A problem with $n = 10^6$ variables is solved with L-BFGS on a machine with 4 GB of RAM. What is the largest memory parameter $m$ that can be used?

**Note.** State whether 4 GB means $4 \cdot 10^9$ or $4 \cdot 2^{30}$ bytes, and whether the vectors $x$, $g$, $p$ and the work vectors of the two loops are counted; the answer changes slightly.